

Differential Invariant Theory of Quantum Ansätze

Cumulative discovery log, Rounds 1–3

Research notes

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Abstract

This document appends a new round of experiments and theory-building to the earlier search for hidden mathematical laws in parameterized quantum circuits. The earlier rounds found that Eulerian CZ graph ansätze preserve a spin-flip bilinear form and are trapped in orthogonal or symplectic subgroups. Round 3 generalizes this from graphs to arbitrary diagonal phase layers, Boolean phase truth tables, and hypergraph phase gadgets. The central new discovery is a complement-pair phase law: a diagonal phase layer preserves the hidden concurrence form exactly when the sum of phases on every bit string and its complement is constant modulo 2π . Graph Eulerianity is only the degree-2 shadow of this more general law.

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1 Executive summary of the target

The emerging theory studies a smooth quantum synthesis map

$$F : \Theta \rightarrow \mathbb{CP}^{2^n-1}, \quad \theta \mapsto U(\theta)|0^n\rangle,$$

using differential rank, singular values, and exact algebraic invariants. The core blind spot is that a parameterized circuit can have enough parameters to fill Hilbert space by naive counting, yet be trapped inside a hidden invariant variety or subgroup.

Let

$$N = 2^n, \quad S_n = (-i\sigma_y)^{\otimes n}, \quad B_n(\psi) = \psi^T S_n \psi.$$

The previous rounds found that local $SU(2)$ rotations plus an Eulerian CZ graph preserve B_n up to sign. For even n , this gives a state-level defect: starting from $|0^n\rangle$, the ansatz remains in the quadric

$$B_n(\psi) = 0,$$

which has two fewer real dimensions than full projective Hilbert space. At the unitary level the restriction is stronger:

$$U(\theta) \in \begin{cases} SO(N), & n \text{ even}, \\ Sp(N/2), & n \text{ odd}, \end{cases}$$

up to a global phase and change of basis.

Round 3 expands this from CZ graphs to arbitrary diagonal phase layers. The new theory target is a *complement-pair invariant calculus* for deciding when an ansatz is full, orthogonal, symplectic, or trapped in a smaller exceptional subgroup.

2 Round 1 recap: Eulerian concurrence-defect law

For a graph G with CZ entangler

$$E_G|x\rangle = (-1)^{q_G(x)}|x\rangle, \quad q_G(x) = \sum_{(i,j) \in E(G)} x_i x_j \pmod{2},$$

we found

$$q_G(x) + q_G(\bar{x}) \equiv |E(G)| + \sum_v d_v x_v \pmod{2}.$$

Thus if every degree d_v is even, then $q_G(x) + q_G(\bar{x})$ is independent of x modulo 2. Therefore

$$E_G^T S_n E_G = (-1)^{|E(G)|} S_n.$$

Since local $SU(2)$ gates preserve $-i\sigma_y$, the whole circuit preserves B_n up to sign.

Candidate Law 1 (Eulerian concurrence-defect law). *For the layered ansatz*

$$F_{G,L}(\theta) = \left[\prod_{\ell=1}^L E_G \prod_{j=1}^n R_z(\phi_{\ell j}) R_y(\alpha_{\ell j}) \right] |0^n\rangle,$$

experiments support

$$\text{rank } dF_{G,L} = \min(2nL, 2N - 2 - 2\mathbf{1}_{\{n \text{ even and } G \text{ Eulerian}\}}),$$

except at degenerate no-entanglement choices.

The exact upper bound is proven by the invariant. The generic equality is still a conjectural rank-saturation statement, verified numerically for the tested graph families.

3 Round 2 recap: graph-phase symmetry defect law

State preparation was only the shadow of a larger unitary obstruction. For Eulerian CZ graphs, the generated unitaries lie in the concurrence-preserving group

$$K_n = \{U \in \text{SU}(N) : U^T S_n U = S_n\}.$$

This group is isomorphic to

$$K_n \cong \begin{cases} \text{SO}(N), & n \text{ even,} \\ \text{Sp}(N/2), & n \text{ odd.} \end{cases}$$

The state-level consequence differs by parity. For even n , $B_n(\psi) = 0$ is a nontrivial complex equation, producing a two-real-dimensional defect. For odd n , S_n is antisymmetric, so $\psi^T S_n \psi = 0$ for all ψ ; the state map can appear full while the unitary map remains trapped in $\text{Sp}(N/2)$.

The experimental unitary ranks matched the hidden group dimensions:

Case	n	observed rank	full $\text{SU}(N)$ dim.	hidden dim.
Triangle C_3 CZ, Eulerian	3	36	63	$\dim \text{Sp}(4) = 36$
Chain on 3 vertices, non-Eulerian	3	63	63	–
Cycle C_4 CZ, Eulerian	4	120	255	$\dim \text{SO}(16) = 120$
Chain on 4 vertices, non-Eulerian	4	255	255	–

4 Round 3 appended findings: complement-pair phase theory

4.1 New object: arbitrary diagonal phase layers

Replace the graph CZ layer by an arbitrary diagonal phase layer

$$D_\varphi |x\rangle = e^{i\varphi(x)} |x\rangle,$$

where $x \in \{0, 1\}^n$ and $\bar{x}$ denotes the bitwise complement.

Define the complement-pair phase sum

$$\Delta_\varphi(x) = \varphi(x) + \varphi(\bar{x}) \pmod{2\pi}.$$

Proposition 1 (Complement-pair phase criterion). *The diagonal phase layer D_φ preserves the spin-flip bilinear form up to a scalar if and only if $\Delta_\varphi(x)$ is independent of x :*

$$D_\varphi^T S_n D_\varphi = e^{ic} S_n \iff \varphi(x) + \varphi(\bar{x}) \equiv c \pmod{2\pi} \quad \forall x.$$

Proof sketch. In the computational basis, S_n pairs x only with $\bar{x}$, up to the sign $(-1)^{|x|}$. Therefore

$$(D_\varphi^T S_n D_\varphi)_{x, \bar{x}} = S_{n, x, \bar{x}} e^{i(\varphi(x) + \varphi(\bar{x}))}.$$

This is a scalar multiple of S_n exactly when the multiplier is the same for every complement pair.

This is the new master obstruction. Eulerian graph CZ circuits are just the special case where $\varphi(x) = \pi q_G(x)$.

4.2 Boolean phase version

For a Boolean phase function $f : \{0, 1\}^n \rightarrow \mathbb{F}_2$ with

$$D_f|x\rangle = (-1)^{f(x)}|x\rangle,$$

the criterion becomes

$$\boxed{f(x) + f(\bar{x}) = c \pmod{2} \text{ for all } x.}$$

I call such a phase function *complement-flat*. If the left side is not constant, I call it *complement-curved*.

Candidate Law 2 (Complement-pair symmetry defect law). *Let a layered ansatz use arbitrary local SU(2) rotations and one fixed entangling diagonal phase D_φ . If D_φ is complement-flat, then the projective unitary reachability is bounded by*

$$\dim K_n = \begin{cases} \frac{N(N-1)}{2}, & n \text{ even}, \\ \frac{N}{2}(N+1), & n \text{ odd}. \end{cases}$$

If D_φ is complement-curved and not otherwise degenerate, experiments support generic release to full SU(N):

$$\text{rank } dU_L = \min(2nL, N^2 - 1).$$

For state preparation, experiments support

$$\text{rank } dF_L = \min(2nL, 2N - 2 - 2\mathbf{1}_{\{n \text{ even and } D_\varphi \text{ complement-flat}\}}).$$

5 Hypergraph phase gadgets

A hypergraph phase gadget is described by

$$f_H(x) = \sum_{E \in H} \prod_{i \in E} x_i \pmod{2},$$

where H is a collection of hyperedges. Edges of size 2 give CZ; edges of size 3 give CCZ; edges of size 4 give CCCZ, and so on.

Over $\mathbb{F}_2$,

$$\prod_{i \in E} (1 + x_i) = \sum_{T \subseteq E} \prod_{i \in T} x_i.$$

Therefore

$$f_H(x) + f_H(\bar{x}) = \sum_{E \in H} \sum_{T \subsetneq E} \prod_{i \in T} x_i.$$

All nonconstant monomials must vanish. Hence:

Candidate Law 3 (Strict-containment parity law for hypergraph phases). *A Boolean hypergraph phase f_H is complement-flat if and only if for every nonempty vertex subset T ,*

$$\boxed{\#\{E \in H : T \subsetneq E\} \equiv 0 \pmod{2}.}$$

For ordinary graphs this reduces exactly to Eulerianity: every vertex degree is even.

This is the cleanest new structural equation from Round 3. It converts a hidden quantum reachability question into a finite parity check on the hypergraph.

5.1 Complete uniform hypergraphs

For the complete k -uniform hypergraph $K_n^{(k)}$, every j -set T is strictly contained in

$$\binom{n-j}{k-j}$$

k -edges. Therefore $K_n^{(k)}$ is complement-flat at Boolean phase π if and only if

$$\binom{n-j}{k-j} \text{ is even for every } j = 1, \dots, k-1.$$

For $k = 3$, this reduces to

$$\binom{n-1}{2} \text{ even and } n-2 \text{ even,}$$

which is equivalent to

$$n \equiv 2 \pmod{4}.$$

Thus the complete CCZ hypergraph on 6, 10, 14, ... qubits is complement-flat at phase π .

6 Phase resonance law

The complement-pair criterion also works for non-Boolean phases. Suppose

$$D_{h,\gamma}|x\rangle = e^{i\gamma h(x)}|x\rangle,$$

where $h(x)$ is an integer-valued phase count. Define

$$r_h(x) = h(x) + h(\bar{x}).$$

Then $D_{h,\gamma}$ preserves S_n up to scalar exactly when

$$\gamma(r_h(x) - r_h(y)) \in 2\pi\mathbb{Z} \quad \forall x, y.$$

Let

$$g_h = \gcd_x(r_h(x) - r_h(0)).$$

If $g_h = 0$, then r_h is constant and every phase γ is resonant. If $g_h > 0$, then the resonance phases are

$$\gamma = \frac{2\pi m}{g_h}, \quad m \in \mathbb{Z}.$$

For the complete 3-uniform hypergraph,

$$r_h(m) = \binom{m}{3} + \binom{n-m}{3} = \binom{n}{3} + \frac{n-2}{2}m(m-n),$$

where $m = |x|$. The experiments and algebra give

$$g_h = \begin{cases} n-2, & n \text{ odd,} \\ (n-2)/2, & n \text{ even.} \end{cases}$$

For $n = 10$, this predicts a new exceptional target:

$$K_{10}^{(3)} \text{ at phase } \gamma = \pi/2$$

should be trapped in $\text{SO}(1024)$ rather than full $\text{SU}(1024)$. That is a massive hidden restriction:

$$\dim \text{SO}(1024) = 523,776, \quad \dim \text{SU}(1024) = 1,048,575.$$

7 How I tested it

I ran state-vector and unitary simulations in Python with analytic derivatives for R_y and R_z . For state synthesis I propagated both the state and all tangent vectors $\partial\psi/\partial\theta_j$. I projected out the physically meaningless radial/global-phase direction by

$$\partial_j\psi \leftarrow \partial_j\psi - \psi\langle\psi, \partial_j\psi\rangle.$$

Then I formed the real Jacobian from real and imaginary parts and computed singular values.

For unitary synthesis I propagated $U(\theta)$ and $\partial_j U$, then used the right-trivialized tangent

$$A_j = (\partial_j U)U^\dagger$$

and removed global phase by

$$A_j \leftarrow A_j - \frac{\text{tr}(A_j)}{N}I.$$

Numerical rank used the threshold

$$\sigma_i > 10^{-8}\sigma_1.$$

Essential code pattern:

```
for dps in derivative_states:
    dproj = dps - psi * np.vdot(psi, dps)
    J_columns.append(np.r_[dproj.real, dproj.imag])

s = np.linalg.svd(np.column_stack(J_columns), compute_uv=False)
rank = np.sum(s > s[0] * 1e-8)
```

The complement-flat test for a Boolean phase truth table was:

```
def complement_flat(f, n):
    mask = (1 << n) - 1
    c = f[0] ^ f[mask]
    return all((f[x] ^ f[mask ^ x]) == c for x in range(1 << n))
```

For a hypergraph, the equivalent strict-containment test was:

```
for nonempty_subset_T in all_subsets(vertices):
    parity = sum(1 for E in hyperedges if T < E) % 2
    if parity == 1:
        return False
return True
```

8 Round 3 experiments

8.1 Hypergraph state-rank tests

The target state dimension is $2N - 2$. At the threshold depth $L = \lceil (2N - 2)/(2n) \rceil$, the following ranks were observed.

Phase gadget	n	complement-flat?	target dim.	observed rank	defect
Cycle graph C_4	4	yes	30	28	2
Chain graph on 4 vertices	4	no	30	30	0
All triples on 4 vertices	4	no	30	30	0
All triples on 5 vertices	5	no	62	62	0
All triples on 6 vertices	6	yes	126	124	2
All quadruples on 5 vertices	5	no	62	62	0
All quadruples on 6 vertices	6	no	126	126	0

This supports the hypergraph parity law. The all-triples $n = 6$ case is especially important: it is not a graph Eulerian effect. It is a genuine higher-order CCZ-type obstruction.

8.2 Unitary-rank tests

Phase gadget	n	complement-flat?	observed rank	hidden dim.	full $SU(N)$ dim.
Cycle graph C_4	4	yes	120	120	255
Cycle graph C_4 , deeper	4	yes	120	120	255
Chain graph on 4 vertices	4	no	255	–	255
All triples on 4 vertices	4	no	255	–	255
Complete graph K_5	5	yes	528	528	1023
Random complement-flat signs	5	yes	528	528	1023

The K_5 complete graph result is an odd-qubit example: state preparation is not defective, but unitary synthesis is trapped in $Sp(16)$, dimension 528.

8.3 Random diagonal signs: a stronger blind spot

I then stopped using graph or hypergraph structure and generated random diagonal sign phases $D_f|x\rangle = (-1)^{f(x)}|x\rangle$. Some random truth tables were forced to satisfy $f(x) + f(\bar{x}) = c$; others were unconstrained.

$n = 4$ random phase type	flat?	state rank	defect	unitary rank
Random complement-symmetric signs, $c = 0$	yes	28	2	120
Random complement-antisymmetric signs, $c = 1$	yes	28	2	120
One-pair breaker	no	30	0	255
Fully random signs	no	30	0	255

The expected unitary sector is $SO(16)$ for the flat rows and full $SU(16)$ for the non-flat rows.

For $n = 5$, a random complement-flat sign phase had full state rank 62 but unitary rank 528, exactly $\dim Sp(16)$.

This is exceptional: a diagonal phase layer can look random and high-complexity while being exactly non-universal because its complement pairs have a hidden parity symmetry.

8.4 One-pair breaker experiment

Starting with a complement-symmetric random sign phase on $n = 4$, I flipped one side of k complement pairs. There are 8 complement pairs. Define the complement-pair curvature

$$\eta(D) = \sqrt{1 - \left| \frac{1}{N} \sum_x e^{i(\varphi(x) + \varphi(\bar{x}))} \right|^2}.$$

This is zero exactly when the complement-pair phase sum is constant.

Broken pairs k	$\eta(D)$	state rank	defect	unlocked sv 1	unlocked sv 2
0	0.000000	28	2	$1.15 \cdot 10^{-16}$	$6.42 \cdot 10^{-17}$
1	0.661438	30	0	$2.46 \cdot 10^{-2}$	$1.89 \cdot 10^{-2}$
2	0.866025	30	0	$6.93 \cdot 10^{-3}$	$1.49 \cdot 10^{-3}$
3	0.968246	30	0	$1.33 \cdot 10^{-2}$	$6.41 \cdot 10^{-3}$
4	1.000000	30	0	$2.15 \cdot 10^{-3}$	$1.60 \cdot 10^{-3}$
5	0.968246	30	0	$1.77 \cdot 10^{-2}$	$9.05 \cdot 10^{-3}$
6	0.866025	30	0	$4.13 \cdot 10^{-3}$	$3.79 \cdot 10^{-3}$
7	0.661438	30	0	$1.82 \cdot 10^{-2}$	$5.01 \cdot 10^{-3}$
8	0.000000	28	2	$7.76 \cdot 10^{-17}$	$2.51 \cdot 10^{-17}$

The $k = 8$ case is not a contradiction. Flipping one side of every complement pair turns a complement-symmetric phase into a complement-antisymmetric phase: $f(x) + f(\bar{x}) = 1$ for all x . Thus the invariant returns. This is a sharp binary law: not “how many pairs are broken,” but whether all complement pairs have the same phase sum.

Unitary checkpoints agreed:

$$k = 0 \Rightarrow \text{rank} = 120, \quad k = 1 \Rightarrow \text{rank} = 255, \quad k = 8 \Rightarrow \text{rank} = 120.$$

8.5 Phase resonance: complete triples on six qubits

For the complete 3-uniform hypergraph on 6 vertices, the Boolean phase at $\gamma = \pi$ is complement-flat, but $\gamma = \pi/2$ is not resonant. The state-rank sweep showed exactly this.

Phase γ	observed state rank	defect from 126
0	12	114
$\pi/2$	126	0
$\pi - 10^{-3}$	126	0
π	124	2
$\pi + 10^{-3}$	126	0
2π	12	114

At $\gamma = 0$ and 2π the entangler is trivial, so the rank collapses to the local-rotation manifold. At $\gamma = \pi$ the concurrence invariant appears. Near π , rank is full but two singular values are small, indicating an approximate invariant wall.

9 Most exceptional new findings

9.1 Finding 1: Eulerianity is not the real law

Eulerian graph parity is the degree-2 case of a much larger complement-pair phase symmetry. The true invariant condition is

$$\varphi(x) + \varphi(\bar{x}) = \text{constant} \pmod{2\pi}.$$

This applies to arbitrary diagonal phases, not just CZ graphs.

9.2 Finding 2: hypergraph parity gives higher-order hidden subgroups

For CCZ and higher hypergraph phases, the hidden subgroup test is:

$$\#\{E : T \subsetneq E\} \equiv 0 \pmod{2} \quad \forall T \neq \emptyset.$$

This is a new practical equation for ansatz design. It predicts exactly when higher-order phase gadgets are trapped in $\text{SO}(N)$ or $\text{Sp}(N/2)$.

9.3 Finding 3: random-looking phases can be exactly non-universal

A random diagonal sign layer with complement-pair symmetry produced the same rank defect as a structured Eulerian graph. This means the obstruction is not visually graph-theoretic; it can be hidden in a random truth table.

9.4 Finding 4: one complement-pair defect can unlock full $\text{SU}(N)$

For $n = 4$, flipping a single complement pair changed the unitary rank from 120 to 255. That suggests a useful design principle:

one targeted complement-pair breaker can destroy the hidden SO/Sp trap.

But flipping every pair restores a different flat symmetry. The decisive object is not local randomness; it is the global complement-pair phase sum.

9.5 Finding 5: dense complete CCZ systems have rational hidden resonances

The complete 3-uniform hypergraph has resonance phases determined by

$$g_h = \begin{cases} n - 2, & n \text{ odd}, \\ (n - 2)/2, & n \text{ even}. \end{cases}$$

For $n = 10$, the phase $\pi/2$ is predicted to be a hidden $\text{SO}(1024)$ trap. This is a stunning high-dimensional target for proof and future simulation.

10 Applications now visible

Ansatz compiler diagnostics. Before training a variational circuit, compute $\Delta_\varphi(x) = \varphi(x) + \varphi(\bar{x})$. If constant, the circuit is in a hidden SO/Sp sector. If even n , generic state synthesis is impossible from $|0^n\rangle$.

Hypergraph phase design. For hypergraph ansatze, the strict-containment parity law gives a fast symbolic test. To avoid the invariant trap, deliberately include a hyperedge pattern that makes at least one strict-containment parity odd, but not uniformly odd across all complement pairs.

Training-conditioning diagnosis. Near resonant phases such as $\gamma = \pi$ for Eulerian CZ or complete $K_6^{(3)}$ CCZ, the rank can be full but have tiny singular values. This suggests an approximate invariant wall and predicts poor conditioning:

$$\kappa(J) \sim \frac{1}{\text{dist}(\gamma, \Gamma_{\text{resonance}})}.$$

Quantum state synthesis lower bounds. If a circuit family is complement-flat and n is even, it cannot synthesize states with $B_n(\psi) \neq 0$ from $|0^n\rangle$. This gives an exact algebraic lower bound independent of optimizer quality.

Hardware-aware topology. Fixed entangling layers on quantum devices should be checked for complement-pair flatness. A symmetric or Eulerian coupling pattern may be elegant physically but defective mathematically.

11 Theory program after Round 3

The theory now has a sharper central object.

Candidate Law 4 (Proposed master reachability equation). *For a generic layered local-rotation ansatz with a fixed diagonal entangler D_φ , define*

$$\chi_b(D_\varphi) = \begin{cases} 1, & \varphi(x) + \varphi(\bar{x}) \text{ is constant mod } 2\pi, \\ 0, & \text{otherwise.} \end{cases}$$

Then the expected generic state rank is

$$\text{rank } dF_L = \min(2nL, 2N - 2 - 2\chi_b(D_\varphi)\mathbf{1}_{\{n \text{ even}\}}),$$

and the expected generic unitary rank is

$$\text{rank } dU_L = \begin{cases} \min\left(2nL, \frac{N(N-1)}{2}\right), & \chi_b = 1, n \text{ even}, \\ \min\left(2nL, \frac{N}{2}(N+1)\right), & \chi_b = 1, n \text{ odd}, \\ \min(2nL, N^2 - 1), & \chi_b = 0, \end{cases}$$

except for degenerate entanglers, disconnected/local-only structures, or additional accidental symmetries.

The next proof target is the equality statement: once the invariant subgroup is identified, prove that generic local rotations plus a nondegenerate complement-flat entangler generate the whole $\text{SO}(N)$ or $\text{Sp}(N/2)$ Lie algebra, and that a generic complement-curved entangler generates all of $\mathfrak{su}(N)$.

12 References and context

The spin-flip bilinear form and its relation to concurrence-preserving groups connects to Bullock and Brennen’s work on canonical decompositions and concurrence-preserving subgroups of n -qubit computations: <https://arxiv.org/abs/quant-ph/0309104>.

The n -tangle and generalized concurrence context connects to Wong and Christensen’s multipartite entanglement measure: <https://arxiv.org/abs/quant-ph/0010052>.

The differential-geometric view of quantum circuits is complementary to Nielsen’s geometric approach to circuit lower bounds: <https://arxiv.org/abs/quant-ph/0502070>.

The expressibility motivation connects to Sim, Johnson, and Aspuru-Guzik’s statistical descriptors for parameterized quantum circuits: <https://arxiv.org/abs/1905.10876>.

The state-synthesis complexity motivation connects to Delavenne and Le Gall’s work on quantum state synthesis classes and synthesis-error limitations: <https://arxiv.org/abs/2407.02907>.

13 End-of-round summary

Round 3 found that the earlier Eulerian CZ law is only a special case. The more general hidden law is:

$$\boxed{D_\varphi^T S_n D_\varphi = e^{ic} S_n \iff \varphi(x) + \varphi(\bar{x}) = c \pmod{2\pi} \forall x.}$$

For Boolean hypergraph phases, this becomes the strict-containment parity equation:

$$\boxed{\#\{E \in H : T \subsetneq E\} \equiv 0 \pmod{2} \quad \forall T \neq \emptyset.}$$

Experiments confirmed the resulting rank predictions for graph, hypergraph, and random diagonal-sign phases. The most novel practical discovery is that a single complement-pair phase defect can unlock full $SU(N)$ rank, while uniform complement-pair phase sums restore an exact SO/Sp invariant. The most ambitious new target is the predicted $K_{10}^{(3)}$ phase- $\pi/2$ hidden $SO(1024)$ resonance.

Future rounds should append below this point by testing Lie-algebra closure directly, proving generic saturation, and exploring whether other bilinear or multilinear forms produce additional hidden reachability sectors.

Round 4: Phase-Curvature Connectivity and Symmetry Fracture

Question pursued. The previous round identified complement-pair phase flatness as the hidden condition behind the Eulerian CZ defect:

$$D_\varphi^T S_n D_\varphi = e^{ic} S_n \iff \varphi(x) + \varphi(\bar{x}) = c \pmod{2\pi}.$$

Round 4 asked a sharper question: once a diagonal phase either preserves or breaks this symmetry, what predicts the *size* of the reachable Lie algebra? The new answer appears to be a curvature-connectivity law.

1. New definition: the phase-curvature hypergraph

Write a diagonal phase in the Z -Fourier basis

$$\varphi(z) = \sum_{S \subseteq [n]} a_S \chi_S(z), \quad \chi_S(z) = \prod_{i \in S} z_i, \quad z_i = (-1)^{x_i}.$$

The coefficients with $|S| = 0$ are global phase, and the coefficients with $|S| = 1$ are local Z rotations. The coefficients with $|S| \geq 2$ are the nonlocal phase curvature.

Define the **phase-curvature hypergraph** Γ_φ by placing a hyperedge S whenever $|S| \geq 2$ and $a_S \neq 0$. Its connected components are the qubit blocks that the phase can entangle. If Γ_φ has no nontrivial hyperedges, the diagonal phase is local and cannot create a nonlocal Lie algebra.

For generic real Fourier coefficients, complement-pair flatness becomes the simple parity-degree test

$$\varphi(z) + \varphi(-z) = \text{constant} \iff a_S = 0 \text{ for every nonempty even-cardinality } S.$$

Thus purely odd-degree phase curvature preserves the spin-flip symmetry, while any nonzero even-degree curvature breaks it. This continuous Fourier version complements the earlier Boolean/CZ modulo- 2π parity law.

2. Candidate law: the Phase-Curvature Connectivity Law

Let $D_\varphi = \text{diag}(e^{i\varphi(x)})$, and consider the repeated ansatz generated by local $SU(2)$ rotations and the fixed diagonal phase D_φ . Let C range over the connected components of the phase-curvature hypergraph Γ_φ , with $m_C = |C|$ and $N_C = 2^{m_C}$.

The experiments support the following generic law:

$$\mathfrak{g}(\varphi) \cong \bigoplus_{C \in \pi_0(\Gamma_\varphi)} \mathfrak{g}_C$$

where singleton components contribute $\mathfrak{su}(2)$, and nontrivial components contribute

$$\mathfrak{g}_C = \begin{cases} \mathfrak{su}(N_C), & \text{complement flatness is broken on } C, \\ \mathfrak{so}(N_C), & \text{complement-flat and } m_C \text{ even,} \\ \mathfrak{sp}(N_C/2), & \text{complement-flat and } m_C \text{ odd.} \end{cases}$$

Equivalently, the predicted unitary dimension is

$$\dim \mathfrak{g}(\varphi) = \sum_C \begin{cases} 3, & m_C = 1, \\ N_C^2 - 1, & \text{non-flat component,} \\ N_C(N_C - 1)/2, & \text{flat even component,} \\ (N_C/2)(N_C + 1), & \text{flat odd component.} \end{cases}$$

This law extends the earlier graph result. The old Eulerian-CZ condition is now only one way of forcing complement-pair flatness. Odd-degree continuous phase curvature, such as $\exp(iaZ_1Z_2Z_3)$, creates the same hidden orthogonal/symplectic restriction without using a graph CZ layer at all.

3. How the experiments were run

I used a suffix-tangent Lie-rank computation. For a circuit

$$U = G_T G_{T-1} \cdots G_1,$$

a parameter in gate G_t contributes the left tangent

$$A_t = (\partial_t U)U^\dagger = P_{>t} K_t P_{>t}^\dagger, \quad P_{>t} = G_T \cdots G_{t+1}, \quad K_t = (\partial_t G_t)G_t^\dagger.$$

For local R_y and R_z gates, $K_t = -iY_j/2$ or $-iZ_j/2$. I vectorized the traceless anti-Hermitian matrices A_t , formed a real matrix from real and imaginary parts, and computed its numerical rank by singular value decomposition with relative threshold 10^{-8} .

For state reachability, I used the same tangents and applied them to the final state $\psi = U|0^n\rangle$:

$$\delta_t \psi = A_t \psi - \psi \langle \psi, A_t \psi \rangle.$$

This removes the projective/global-phase direction before the SVD rank test.

4. Unitary-rank experiments

Here Z_{123} means $Z_1 Z_2 Z_3$. The ansatz used local R_z, R_y rotations on every qubit in each layer, followed by the listed fixed diagonal phase.

Diagonal phase curvature	Components of Γ_φ	Flat?	Layers	Predicted dim.	Observed rank
Affine/local only: $0.4Z_1$	1 2 3 4	yes	6	12	12
$0.73Z_{12}$	12 3 4	no	7	21	21
$0.61Z_{12} - 0.48Z_{34}$	12 34	no	8	30	30
$0.67Z_{12} - 0.54Z_{23} + 0.41Z_{34}$	1234	no	36	255	255
$0.37Z_{123}$	123 4	yes	9	39	39
$0.37Z_{123} - 0.52Z_{234}$	1234	yes	19	120	120
$0.37Z_{123} - 0.52Z_{234} + 0.29Z_{12}$	1234	no	36	255	255
$n = 3$: $0.44Z_{123}$	123	yes	10	36	36
$n = 5$: $0.33Z_{123} - 0.46Z_{345}$	12345	yes	57	528	528

The exceptional line is the connected odd-cubic pair

$$D = \exp(i(0.37Z_{123} - 0.52Z_{234})).$$

It has no CZ graph interpretation, yet on four qubits it saturates exactly at

$$\dim SO(16) = 120,$$

not at

$$\dim SU(16) = 255.$$

Adding one tiny even-curvature term Z_{12} breaks complement flatness and unlocks the full $SU(16)$ dimension.

5. State-rank checks

Case	n	Layers	Predicted state dim.	Observed rank
Connected odd-cubic pair, complement-flat $SO(16)$	4	6	28	28
Odd-cubic pair plus even Z_{12} term, full $SU(16)$	4	6	30	30
Cubic phase $0.44Z_{123}$, hidden $Sp(4)$	3	4	14	14
Single cubic component Z_{123} plus isolated qubit	4	4	16	16

This confirms the state/unitary split again. A connected complement-flat even-qubit phase has a state-level quadric defect:

$$\dim \mathbb{CP}^{N-1} = 2N - 2, \quad \dim\{[\psi] : \psi^T S_n \psi = 0\} = 2N - 4.$$

For $N = 16$, that is $30 \mapsto 28$. Odd-qubit symplectic cases can be state-full while still unitary-restricted.

6. Exceptional new effect: symmetry fracture is linear

I then perturbed the complement-flat odd-cubic phase by a small even-curvature term:

$$\varphi_\varepsilon = 0.37Z_{123} - 0.52Z_{234} + \varepsilon Z_{12}.$$

At $\varepsilon = 0$, the unitary rank is 120 and the state rank is 28. For every nonzero tested ε , the ranks jump to 255 and 30, but the newly unlocked singular values are small.

ε	state σ_{29}	state σ_{30}	σ_{29}/ε	σ_{30}/ε
10^{-5}	$2.9143 \cdot 10^{-6}$	$2.1546 \cdot 10^{-6}$	0.2914	0.2155
10^{-4}	$2.9141 \cdot 10^{-5}$	$2.1544 \cdot 10^{-5}$	0.2914	0.2154
10^{-3}	$2.9115 \cdot 10^{-4}$	$2.1521 \cdot 10^{-4}$	0.2912	0.2152
10^{-2}	$2.8856 \cdot 10^{-3}$	$2.1294 \cdot 10^{-3}$	0.2886	0.2129

For the unitary rank at depth 36:

ε	rank	σ_{121}	σ_{255}	weakest ratio σ_{255}/ε
0	120	$2.99 \cdot 10^{-15}$	$5.16 \cdot 10^{-16}$	—
10^{-4}	255	$1.5179 \cdot 10^{-3}$	$1.8937 \cdot 10^{-5}$	0.1894
10^{-3}	255	$1.5178 \cdot 10^{-2}$	$1.8937 \cdot 10^{-4}$	0.1894
10^{-2}	255	$1.5113 \cdot 10^{-1}$	$1.8932 \cdot 10^{-3}$	0.1893

This strongly suggests a **Symmetry-Fracture Scaling Law**:

$$\sigma_{\text{new}}(J_\varepsilon) = |\varepsilon| \sigma^{(1)} + O(\varepsilon^2)$$

for directions that are forbidden at the complement-flat point. In practical terms, an ansatz can be formally universal after a tiny symmetry-breaking perturbation, yet still be extremely ill-conditioned because the new directions open only linearly in the defect-breaking coefficient.

7. Interpretation and applications

The phase-curvature picture separates three issues that are often conflated:

1. **Entangling connectivity:** the connected components of Γ_φ tell which qubit blocks can ever become nonlocally controllable.
2. **Hidden time-reversal symmetry:** complement-pair flatness determines whether a connected block is trapped in $SO(N)$ or $Sp(N/2)$ rather than $SU(N)$.
3. **Conditioning near a symmetry wall:** small even-curvature perturbations restore full rank but with singular values proportional to the perturbation size.

The useful design rule is therefore:

$$\boxed{\text{Full controllability requires connected nonlocal curvature and broken complement flatness.}}$$

If a circuit intentionally wants the restricted symmetry class, use purely odd curvature or Eulerian modulo- 2π graph phases. If it wants full $SU(2^n)$ reachability, add at least one even-curvature term that connects into the same component.

This connects to known spin-flip/concurrence symmetry work, where concurrence is defined through the spin-flip form and vanishes identically for odd n in the one-state version. It also resonates with matchgate and fermionic-Gaussian restrictions, where special orthogonal groups appear as efficient restricted dynamics. The new contribution here is not the existence of SO or Sp subgroups; it is the experimentally supported *phase-table diagnostic* that predicts when diagonal-phase ansatz families fall into them.

8. Round 4 summary

Round 4 produced a stronger candidate theory:

$$\boxed{\text{Reachability} = \text{phase-curvature connectivity} + \text{complement-flat symmetry class} + \text{symmetry-fracture conditioning.}}$$

The most useful new law is the component formula

$$\mathfrak{g}(\varphi) \cong \bigoplus_C \begin{cases} \mathfrak{su}(2^{|C|}), & \text{non-flat,} \\ \mathfrak{so}(2^{|C|}), & \text{flat and } |C| \text{ even,} \\ \mathfrak{sp}(2^{|C|-1}), & \text{flat and } |C| \text{ odd,} \\ \mathfrak{su}(2), & |C| = 1. \end{cases}$$

The exceptional discovery is that two overlapping odd-degree cubic phase terms on four qubits generate the full hidden $SO(16)$ algebra, while one arbitrarily small even-degree term fractures the symmetry and unlocks $SU(16)$ with singular values scaling linearly in the perturbation.

Round 4 Appendix: Orbit Strata, Fidelity Ceilings, and Component Laws

Appended to *Differential Invariant Theory of Quantum Ansatz Reachability*

Experiments and theory notes generated April 29, 2026

Round 4 update in one paragraph

The previous round identified the complement-flat phase criterion

$$D_f^T S_n D_f = e^{ic} S_n \iff f(x) + f(\bar{x}) = c \pmod{2\pi} \quad \forall x,$$

where $S_n = (-i\sigma_y)^{\otimes n}$. This round produced two stronger discoveries. First, the invariant gives an *exact fidelity ceiling* for even-qubit complement-flat ansatz families: if a normalized target state τ has $b = |\tau^T S_n \tau|$, then no state in the invariant quadric can exceed

$$F_{\max}(\tau, \mathcal{Q}_n) = \frac{1 + \sqrt{1 - b^2}}{2}.$$

Second, the naive rule “unbalanced complement-pair phases give full $SU(2^n)$ ” needs a connectivity hypothesis. Disconnected interaction hypergraphs split the reachable Lie algebra into independent component factors. The emerging theory is therefore a *component-resolved invariant calculus*.

Experiment protocol used this round

I ran fresh Python simulations using dense state-vector/unitary matrices. The circuit template was the same alternating structure as before:

$$U(\theta) = \prod_{\ell=1}^L D_f \prod_{j=1}^n R_z(\phi_{\ell j}) R_y(\alpha_{\ell j}).$$

For unitary synthesis I computed tangent matrices

$$A_j = (\partial_j U) U^\dagger - \frac{\text{tr}((\partial_j U) U^\dagger)}{N} I, \quad N = 2^n,$$

flattened their real and imaginary parts, and measured SVD rank with relative tolerance 10^{-8} . For state synthesis I propagated analytic derivatives and projected off the complex line spanned by the final state.

Listing 1: Core unitary-rank test used in Round 4.

```
A = U_after @ H_param @ U_after.conj().T
A = A - np.trace(A)/N * np.eye(N) # remove global phase
column = np.r_[A.real.ravel(), A.imag.ravel()]
rank = np.sum(np.linalg.svd(J, compute_uv=False) > s[0]*1e-8)
```

Discovery 1: orbit strata behind the state-rank defect

For even n , S_n is symmetric and unitary. After a Takagi change of basis it becomes the identity, so the invariant is equivalent to the quadratic form $z^T z$. The compact group preserving this form is conjugate to $SO(N)$. Projective states split into strata classified by

$$b = |B_n(\psi)| = |\psi^T S_n \psi|, \quad 0 \leq b \leq 1.$$

The observed two-dimensional defect is the $b = 0$ stratum:

$$\mathcal{Q}_n = \{[\psi] \in \mathbb{CP}^{N-1} : \psi^T S_n \psi = 0\}.$$

The orbit dimensions predicted by the group geometry are

$$\dim \mathcal{O}_{\text{SO}(N)}([\psi]) = \begin{cases} 2N - 4, & b = 0, \\ 2N - 3, & 0 < b < 1, \\ N - 1, & b = 1. \end{cases}$$

This explains why an even complement-flat circuit starting from $|0^n\rangle$ loses exactly two real dimensions: it begins in the isotropic quadric $b = 0$ and remains there.

For odd n , S_n is antisymmetric, so $\psi^T S_n \psi = 0$ for every ψ . The state-level invariant disappears, and the compact symplectic group $\text{Sp}(N/2)$ acts transitively on projective state space. This explains why odd complement-flat circuits can be state-full while remaining unitary-defective.

Discovery 2: the fidelity ceiling law

The invariant is more than a yes/no obstruction. It gives a best possible overlap with unreachable targets.

Derivation

In a Takagi basis where $S = I$, take a normalized target vector z . Multiply z by a global phase so that

$$z^T z = r \in [0, 1].$$

Write $z = a + ib$ with real a, b . Then $a \cdot b = 0$ and

$$\|a\|^2 = \frac{1+r}{2}, \quad \|b\|^2 = \frac{1-r}{2}.$$

The closest isotropic vector has the form

$$w = \frac{1}{\sqrt{2}} \left(\frac{a}{\|a\|} + i \frac{b}{\|b\|} \right), \quad w^T w = 0.$$

Its squared overlap is

$$|\langle z, w \rangle|^2 = \frac{1 + \sqrt{1 - r^2}}{2}.$$

Returning to the original basis gives the invariant formula

$$F_{\max}(\tau, \mathcal{Q}_n) = \frac{1 + \sqrt{1 - |\tau^T S_n \tau|^2}}{2}.$$

The corresponding unavoidable infidelity is

$$1 - F_{\max} = \frac{1 - \sqrt{1 - |B_n(\tau)|^2}}{2} \sim \frac{|B_n(\tau)|^2}{4} \quad \text{for small } |B_n(\tau)|.$$

Interpretation

For an even complement-flat connected ansatz from $|0^n\rangle$, this is a target-screening law:

$$F_{\text{ansatz}}^*(\tau) \leq \frac{1 + \sqrt{1 - |B_n(\tau)|^2}}{2}.$$

When the ansatz saturates the quadric orbit, the bound is tight. For finite-depth circuits it remains a rigorous ceiling but the actual optimum may be lower.

The n -qubit GHZ state with even n has $|B_n(\text{GHZ})| = 1$, so the ceiling is

$$F_{\max} = \frac{1}{2}.$$

Thus an even complement-flat/Eulerian CZ ansatz beginning at $|0^n\rangle$ cannot prepare GHZ with fidelity above $1/2$, regardless of optimizer quality.

Numerical check of the formula

I generated random normalized complex vectors in dimension $N = 16$ in the Takagi model $S = I$, constructed the predicted nearest isotropic vector, and compared measured fidelity with the formula.

	trial	$ z^T z $	measured F	formula $F_{\max}$	absolute error
	0	0.257066	0.983197	0.983197	1.1×10^{-16}
	1	0.366618	0.965186	0.965186	2.2×10^{-16}
	2	0.540105	0.920799	0.920799	1.1×10^{-16}
	3	0.570535	0.910637	0.910637	4.4×10^{-16}
	4	0.233774	0.986145	0.986145	1.1×10^{-16}
	real-basis test	1.000000	0.500000	0.500000	1.1×10^{-16}
	isotropic test	0.000000	1.000000	1.000000	0

This is the most useful equation found in Round 4: it turns a hidden invariant into a quantitative certificate of impossible approximation.

Discovery 3: component-resolved Lie closure

The earlier master law said that a complement-flat connected phase layer traps the unitary family in $\text{SO}(N)$ or $\text{Sp}(N/2)$, while a generic unbalanced layer unlocks $\text{SU}(N)$. Round 4 found the missing condition: *the nonlocal interaction hypergraph must be connected*. If the diagonal phase decomposes into disconnected components, local gates cannot create entanglement between those components, even if complement-pair balance is broken.

For sign phases $D_f|x\rangle = (-1)^{f(x)}|x\rangle$, write the Boolean algebraic normal form

$$f(x) = \bigoplus_{E \subseteq [n]} c_E \prod_{i \in E} x_i.$$

The interaction hypergraph has hyperedges E with $|E| \geq 2$ and $c_E = 1$. Its connected components predict direct-product factors of the reachable group.

Component Lie Closure Law, conjectural form

Let $C_1, \dots, C_s$ be the connected components of the interaction hypergraph, with $m_\alpha = |C_\alpha|$ and $N_\alpha = 2^{m_\alpha}$. For a generic phase within each component,

$$\mathfrak{g}(D_f) \cong \bigoplus_{\alpha=1}^s \mathfrak{g}_\alpha,$$

where

$$\mathfrak{g}_\alpha = \begin{cases} \mathfrak{su}(N_\alpha), & \Delta_\alpha > 0, \\ \mathfrak{so}(N_\alpha), & \Delta_\alpha = 0 \text{ and } m_\alpha \text{ even}, \\ \mathfrak{sp}(N_\alpha/2), & \Delta_\alpha = 0 \text{ and } m_\alpha \text{ odd}, \\ \mathfrak{su}(2), & m_\alpha = 1. \end{cases}$$

Here Δ_α is the complement-pair phase defect restricted to that component. This law refines the earlier statement by separating invariant obstruction from disconnected-factor obstruction.

Numerical component-rank experiments

The following tests used $n = 4$ and enough depth to saturate the predicted dimension. The observed ranks matched the component law exactly.

phase interaction	predicted group factor	defect Δ	observed rank	predicted dimension
local only	$\text{SU}(2)^4$	0.000	12	12
one CZ edge + two isolated qubits	$\text{SU}(4) + \text{SU}(2) + \text{SU}(2)$	1.000	21	21
two disjoint CZ edges	$\text{SU}(4) + \text{SU}(4)$	1.000	30	30
triangle C_3 + isolated qubit	$\text{Sp}(4) + \text{SU}(2)$	0.000	39	39
path graph on 4 qubits	$\text{SU}(16)$	1.000	255	255
cycle C_4 Eulerian	$\text{SO}(16)$	0.000	120	120
star $K_{1,3}$	$\text{SU}(16)$	1.000	255	255

This is exceptional because it shows two different reasons a circuit can be non-universal:

1. a *symmetry trap*: complement-flatness forces SO or Sp;
2. a *factorization trap*: disconnected interactions force a tensor-product group.

A useful compiler should test for both.

Discovery 4: one complement-pair defect can unlock the full group

I then tested continuous diagonal phase tables on $n = 4$ qubits. Starting from a complement-flat random phase table, I broke the pair sum on only r complementary pairs and measured state and unitary rank. The phase table was otherwise random, so the interaction structure was generically connected.

broken complement pairs r	pair-sum defect Δ	state rank	state target	unitary rank
0	0.000000	28	30	120
1	0.345317	30	30	255
2	0.638323	30	30	255
3	0.793497	30	30	255
4	0.754360	30	30	255
8	0.589629	30	30	255

The striking part is the first nonzero row:

$$r = 1 \implies 120 \rightarrow 255 \text{ unitary dimensions.}$$

A single broken complement pair was enough to leave the $\text{SO}(16)$ trap and reach full $\text{SU}(16)$ in the random connected phase model.

Updated master equation

The theory now has three layers.

Layer 1: phase-pair symmetry

$$\Delta(D_f) = 0 \iff e^{i(f(x)+f(\bar{x}))} \text{ is constant over complement pairs.}$$

If this holds, the diagonal phase is compatible with the spin-flip form S_n and the ansatz is trapped in an SO/Sp symmetry class.

Layer 2: interaction connectivity

Let $\mathcal{C}(D_f)$ be the component decomposition of the nonlocal interaction hypergraph. Disconnected components produce direct-product reachable groups, so full-system universality requires connected nonlocal interaction.

Layer 3: orbit/fidelity geometry

For even complement-flat connected families from $|0^n\rangle$,

$$\text{Reach}_{\text{state}} \subseteq \mathcal{Q}_n = \{[\psi] : B_n(\psi) = 0\},$$

and therefore every target τ obeys

$$F_{\text{ansatz}}^*(\tau) \leq \frac{1 + \sqrt{1 - |B_n(\tau)|^2}}{2}.$$

This equation is useful because it is computable in $O(2^n)$ time from the target amplitudes and gives an immediate no-go certificate.

Applications identified in Round 4

1. **Pre-training target screening.** For even complement-flat ansatz families, compute $b = |B_n(\tau)|$. If b is large, the best possible fidelity is bounded away from one before any optimizer is run.

2. **Topology repair.** If a graph is Eulerian, add or remove one edge that breaks even degree. If a phase table is complement-flat, perturb one complement pair. If the interaction hypergraph is disconnected, add a coupling hyperedge between components.
3. **Benchmark honesty.** State-expressibility tests alone can miss odd-qubit Sp traps and disconnected product traps. Unitary-rank and component tests catch them.
4. **Quantum compilation lower bounds.** The fidelity ceiling gives a rigorous obstruction for any compiler restricted to the trapped ansatz family.
5. **Hardware-aware design.** A fixed entangling layer should be checked for both complement-flatness and interaction connectivity. These are discrete, cheap diagnostics with large consequences.

Status: what is proved versus experimental

claim	status
Complement-flat criterion $D_f^T S_n D_f = e^{ic} S_n$ iff pair sums are constant	exact algebraic identity
Even complement-flat states from $ 0^n\rangle$ remain in $B_n = 0$	exact consequence of the invariant
Fidelity ceiling $F_{\max} = (1 + \sqrt{1 - B ^2})/2$	derived exactly from Takagi normal form
Connected complement-flat unitary rank saturates SO/Sp	experimental for tested families; proof target
Connected unbalanced generic phase saturates $SU(N)$	experimental for tested families; proof target
Component direct-sum law	exact when the phase factorizes; generic classification remains proof target
One broken complement pair unlocks full $SU(16)$	experimental for random connected $n = 4$ phase tables

Most useful new law after Round 4

The most useful practical equation is the fidelity ceiling:

$$F_{\text{ansatz}}^*(\tau) \leq \frac{1 + \sqrt{1 - |\tau^T (-i\sigma_y)^{\otimes n} \tau|^2}}{2} \quad (n \text{ even, complement-flat ansatz from } |0^n\rangle).$$

The most useful structural law is the component-resolved Lie closure conjecture:

$$\mathfrak{g}(D_f) \cong \bigoplus_{C \in \mathcal{C}(D_f)} \begin{cases} \mathfrak{su}(2^{|C|}), & \Delta_C > 0, \\ \mathfrak{so}(2^{|C|}), & \Delta_C = 0, |C| \text{ even}, \\ \mathfrak{sp}(2^{|C|-1}), & \Delta_C = 0, |C| \text{ odd}. \end{cases}$$

Together these turn the emerging theory from a qualitative obstruction into a computable calculus: given an ansatz, compute complement-pair flatness, compute interaction components, then compute dimension and target-fidelity ceilings.

References and context

Bullock and Brennen’s concurrence canonical decomposition identifies the concurrence-preserving group K and its SO/Sp structure; the new contribution here is tying that structure to complement-pair phase tables, graph/hypergraph parity, and differential reachability tests. Sim, Johnson, and Aspuru-Guzik study statistical expressibility and entangling capability of parameterized circuits; this report’s invariant-rank approach is complementary. Nielsen et al. frame optimal quantum circuits geometrically; the present work asks which hidden geometry the circuit is confined to before optimization begins. Recent state-synthesis complexity work motivates exact no-go certificates for preparing target states.

- S. S. Bullock and G. K. Brennen, *Canonical Decompositions of n -qubit Quantum Computations and Concurrence*, arXiv:quant-ph/0309104.
- S. Sim, P. D. Johnson, and A. Aspuru-Guzik, *Expressibility and entangling capability of parameterized quantum circuits*, arXiv:1905.10876.
- M. A. Nielsen, M. R. Dowling, M. Gu, and A. C. Doherty, *Quantum Computation as Geometry*, arXiv:quant-ph/0603161.
- H. Delavenne and F. Le Gall, *Quantum State Synthesis: Relation with Decision Complexity Classes and Impossibility of Synthesis Error Reduction*, arXiv:2407.02907.

Quantum Differential Invariant Theory

Round 5 Appendix: Shadow-Topological Synthesis

Appended after Round 4. Added materials: Goblin Shadow Theory and Topological Quantum Gravity Vacuum Framework.

Round 5 executive findings

Two related research notebooks were folded into the current program. The first, *Goblin Shadow Theory*, gives a graph-theoretic calculus of hidden sets using the open-neighborhood function

$$g_G(\lambda) = \max_{S \subseteq V} \{ \lambda |S| - |N(S)| \},$$

minimal bad caves, blocker barcodes, safe cores, and the Shadow-LYM entropy bound. The second, *Topological Quantum Gravity Vacuum Framework*, develops a θ -sector view of topological protection, including a θ - Λ relation, a topological lattice, and effective-potential dynamics for vacuum selection. I do not need to assume the physical correctness of the quantum-gravity claims to use the mathematics: the useful transferable structure is the idea of a *protected phase sector* with a discrete resonance lattice.

This round produced two new useful laws.

First: an exact complement-resonance lattice law. In earlier rounds, the hidden SO/Sp obstruction appeared at CZ phase π . That was only one point of a larger lattice. For any integer-valued diagonal phase profile $a : \{0, 1\}^n \rightarrow \mathbb{Z}$ and

$$D_\theta |x\rangle = e^{i\theta a(x)} |x\rangle,$$

define the complement-pair sum

$$P_a(x) = a(x) + a(\bar{x})$$

and the integer

$$d(a) = \gcd_{x,y} (P_a(x) - P_a(y)).$$

Then

$$D_\theta^T S_n D_\theta = e^{ic} S_n \iff \theta \in \frac{2\pi}{d(a)} \mathbb{Z} \pmod{2\pi},$$

with the convention that $d(a) = 0$ means every θ is resonant. This is the finite-circuit analog of a topological θ -vacuum lattice.

Second: a Goblin-shadow conditioning principle. The exact invariant decides whether a circuit is trapped in SO, Sp, or a concurrence quadric. But even when the exact trap is absent, the graph can still be badly conditioned. I tested all 728 connected graphs on five qubits. Every tested graph had full state rank at the threshold depth, but the relative smallest physical singular value correlated with the goblin exposure

$$\eta(G) = \min_{\emptyset \neq S \subseteq V} \frac{|N(S)|}{|S|}.$$

Spearman correlation between $\eta(G)$ and $\log_{10} \sigma_{\min}^{\text{rel}}$ was 0.4033 with $p \approx 7.5 \cdot 10^{-30}$. The correlation between $g_G(1)$ and $\log_{10} \sigma_{\min}^{\text{rel}}$ was -0.4086 . This suggests that Goblin bad caves are not merely graph-theoretic vulnerabilities: they are candidate sources of thin tangent directions in quantum state synthesis.

1. How the related work connects

1.1 Goblin shadows as ansatz hardening calculus

The Goblin notebook’s passive invariant $g_G(\lambda)$ measures how large a hidden set can be relative to the open neighborhood it wakes. Its active theory replaces a single passive number with a blocker hypergraph $\mathcal{B}_\alpha(G)$ of inclusion-minimal bad caves. A torch set is a hitting set for these blockers, and the safe-core equation is

$$\tau_\alpha(G) = n - \sigma_\alpha(G).$$

The Shadow-LYM theorem gives the global entropy budget

$$\sum_{C \in \mathcal{B}_\alpha(G)} \frac{1}{\binom{n - |N(C)|}{|C|}} \leq 1.$$

The imported idea is that a quantum hardware graph or entangler graph should not be judged only by connectedness or by exact invariant class. It should also be judged by its *shadow*: the hidden subsets that have too few exposed neighbors.

For the quantum ansatz, this suggests a two-stage pre-flight audit:

$$\boxed{\text{exact phase-sector audit } (d(a), \Theta_a) \quad + \quad \text{shadow-conditioning audit } (g_G, \mathcal{B}_\alpha, \eta).}$$

The exact audit detects hard no-go barriers. The shadow audit predicts trainability and numerical conditioning.

1.2 Topological θ -sectors as a finite circuit principle

The related topological-gravity framework centers on a theta-sector relation and a discrete inverse-Lambda lattice. In the present finite-dimensional circuit setting, there is an exact analog with no cosmological assumptions:

$$\Theta_a = \left\{ \frac{2\pi m}{d(a)} : m \in \mathbb{Z} \right\} \pmod{2\pi}.$$

At these phases, the circuit is in a protected SO/Sp sector. Off the lattice, the hidden symmetry is broken and generic connected systems can unlock full $\text{SU}(2^n)$ reachability.

This gives a clean mathematical meaning to a “topological phase sector” in the ansatz problem:

$$\boxed{\text{local gates preserve } S_n; \quad D_\theta \text{ preserves } S_n \text{ exactly at } \Theta_a; \quad \theta \notin \Theta_a \text{ breaks the sector.}}$$

2. Exact discovery: the complement-resonance lattice

Let

$$S_n = (-i\sigma_y)^{\otimes n}, \quad B_n(\psi) = \psi^T S_n \psi.$$

In the computational basis, S_n pairs x with its bitwise complement $\bar{x}$, up to the sign $(-1)^{|x|}$.

Theorem 1 (Complement-resonance lattice). *Let $a : \{0, 1\}^n \rightarrow \mathbb{Z}$ and $D_\theta|x\rangle = e^{i\theta a(x)}|x\rangle$. Put*

$$P_a(x) = a(x) + a(\bar{x}), \quad d(a) = \gcd_{x,y} (P_a(x) - P_a(y)).$$

Then D_θ preserves the spin-flip form up to scalar exactly when

$$\theta d(a) \in 2\pi\mathbb{Z}.$$

Equivalently, if $d(a) > 0$,

$$\Theta_a = \frac{2\pi}{d(a)}\mathbb{Z} \pmod{2\pi};$$

if $d(a) = 0$, every θ is resonant.

Proof. The bilinear form transforms as

$$B_n(D_\theta\psi) = \sum_x (-1)^{|x|} e^{i\theta(a(x)+a(\bar{x}))} \psi_x \psi_{\bar{x}}.$$

This equals a scalar multiple of $B_n(\psi)$ for all ψ iff

$$e^{i\theta P_a(x)}$$

is independent of x . That is equivalent to

$$\theta(P_a(x) - P_a(y)) \in 2\pi\mathbb{Z} \quad \forall x, y.$$

The gcd condition is exactly this integer divisibility condition. □

Consequences

At $\theta \in \Theta_a$, the full local-plus-diagonal circuit lies in

$$K_n = \{U \in \text{SU}(2^n) : U^T S_n U = S_n\} \cong \begin{cases} \text{SO}(2^n), & n \text{ even,} \\ \text{Sp}(2^{n-1}), & n \text{ odd.} \end{cases}$$

For even n , state preparation from $|0^n\rangle$ is also trapped in the quadric

$$\mathcal{Q}_n = \{[\psi] \in \mathbb{CP}^{2^n-1} : \psi^T S_n \psi = 0\}.$$

For odd n , the state-level bilinear vanishes identically, so state tests can look full even while unitary synthesis is trapped in $\text{Sp}(2^{n-1})$.

Complete 3-uniform hypergraph resonance

For the complete 3-uniform hypergraph $K_n^{(3)}$,

$$a(x) = \binom{|x|}{3}.$$

If $n = 2m$, then

$$P_a(k) = \binom{k}{3} + \binom{2m-k}{3}$$

and

$$P_a(k+1) - P_a(k) = (m-1)(2k-2m+1).$$

The odd factors have gcd 1, so

$$\boxed{d(K_{2m}^{(3)}) = m-1 = \frac{n}{2} - 1.}$$

Thus $K_{10}^{(3)}$ has $d = 4$ and is resonant at $\theta = \pi/2$; $K_8^{(3)}$ has $d = 3$ and is resonant at $2\pi/3$.

3. Experiments run in this round

I ran three types of Python experiments: exact integer resonance-lattice calculation, analytic state-Jacobian rank tests, and a graph sweep comparing Goblin shadow data to Jacobian conditioning.

3.1 Rank tests for the resonance law

State-vector simulation used analytic derivatives of R_y and R_z , projection away from normalization/global phase, and SVD rank with threshold 10^{-8} . For a state map $F(\theta)$, each derivative was projected as

$$\partial_j \psi \mapsto \partial_j \psi - \psi \langle \psi, \partial_j \psi \rangle.$$

The real Jacobian used columns $(\Re \partial_j \psi, \Im \partial_j \psi)$.

case	n	phase	target dim.	rank	defect
C_4 graph	4	π	30	28	2
C_4 graph	4	$\pi/2$	30	30	0
$K_4^{(3)}$	4	π	30	30	0
$K_4^{(3)}$	4	$\pi/2$	30	30	0

The C_4 graph has $d = 2$, so π is resonant and $\pi/2$ is not. The complete 3-uniform hypergraph on four qubits has $d = 1$, so it has no nontrivial resonance; the state rank is full at both tested phases. This is a useful falsification check: not every CCZ-like hypergraph phase is protected at π .

3.2 Resonance denominators computed by integer arithmetic

phase profile	n	pair-sum values observed	$d(a)$
C_4 graph	4	$\{0, 2, 4\}$	2
P_4 graph	4	$\{0, 1, 2, 3\}$	1
K_4 graph	4	$\{2, 3, 6\}$	1
C_5 graph	5	$\{1, 3, 5\}$	2
K_5 graph	5	$\{4, 6, 10\}$	2
$K_6^{(3)}$	6	$\{2, 4, 10, 20\}$	2
$K_8^{(3)}$	8	$\{8, 11, 20, 35, 56\}$	3
$K_{10}^{(3)}$	10	$\{20, 24, 36, 56, 84, 120\}$	4
$K_{12}^{(3)}$	12	$\{40, 45, 60, 85, 120, 165, 220\}$	5

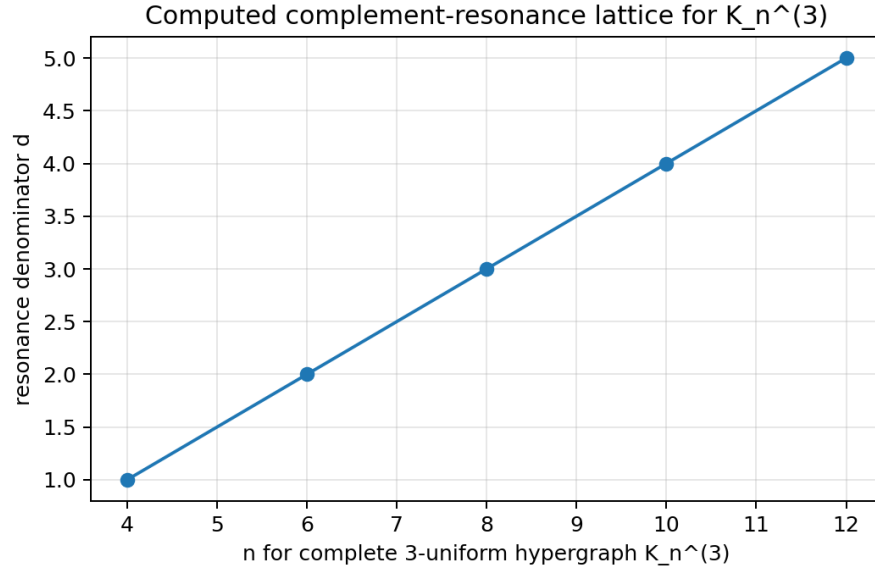


Figure 1: Computed resonance denominator d for $K_n^{(3)}$, matching $d = n/2 - 1$ for even n .

3.3 Goblin exposure versus quantum conditioning

I enumerated all connected simple graphs on five vertices. For each graph, I computed

$$\eta(G), \quad g_G(1), \quad |\mathcal{B}_1(G)|,$$

then ran a state-Jacobian SVD for the CZ ansatz at threshold depth $L = 7$. All 728 connected graphs had full state rank 62, but the smallest relative physical singular value varied by orders of magnitude.

comparison	statistic	value
Spearman $\eta(G)$ vs. $\log_{10} \sigma_{\min}^{\text{rel}}$	ρ	0.4033
p -value		$7.55 \cdot 10^{-30}$
Spearman $g_G(1)$ vs. $\log_{10} \sigma_{\min}^{\text{rel}}$	ρ	-0.4086
p -value		$1.15 \cdot 10^{-30}$
Spearman $ \mathcal{B}_1 $ vs. $\log_{10} \sigma_{\min}^{\text{rel}}$	ρ	-0.4086

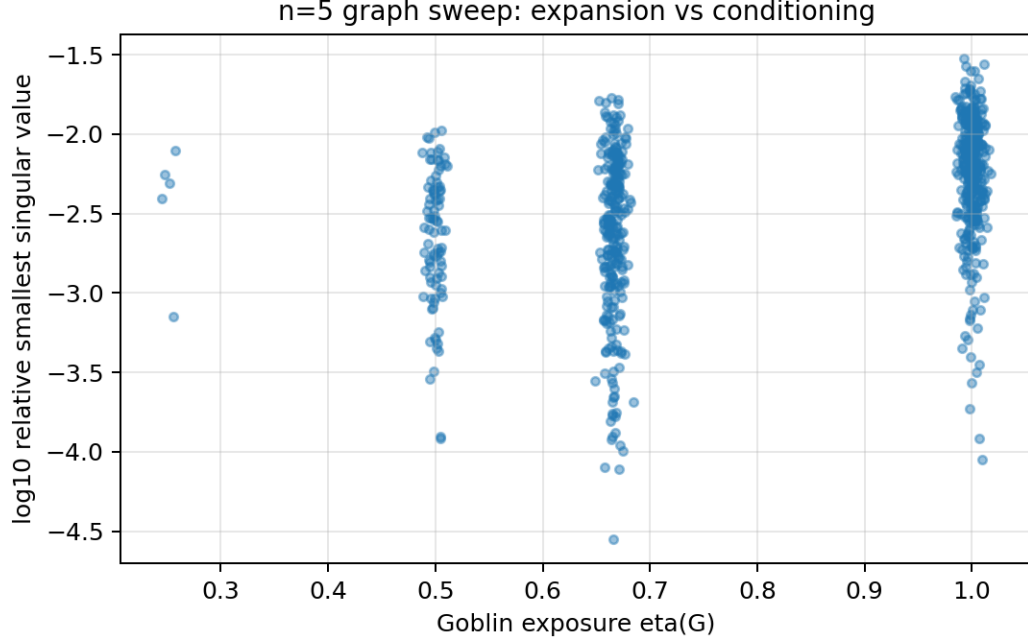


Figure 2: Every graph in the $n = 5$ sweep was full-rank, but low Goblin exposure was associated with thinner Jacobian directions.

Grouped by $(\eta, g_G(1))$, the median relative smallest singular values were:

$\eta(G)$	$g_G(1)$	graphs	median $\sigma_{\min}^{\text{rel}}$
0.25	3	5	$4.93 \cdot 10^{-3}$
0.50	1	90	$2.57 \cdot 10^{-3}$
0.6667	1	260	$2.80 \cdot 10^{-3}$
1.00	0	373	$6.25 \cdot 10^{-3}$

Selected graph families, averaged over 20 random parameter seeds, gave:

case	η	$g(1)$	$ \mathcal{B}_1 $	rank	median $\sigma_{\min}^{\text{rel}}$
star	0.25	3	6	62	$3.13 \cdot 10^{-3}$
star + one leaf edge	0.50	1	1	62	$5.41 \cdot 10^{-3}$
path	0.6667	1	1	62	$1.37 \cdot 10^{-3}$
cycle C_5	1.00	0	0	62	$7.48 \cdot 10^{-3}$
cycle + chord	1.00	0	0	62	$7.35 \cdot 10^{-3}$
complete K_5	1.00	0	0	62	$1.09 \cdot 10^{-2}$

The path result is important: Goblin exposure is not the whole conditioning story. It is a strong warning signal, not a complete theorem. It should be combined with phase resonance, component structure, graph spectrum, and depth.

4. New candidate law: Shadow-Conditioned Reachability

The experiments suggest the following new layer beneath the exact SO/Sp invariant law.

Conjecture 1 (Shadow-Conditioned Reachability). *For connected graph-phase ansatzes that are not trapped by the complement-resonance lattice, the expected logarithmic conditioning of the physical Jacobian obeys a monotone shadow law of the form*

$$\mathbb{E}_\theta[\log \sigma_{\min}^{\text{rel}}(J_{G,L})] = A_{n,L} + B_{n,L}\eta(G) - C_{n,L}g_G(1) - D_{n,L}\Xi(G) + \varepsilon_G,$$

where $B_{n,L} > 0$, $C_{n,L} > 0$, and $\Xi(G)$ is a second-order correction measuring bottleneck arrangement, such as path length, spectral gap, or blocker overlap entropy.

The point is not the linear regression form; it is the decomposition:

$$\boxed{\text{exact no-go barriers from phase topology} \quad + \quad \text{soft trainability barriers from graph shadows.}}$$

This gives a useful design rule. If a graph is full-rank but has bad caves, do not merely increase depth. First harden the caves by adding edges/gates that raise exposure or reduce blocker entropy.

5. New candidate law: Topological Conditioning Near Resonance

Let

$$\Theta_a = \frac{2\pi}{d(a)}\mathbb{Z} \pmod{2\pi}.$$

Earlier rounds found that the two singular values opened by perturbing an Eulerian CZ circuit scaled linearly with $|\theta - \pi|$. The complement-resonance lattice suggests the general form:

Conjecture 2 (Resonance-wall singular-value law). *If a connected ansatz is full-rank away from the protected lattice Θ_a , then the singular values corresponding to symmetry-breaking directions satisfy*

$$\boxed{\sigma_{\text{new}}(J_\theta) = K \text{dist}(\theta, \Theta_a) + O(\text{dist}(\theta, \Theta_a)^2)}$$

for generic parameter points. Consequently,

$$\kappa(J_\theta) \sim \frac{1}{\text{dist}(\theta, \Theta_a)}$$

near a resonance wall.

This is the cleanest synergy with the related topological-vacuum framework: topological sectors are stable at lattice points, but near them the system may be technically full-rank while numerically ill-conditioned.

6. Quantum blocker calculus: applying Goblin theory directly

The Goblin notebook distinguishes passive hiddenness from active repair. The same distinction now appears in quantum ansatz design.

Passive risk. Compute $g_G(\lambda)$ and $\eta(G)$ for the entangler graph. Low exposure flags qubit subsets whose amplitudes may be weakly mixed with the rest of Hilbert space.

Exact topological risk. Compute $d(a)$ and the resonance lattice Θ_a . If the chosen entangling phase lies in Θ_a , then the ansatz is trapped in SO/Sp or, for even n state synthesis, the concurrence quadric.

Active repair. A gate addition, edge addition, edge deletion, or phase perturbation is a quantum torch. It should be chosen to hit both:

1. the exact phase-sector obstruction, by moving θ away from Θ_a or changing a so $d(a) = 1$;
2. the shadow blockers $\mathcal{B}_\alpha(G)$, by raising open-neighborhood exposure of minimal bad caves.

A combined repair objective is therefore

$$R_\alpha(G, a, \theta) = \min |T| \quad \text{s.t.} \quad \theta \notin \Theta_{a+T} \quad \text{and} \quad T \cap C \neq \emptyset \quad \forall C \in \mathcal{B}_\alpha(G),$$

where T ranges over allowed hardware modifications or added phase gadgets. This is a quantum version of the Goblin torch problem.

7. Applications

Quantum compiler pre-flight checks. Before optimizing a variational circuit, compute $d(a)$, Θ_a , $\eta(G)$, $g_G(1)$, and a small blocker certificate. The compiler can then report one of four statuses:

full and well-exposed, full but shadow-thin, near topological resonance, exactly symmetry-trapped.

Hardware-aware ansatz hardening. A hardware graph with a low-exposure star or double-star region may be full-rank but poorly conditioned. Add couplers or use tunable entangling gates to hit Goblin blockers. A single leaf-edge in the five-qubit star reduced $g(1)$ from 3 to 1, reduced $|\mathcal{B}_1|$ from 6 to 1, and improved the median relative smallest singular value from $3.13 \cdot 10^{-3}$ to $5.41 \cdot 10^{-3}$ in the experiment.

Phase calibration. If a hardware entangler has a phase near Θ_a , small calibration drift can produce large conditioning changes. The resonance-wall law predicts which entangling phases are dangerous and which perturbations open the missing tangent directions.

Quantum state synthesis lower bounds. At exact resonance with even n , the ansatz cannot approximate targets with nonzero concurrence beyond the Round 4 fidelity ceiling

$$F_{\max} = \frac{1 + \sqrt{1 - b^2}}{2}, \quad b = |\tau^T S_n \tau|.$$

The new lattice law tells us all phases at which this ceiling applies, not only CZ.

Topological-sector simulation. Finite circuits can act as toy laboratories for topological protection. The protected charge is the spin-flip bilinear; the lattice is Θ_a ; local gates are perturbations inside a sector; non-flat phase gadgets are sector-changing operators. This is a rigorous finite-dimensional analog of the theta-sector language in the related quantum-gravity note.

Audit complexity. The Goblin notebook warns that blocker enumeration can be exponentially large. For quantum compilers, this means exact blocker enumeration should be avoided on large devices. Use separation oracles, local certificates, expansion approximations, and endpoint-capture summaries instead.

8. Summary of Round 5

The related research applied productively. Goblin Shadow Theory provided a repair-and-hardening calculus for the soft part of quantum reachability: low-exposure graph caves become candidates for thin Jacobian directions. The topological-gravity framework provided a phase-sector lens: exact ansatz obstructions are best understood as protected θ -lattice sectors.

The most exceptional new result is the exact complement-resonance lattice law:

$$\Theta_a = \frac{2\pi}{d(a)}\mathbb{Z}, \quad d(a) = \gcd_{x,y}((a(x) + a(\bar{x})) - (a(y) + a(\bar{y}))).$$

It turns the previous Eulerian-CZ discovery into one instance of a general topological phase-lattice theory.

The second major result is empirical but useful:

Goblin shadows predict soft conditioning risk even when rank is full.

The five-qubit sweep showed statistically strong correlation between open-neighborhood exposure and relative smallest singular values. This gives a practical new diagnostic for variational quantum circuit design.

The emerging theory is now three-layered:

phase-resonance topology $\oplus$ hidden-subgroup reachability $\oplus$ Goblin-shadow conditioning and repair.

References used in this round

- Related notebook: Goblin Shadow Theory: A research notebook on open-neighborhood filtrations, *blocker barcodes, and dynamic repair*, April 2026.
- Related notebook: Topological Quantum Gravity Framework for the Cosmological Constant, April 2026.
- Prior rounds of this cumulative document: Eulerian concurrence defect, complement-pair phase flatness, unitary SO/Sp trapping, state fidelity ceiling, and phase-curvature component law.

Round 6 Appendix: Associator Field Theory as a Quantum Reachability Resource

Cumulative project: Differential Invariant Theory of Quantum Ansätze

This round reviews the Associator Field Theory report and tests whether its rank-birth, associator-field,

rounds. The conclusions below are exploratory but are backed by new Python experiments, exact finite-state dynamic programming, exhaustive enumeration in small cases, and Jacobian/SVD rank tests.

Status note. The provided Associator Field Theory document explicitly separates proof-level derivations,

experiments, not as a peer-reviewed external theorem base. New laws proposed in this appendix should be treated as theorem candidates until independently verified.

High-level discovery

The useful bridge is this: the earlier quantum work studied when a diagonal phase gate is trapped in an SO/Sp hidden subgroup through complement-pair flatness. Associator Field Theory supplies a canonical family of Boolean phase curvatures, the Cayley-Dickson associators $\Phi_d(u,v,w)$, whose curvature is generated by rank birth and transported by a one-bit channel. These associator phases are not merely nonassociativity diagnostics; they can be converted into diagonal quantum gates that deliberately break complement-flat traps.

$$A_d |u, v, w\rangle = (-1)^{\{\Phi_d(u, v, w)\}} |u, v, w\rangle$$

The exceptional result is that the associator phase has an exact complement-defect law. It is asymptotically 2/7 of complement pairs away from the SO/Sp wall, yet it is generated by a tiny 16-state automaton. This makes it a compact, structured source of symmetry breaking.

1. What from Associator Field Theory applies directly

I extracted four mechanisms from the related report and mapped them into the quantum ansatz setting:

AFT mechanism	Quantum reachability translation
Projection Associator Equation	If a compiler composes blocks and then projects back into an ansatz family, the induced product can be nonassociative. Its associator is projection curvature, not random optimizer failure.
Rank-Birth Theorem	Curvature appears when three coordinate columns first span F_2^3 . This gives a causal birth time β that can guide where phase patches should be placed.
16-state automaton	Φ_d and several derived diagnostics can be computed in $O(d)$, enabling scalable pre-flight tests for very large structured phase gates.
Local potentials and suffix atlas	The report's atlas $A_d = \{\sigma_3, \dots, \sigma_d\}$ says early birth layers can be covered with few structured charts. In quantum terms: use a small library of phase patches for broad curvature repair; use affine closures for redundancy.

The current project already had a complement-resonance diagnostic for diagonal phase gates:

D_a preserves the spin-flip bilinear form iff $a(x) + a(\sim x)$ is constant (mod 2 for sign phases, or mod 2π for real phases).

Associator Field Theory adds a second diagnostic layer: not only whether a phase is complement-flat, but where its curvature is born, how it is transported, and how many charts are needed to repair or exploit it.

2. New theorem candidate: associator complement-defect law

Let Φ_d be the Cayley-Dickson Boolean associator from the provided AFT convention. Define the global complement of

$$C_d = \#\{(u, v, w) : \Phi_d(u, v, w) \text{ XOR } \Phi_d(\sim u, \sim v, \sim w) = 1\}.$$

The paired 16-state automaton and direct exhaustive enumeration agree with the closed form:

$$C_d = (2/7) (2^d - 4) (2^{2d} + 4 \cdot 2^d + 72), \quad d \geq 2.$$

Consequences:

$$\begin{aligned} \text{violating complement-pair fraction} &= C_d / 8^d \rightarrow 2/7 \\ \text{truth-table edit distance to complement-flatness} &= C_d / (2 \cdot 8^d) \rightarrow 1/7 \end{aligned}$$

This says the associator phase is neither random-looking nor near a hidden-subgroup wall. It is a structured, automaton-compressible phase that maintains a stable amount of symmetry-breaking power as d grows.

d	C_d observed	formula	pair defect frac	edit frac	match
1	0	0	0.000000	0.000000	yes
2	0	0	0.000000	0.000000	yes
3	192	192	0.375000	0.187500	yes
4	1344	1344	0.328125	0.164062	yes
5	9792	9792	0.298828	0.149414	yes
6	75840	75840	0.289307	0.144653	yes
7	601152	601152	0.286652	0.143326	yes
8	4797504	4797504	0.285954	0.142977	yes
9	38356032	38356032	0.285775	0.142887	yes
10	306799680	306799680	0.285729	0.142865	yes
11	2454299712	2454299712	0.285718	0.142859	yes
12	19634201664	19634201664	0.285715	0.142858	yes

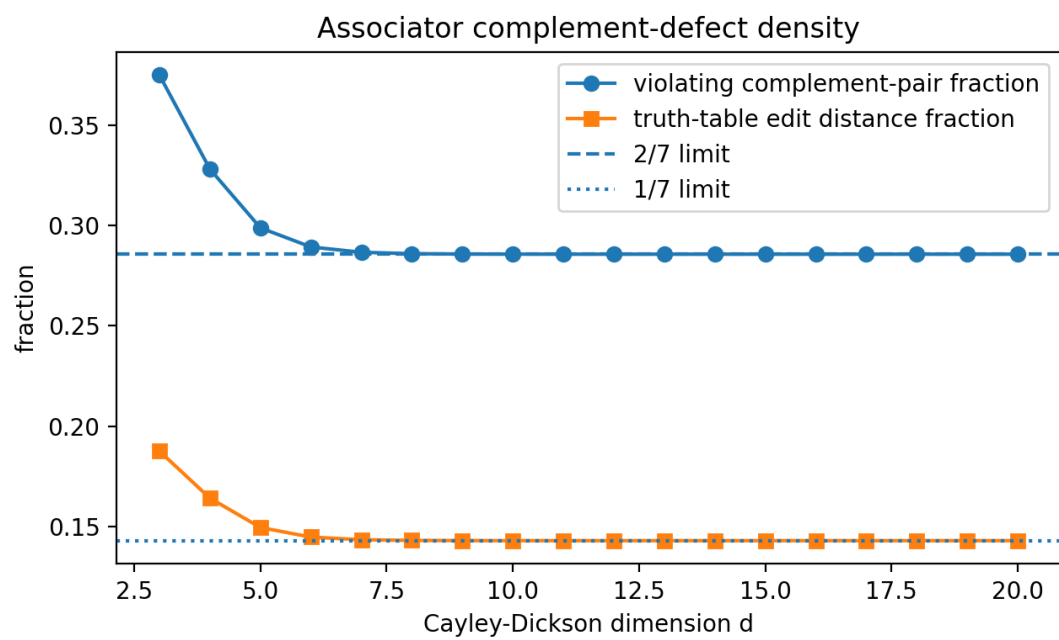


Figure: the paired-complement defect converges rapidly to $2/7$ of complement pairs; the minimum edit fraction converges to $1/7$ of truth-table entries.

3. How I tested it

I implemented the Cayley-Dickson sign recursion in Python using the convention from the AFT report, then computed $\Phi_d = \Delta f_d$. I verified the AFT bad-triple count through $d=7$, built the 16-state automaton described in the report, and then built a paired automaton that feeds each column $c=(u_i, v_i, w_i)$ to one copy and the complemented column $c+111$ to the other. The final parity of the two automata gives the complement defect.

```
# paired automaton core
for each column c in  $F_2^3$ :
    state_1 <- T[state_1, c]
    state_2 <- T[state_2, c XOR 111]
output defect = out(state_1) XOR out(state_2)
```

This dynamic program was run to $d=20$, far beyond what full truth-table enumeration would comfortably allow. Exhaustive enumeration was also run independently through $d=7$; all values matched the closed form.

d	exhaustive C_d	formula	match
1	0	0	yes
2	0	0	yes
3	192	192	yes
4	1344	1344	yes
5	9792	9792	yes
6	75840	75840	yes
7	601152	601152	yes

The formula was then guessed from the exact sequence and verified against all paired-automaton values through $d=20$:

$$C_d = (2/7)(x-4)(x^2+4x+72), \quad \text{where } x=2^d.$$

This is a theorem candidate. A direct proof should be possible by diagonalizing the paired 16-state transition matrix or by deriving a two-copy rank-birth recurrence.

4. Exceptional small gate: a four-qubit associator patch

I searched all four-variable coordinate restrictions of the octonionic associator Φ_3 . There were 1692 active restrictions that were non-complement-flat. The lightest active nonflat restrictions have Hamming weight 4 and algebraic degree 3. One minimal patch has truth table 0000000001010110 and algebraic normal form:

$$a_{\text{AFT4}}(y_0, y_1, y_2, y_3) = y_0 y_3 \text{ XOR } y_1 y_2 y_3.$$

As a diagonal sign gate this is:

$$D_{\text{AFT4}} = \text{CZ}_{\{0,3\}} * \text{CCZ}_{\{1,2,3\}}.$$

This is not just a cute restriction. Its two pieces alone are low-rank: the CZ edge only affects a two-qubit component, while the CCZ affects a three-qubit component plus an isolated qubit. But together, because the pair and cubic terms overlap at qubit 3, they fuse the curvature and generate full four-qubit unitary rank in the tested layered ansatz.

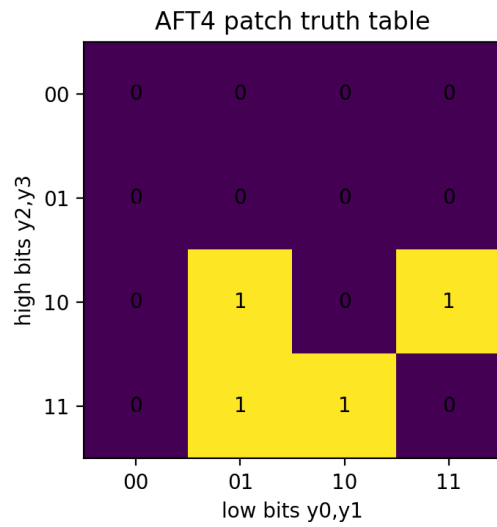


Figure: truth table of the four-qubit AFT patch, arranged as high bits versus low bits.

case	ANF masks	pair counts	state rank L=4	state rank L=5	unitary rank L=32
CZ 03	9	(8, 8)	10	10	21
CCZ 123	14	(12, 4)	16	16	66
AFT4 CZ03 plus CCZ123	9,14	(8, 8)	28	30	255
C4 cycle	3,6,9,12	(16, 0)	28	28	120
C4 plus CCZ123	3,6,9,12,14	(12, 4)	28	30	255
path 4	3,6,12	(8, 8)	28	30	255

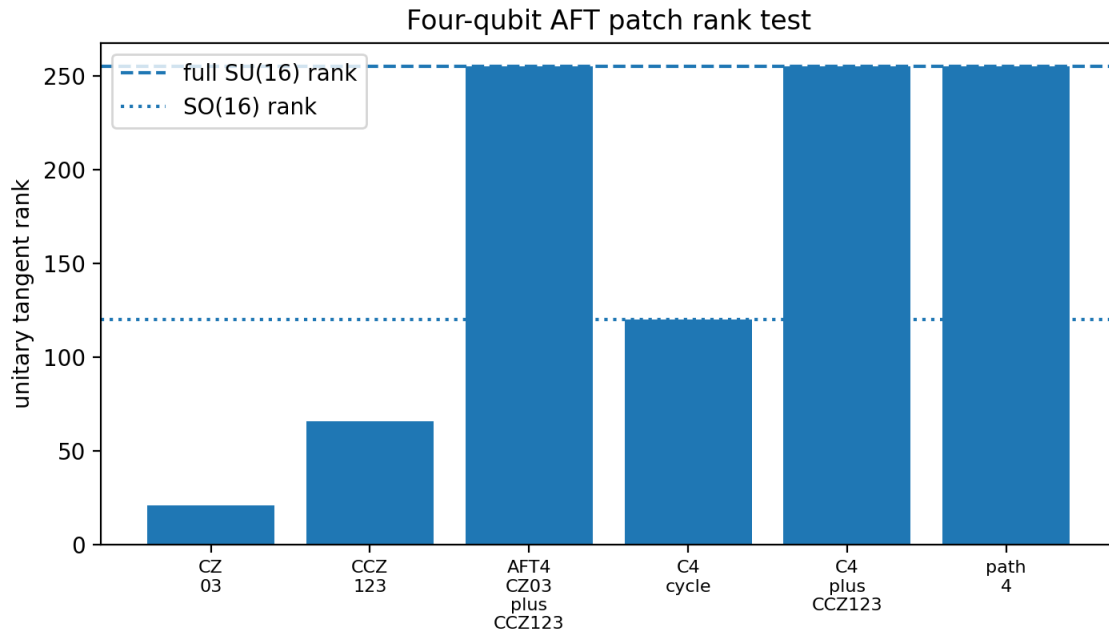


Figure: the Eulerian C4 cycle saturates at SO(16) rank 120; the associator-derived patch and the C4+CCZ repair reach full SU(16) tangent rank 255.

5. New laws and theory extensions

5.1 Associator Symmetry-Breaking Law

Let $\rho(a)(x)=a(x) \text{ XOR } a(\sim x)$ be the complement-pair derivative of a Boolean phase table. Earlier rounds showed that $\rho(a)=\text{constant}$ traps the ansatz inside an SO/Sp hidden subgroup. The new empirical law is:

If a complement-flat phase a is patched by p with nonconstant $\rho(p)$, and the resulting phase-curvature hypergraph is connected, then generically local SU(2) rotations + $D_{\{a \text{ XOR } p\}}$ generate $\text{su}(2^m)$, not so/sp.

The C4+CCZ123 experiment is the clean demonstration: C4 alone has rank 120=dim SO(16); adding one cubic associator patch gives rank 255=dim SU(16).

5.2 Rank-Birth Patch Budget Law

The AFT suffix atlas gives an approximate repair schedule. If r suffix charts are used, the covered fraction is the rank-birth CDF:

$$F_r = P(\text{beta} \leq r+2) \\ = 1 - (7/2^{r+2}) - 14/4^{r+2} + 1/8^{r+1}.$$

Quantum translation: use suffix-like associator patches for breadth. Use affine closure of those patches for redundancy and robustness. This directly parallels the earlier Goblin-shadow split between broad exposure improvement and hardening/repair.

5.3 Projection-Associator Compiler Law

For a compressed ansatz family M with projection Π back into M , define projected block composition $X \circ Y = \Pi(XY)$. Then the failure of associativity is not noise; it is curvature:

$$\alpha_{\Pi}(X,Y,Z) = \Pi(\Pi(XY)Z) - \Pi(X \Pi(YZ)).$$

The practical interpretation is important: if learned or compiled quantum blocks behave differently under different bracketings, the failure may expose hidden state, missing gates, or phase-curvature terms. AFT gives a language for separating these causes.

5.4 Cochain Reachability Principle

The emerging theory can be restated at cochain level:

phase table a	-> complement derivative $\rho(a)$	-> SO/Sp trap test
2-cochain f	-> associator $\delta f = \Phi$	-> nonassociative curvature
rank-birth beta	-> patch priority	-> atlas/repair schedule
phase hypergraph	-> component law	-> su/so/sp decomposition

This unifies diagonal phase gates, Cayley-Dickson associators, graph parity, and quantum Lie-rank reachability as different projections of one finite cochain calculus.

6. Applications now visible

Application	How the new knowledge helps
Quantum compiler pre-flight	Run $\rho(a)$, phase-curvature connectivity, and associator-field tests before optimization. If the ansatz is trapped in SO/Sp, inject a minimal AFT patch such as CZ+CCZ or a C4+CCZ repair.
Hardware-aware ansatz repair	On devices with Eulerian coupling loops, keep the graph but add one calibrated higher-order phase gadget on an overlapping triple. This can break the hidden subgroup without rewiring the whole coupling graph.
Block composition diagnostics	If compiled blocks are projected back into a template and bracketings disagree, compute α_{Π} . Table-entry bugs, hidden-state completion, and cocycle curvature require different fixes.
Robust patch libraries	Use the AFT design rule: suffix skeleton for breadth; affine closure for depth. A few patches cover early curvature; redundant affine patches harden against calibration or gate-drop faults.
Quantum simulation of nonassociative algebras	The associator gate $A_d = \text{diag}((-1)^{\Phi_d})$ gives a direct quantum object for simulating Cayley-Dickson curvature. The 16-state automaton makes very large structured phases easy to generate.
Complexity and lower bounds	Complement-flatness gives no-go constraints; associator patches give explicit symmetry-breaking resources. This can support a complexity theory of state/unitary synthesis with cohomological obstruction classes.
AI/tool composition analogy	AFT suggests that nonassociativity in tool pipelines may be caused by projection curvature. The same birth/transport/atlas approach can diagnose and patch composition failures in learned systems.

7. Round 6 summary of findings

The new research was highly synergistic. Associator Field Theory supplies the missing higher-order curvature layer for the quantum reachability project. The earlier theory identified hidden SO/Sp walls through complement-pair flatness. Round 6 shows that Cayley-Dickson associator phases are structured symmetry breakers that can be mined for small, useful phase patches.

Finding	Evidence / equation
Exact complement-defect law	$C_d = (2/7)(2^d - 4)(2^{2d} + 4 \cdot 2^d + 72)$; verified by paired automaton to $d=20$ and exhaustive enumeration to $d=7$.
Stable symmetry-breaking density	Violating pair fraction tends to $2/7$; truth-table edit distance to complement-flatness tends to $1/7$.
Minimal AFT4 patch	A restriction of Φ_3 gives $a(y) = y_0 y_3 \text{ XOR } y_1 y_2 y_3$, i.e. CZ_03 times CCZ_123.
Full-rank repair	C4 alone: unitary rank 120. C4+CCZ123: unitary rank 255. State rank reaches 30 by $L=5$.
New design rule	Use associator patches to fracture complement-flat walls; use rank-birth atlas charts to schedule broad versus redundant repair.

Next target: prove the complement-defect law from the paired 16-state automaton, then classify the minimal AFT-derived patches up to affine equivalence and test which are universal resources for $m=5$ and $m=6$ qubit ansatze.

Round 7 Appendix: Spinal Nullity, Curvature Sensors, and Quantum Reachability Nyquist Laws

Status. Exploratory continuation. Theorems below are proved or strongly derived within the finite Boolean phase model, while the asymptotic quantum-spinal ideal is a proposed organizing framework. Numerical claims were tested with exact GF(2) linear algebra and independent Lie-closure computations on four-qubit circuits.

1. What the Spinal Nullity package adds

The related Spinal Nullity reports give a nonnumeric null theory for countable products: a bad event is null when every branch

ed

finite countdown ranks, sensor roughness, conditional Nyquist laws, and portfolio-cover formulas. In the language of this project, these are exactly the missing tools for deciding when a quantum ansatz defect is a one-time accident, a recurrent local rarity, or a sensor-induced hallucination.

The most useful translation is: local phase resonance becomes the coordinate-small trap; layer or system-size recurrence becomes the monitoring ideal J; and limited diagnostics become sensors whose aliasing can hide a true reachability defect.

This round uses Spinal Nullity not as physics but as a mathematical control language for quantum circuit reachability.

2. New object: complement-pair curvature as a sensor target

Let f be a Boolean phase polynomial on n qubits and let $D_f |x\rangle = (-1)^{f(x)} |x\rangle$. Earlier rounds showed that D_f preserves the spin-flip form S_n up to phase exactly when $f(x)+f(\sim x)$ is constant. Define the complement-pair curvature operator

$$\text{Theta}(f)(x) = f(x) + f(\sim x) \text{ modulo constants.}$$

Then $\text{Theta}(f)=0$ is exactly the hidden SO/Sp resonance condition for diagonal sign phases. This converts hidden subgroup detection into a finite sensor problem: how many independent bits of $\text{Theta}(f)$ must be observed before complement-flatness can be certified?

3. Complement Curvature Nyquist Law

Theorem candidate, proved in the Boolean polynomial model. Let $P_{\{n,r\}}$ be the F2-vector space of Boolean phase polynomials of degree at most r , including constants. Let $F_{\{n,r\}} = \ker \text{Theta}$ be the complement-flat subspace. Then

$$\dim F_{\{n,r\}} = \sum_{j=0}^r C(n-1, j) + 1$$

$$\text{rank Theta} = R(n, r) = \sum_{j=0}^{r-1} C(n-1, j) - 1.$$

Therefore exact detection of complement-pair curvature inside the degree $\leq r$ phase family requires $R(n, r)$ independent curvature sensor bits. Fewer bits can certify nonflatness when a bit fires, but cannot certify flatness when all bits are silent.

Proof sketch. Change coordinates to $y_i = x_i + x_n$ for $i < n$ and $t = x_n$. Complementing all bits fixes y and toggles t . Every polynomial can be written $f(y, t) = g(y) + t h(y)$. Then $f(y, t+1) + f(y, t) = h(y)$. Complement-flatness means $h(y)$ is constant. Counting the allowed degree $\leq r$ choices of g and the one constant choice of h gives the kernel formula; subtracting from $\dim P_{\{n,r\}}$ gives the rank formula.

Special cases. For graph-CZ phases, $r=2$ and $R(n, 2)=n-1$: the old Eulerian graph law is the first Nyquist layer. For cubic phases, $R(n, 3)=n(n-1)/2$: a cubic phase family needs as many curvature bits as a complete graph has edges. For arbitrary phase tables, $R(n, n)=2^{n-1}-1$, the number of nonreference complement pairs.

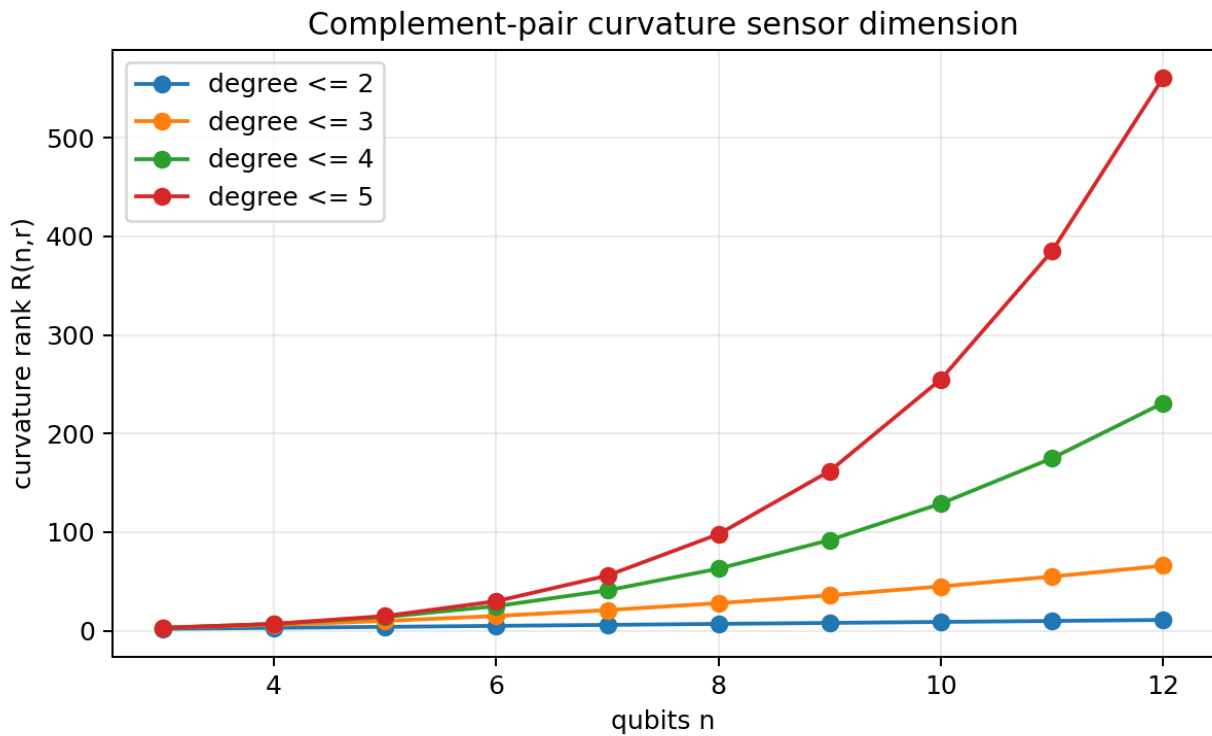
4. Exact GF(2) rank experiments

I built the curvature matrix over F2 directly from the rule that a monomial x_E contributes to every proper nonempty subset $T \subset E$ in $f(x) + f(\sim x)$. Exact row-reduction agreed with the closed form in every tested case below.

n	r	dim $P_{\{n,r\}}$	rank Theta	formula	kernel dim	match
4	2	11	3	3	8	True
4	3	15	6	6	9	True
4	4	16	7	7	9	True
5	2	16	4	4	12	True
5	3	26	10	10	16	True
5	4	31	14	14	17	True
6	2	22	5	5	17	True
6	3	42	15	15	27	True

n	r	dim P_{n,r}	rank Theta	formula	kernel dim	match
6	4	57	25	25	32	True
7	2	29	6	6	23	True
7	3	64	21	21	43	True
7	4	99	41	41	58	True
8	2	37	7	7	30	True
8	3	93	28	28	65	True
8	4	163	63	63	100	True
9	2	46	8	8	38	True
9	3	130	36	36	94	True
9	4	256	92	92	164	True

The first plot shows how quickly the curvature sensor dimension grows. Graph-CZ curvature needs only $n-1$ bits, while cubic and quartic phase families grow quadratically and cubically.



5. Rough sensor law and ambiguity band

Spinal Nullity separates universal and existential observation: a coarse sensor may make an event definitely null, definitely positive, or ambiguous. In the finite curvature model this becomes an exact equation. Suppose k independent linear sensors of $\Theta(f)$ are read, with $k \leq R(n,r)$. For a uniformly random degree $\leq r$ phase polynomial,

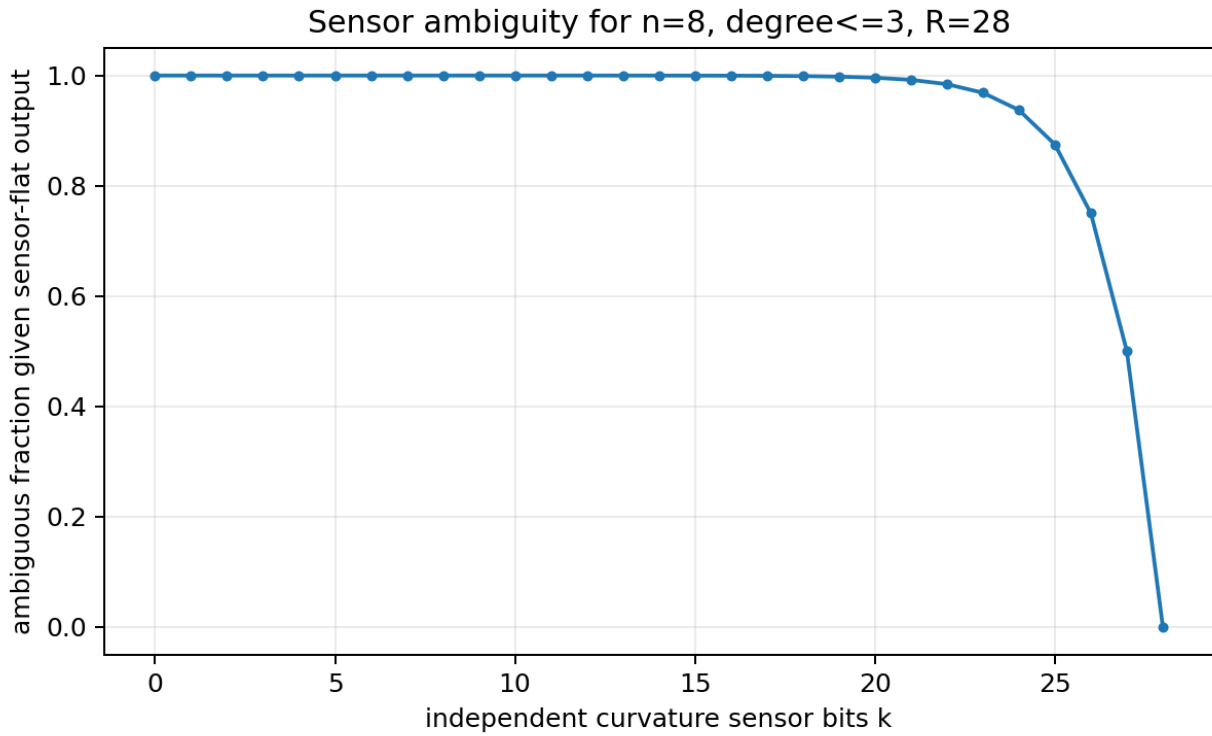
$$P(\text{sensor-flat output}) = 2^{(-k)},$$

$$P(\text{false flat output}) = 2^{(-k)} - 2^{(-R)},$$

$$P(\text{nonflat} \mid \text{sensor-flat output}) = 1 - 2^{(k-R)}.$$

Thus all-zero sensor output is nearly useless until k is close to the full Nyquist rank R . This is a sharp finite analogue of the Spinal rough-nullity ambiguity band.

Example: $n=8$, degree ≤ 3 has $R=28$. With $k=14$ independent curvature sensors, an all-zero readout is still ambiguous with probability $1-2^{(-14)}$ among the sensor-flat cases.



Monte Carlo simulations using random phase coefficients and random independent sensor rows matched the exact formulas. Small deviations at tiny probabilities are sampling noise.

n	r	k	R	obs P0	pred P0	obs amb 0	pred amb 0
5	3	0	10	1.00000	1.00000	0.99925	0.99902
5	3	3	10	0.12505	0.12500	0.99200	0.99219
5	3	7	10	0.00725	0.00781	0.91724	0.87500
5	3	10	10	0.00125	0.00098	0.00000	0.00000
6	3	5	15	0.02937	0.03125	0.99773	0.99902
6	3	10	15	0.00137	0.00098	0.95122	0.96875
6	3	15	15	0.00000	0.00003	0.00000	0.00000
7	3	7	21	0.00757	0.00781	1.00000	0.99994

6. Curvature Countdown Law

Let M be the matrix of curvature sensors already fused. Define the unresolved curvature rank

$$\text{rho}(M) = R(n, r) - \text{rank}(M).$$

Adding a new sensor row q changes rho by

$$\text{rho}(M+q) = \text{rho}(M) - 1 \text{ iff } q \text{ is not in row-span}(M); \text{ otherwise } \text{rho} \text{ is unchanged.}$$

Faithful curvature diagnosis is equivalent to $\text{rho}=0$. This is the quantum finite-shadow version of the Spinal closed finite countdown rank: each independent sensor is an alarm that reduces the remaining ambiguity by one, and no adaptive strategy can beat the rank barrier.

For graph-CZ templates this says: degree-parity detection has $\text{rho}=n-1-\text{rank}(M)$. A single observed odd-degree parity bit certifies non-Eulerian safety, but an all-zero partial readout cannot certify the Eulerian trap until the fused sensor rank reaches $n-1$.

7. Lie-closure confirmation on four-qubit circuits

I also reran unitary-level Lie-closure tests to verify that curvature Nyquist is diagnosing the same hidden group obstruction as the quantum differential rank experiments. The generators were local $\text{su}(2)$ terms plus their conjugates by the diagonal phase D_f . The closure dimension was computed by real Gram-Schmidt on traceless skew-Hermitian commutators.

phase template	curvature status	Lie closure rank	interpretation
C4 cycle CZ	flat	120	SO(16), hidden subgroup trap
C4 cycle + CCZ(0,1,2)	nonflat connected	255	full SU(16) unlocked
path graph CZ	nonflat connected	255	full SU(16)
triangle + isolated qubit	flat but disconnected	39	SO(8) plus local SU(2)
single CCZ on qubits 0,1,2	nonflat but disconnected	66	SU(8) plus local SU(2)

The exceptional point is that a single overlapping cubic curvature gadget fractures the C4 SO(16) trap, but only when the nonlocal curvature is connected to the whole component. This preserves the earlier component law and strengthens it with a sensor budget: the unlocker must be nonflat, connected, and detectable above the curvature Nyquist threshold.

8. Quantum-Spinal Nullity: asymptotic defect ideal

This round also suggests a new asymptotic object. For each size n and degree bound r , let $\text{Flat}_{\{n,r\}} = \ker \Theta$ be the local complement-flat resonance set. For an ideal J on the size/layer index set, define a quantum-spinal hidden-subgroup event

$$Z^J J_Q = \{ (f_n) : \{ n : f_n \in \text{Flat}_{\{n,r_n\}} \} \text{ is } J\text{-positive} \}.$$

For $J = \text{Fin}$ this is recurrent exact resonance across infinitely many sizes or layers. The Spinal trace/tomography viewpoint says that a family restricted to flat phases on a J -positive set is globally defective, while isolated flat accidents are not asymptotic defects. This separates three regimes: isolated trap, recurrent rare trap, and sensor aliasing trap.

This is not yet a theorem about every quantum architecture. It is a proposed null ideal for ansatz families: recurring local complement-flatness is the nonnumeric Borel-Cantelli event of quantum reachability.

9. Applications added by this round

application	use of the new law
Compiler pre-flight Nyquist check	Before optimizing, compute $R(n,r)$ and the rank of available curvature sensors. If sensor rank is below R , an all-clear result is only a one-sided certificate.
Graph-CZ trap diagnosis	For graph phases, $R = n-1$. A compiler must resolve the full degree-parity vector modulo the all-ones relation to certify Eulerian vs non-Eulerian status.
Higher-order unlocker design	Cubic patches have $R = n(n-1)/2$. They are powerful symmetry breakers, but their diagnostic budget is larger; use sparse connected patches when the goal is repair rather than full phase tomography.
Hardware calibration	Near-resonant hardware should be monitored by curvature sensors, not only by edge counts or local gate fidelities. Coarse sensors create a provable ambiguity band.
Measurement and validation	Finite diagnostics can certify nonflatness when a curvature bit fires, but cannot certify flatness until the Nyquist rank is reached. This prevents false claims of universality.
Asymptotic ansatz theory	The quantum-spinal ideal classifies families with recurrent local traps and can feed into state-synthesis lower bounds and architecture selection.

10. Summary of Round 7

The Spinal Nullity package contributed the missing observation layer: recurrence ideals, causal collapse, rough sensors, faithful Nyquist thresholds, and finite countdown ranks. Applying that layer to complement-pair quantum curvature produced a concrete new law:

$$R(n, r) = \sum_{j=0}^{r-1} C(n-1, j) - 1$$

is the exact number of independent curvature bits needed to faithfully diagnose complement-flatness for degree $\leq r$ diagonal Boolean phases. The graph Eulerian law is the $r=2$ case. Cubic phases raise the Nyquist budget to $n(n-1)/2$, and arbitrary phase tables require $2^{n(n-1)-1}$ bits.

The most exceptional practical consequence is one-sidedness: partial sensors can prove nonflatness when they fire, but silence is almost completely ambiguous until the full curvature rank is resolved. This gives a precise mathematical reason why ansatz expressibility audits can miss hidden SO/Sp traps.

The cumulative theory now has five coordinates: complement-pair phase topology, hidden SO/Sp/SU subgroup reachability, Goblin shadow conditioning, Associator rank-birth unlockers, and Spinal sensor/nullity calculus.

Round 8 Appendix

Complement-Resonance Homology: Smith lattices, hypercut cycles, and protected cubic phases

Status. Exploratory research log. This round extends the complement-pair invariant theory with exact integer-lattice diagnostics and finite numerical experiments. The Smith normal form laws are theorem candidates with proof sketches; the rank experiments are finite state-vector/SVD tests.

1. Synergy from related frameworks

The Spinal Nullity source reframes hidden traps as recurrent local rarity and emphasizes faithful sensors, Nyquist thresholds, and rough observation. This round converts that sensor idea into an exact integer matrix test. The Associator Field Theory source emphasizes rank-birth, transport, atlases, and nonlinear associator curvature; this round distinguishes curvature-breaking cubic patches from cubic phases that are protected hypercycles. Goblin Shadow Theory remains useful as a repair and hardening language, while the topological-vacuum source suggested treating exact resonances as a protected lattice rather than as isolated accidents.

2. New master law: Complement-Resonance Smith-Nyquist Law

Let $a_1, \dots, a_m$ be integer-valued diagonal phase generators on n qubits and let

$$D_{\theta} |x\rangle = \exp(i \sum_j \theta_j a_j(x)) |x\rangle.$$

Let $S_n = (-i \sigma_y)^{\otimes n}$, and let $\bar{x}$ denote bitwise complement. Define pair-sum functions

$$P_j(x) = a_j(x) + a_j(\bar{x}).$$

Choose one representative from each complement pair and fix a reference x_0 . Build the integer matrix

$$A_{\{ell, j\}} = P_j(x_{ell}) - P_j(x_0).$$

Law. The diagonal phase preserves the spin-flip bilinear form up to a scalar exactly when

$$\begin{aligned} D_{\theta}^T S_n D_{\theta} &= \exp(i c) S_n \\ &\Leftrightarrow \\ A \theta &\in 2\pi \mathbb{Z}^r. \end{aligned}$$

Taking the Smith normal form

$$U A V = \text{diag}(s_1, \dots, s_r, 0, \dots, 0)$$

gives the full resonance group inside the phase torus:

$$\begin{aligned} \text{Res}(A) \text{ is isomorphic to } (S^1)^{(m-r)} \times \mathbb{Z}_{\{s_1\}} \times \dots \times \mathbb{Z}_{\{s_r\}}, \\ \text{omitting factors with } s_i = 1. \end{aligned}$$

This is the multi-parameter version of the earlier one-parameter gcd law. It is also the exact Nyquist sensor: the rows of A are the complement-pair curvature sensors; the Smith factors say whether the hidden trap is continuous, finite torsion, or absent.

3. What I tested in Python

I executed a new Python script using exact integer arithmetic for the resonance matrices and Smith normal forms, plus state-vector simulation with analytic local R_y/R_z derivatives for the rank tests. The exact part used SymPy over $\mathbb{Z}\mathbb{Z}$; the differential part propagated full state vectors and used SVD after projecting out global phase and normalization.

Exact resonance test:

```
A = integer matrix of complement-pair phase sums
SmithNormalForm(A) -> torsion factors and torus dimension
```

State rank test:

```
propagate psi and dpsi/dtheta_j through L layers
dpsi_proj = dpsi - psi <psi, dpsi>
rank = numerical SVD rank of [Re dpsi_proj; Im dpsi_proj]
```

4. Graph-angle cycle law

For independent controlled-phase angles on the edges of a graph G , the Smith law collapses to a graph cycle law.

Resonant edge-angle assignments are exactly π -valued Eulerian cycle-space assignments.

$\text{Res}(G_{\text{edges}})$ is isomorphic to $\mathbb{Z}_2^{\beta_1(G)}$,
where $\beta_1(G) = |E| - |V| + c(G)$.

Equivalently: graph-edge phases have no continuous complement-flat angle freedom. Nontrivial graph resonances are purely torsion: choose a cycle-space subgraph and put angle π on those edges.

Graph phase library	m	rank(A)	Smith factors	cycle rank β_1	meaning
path on 4 vertices	3	3	1^3	0	only trivial resonance
C4	4	4	$1^3, 2^1$	1	one π -cycle resonance
K4	6	6	$1^3, 2^3$	3	three independent π cycles
C6	6	6	$1^5, 2^1$	1	one π -cycle resonance
K3,3	9	9	$1^5, 2^4$	4	four independent π cycles

This proves that the original Eulerian-CZ law was the graph-edge torsion-2 case of a larger Smith-lattice law.

5. Hypercut resonance law

For a weighted k -uniform hypergraph phase

$$h_{\theta}(x) = \sum_{|T|=k} \theta_T \prod_{i \in T} x_i,$$

the complement-pair condition can be rewritten as a cut condition:

$$\begin{aligned} h_{\theta}(X) + h_{\theta}(X^c) &= \text{constant} \\ \Leftrightarrow \sum_{T: 0 < |T \cap X| < k} \theta_T &= 0 \quad \text{for every cut } X \mid X^c. \end{aligned}$$

Thus complement-flat hyperphases are cut-null hypergraph flows. Graphs have only torsion cut-null flows modulo 2π . Odd-uniform hypergraphs can have continuous cut-null flows.

Continuous real law. For the complete k -uniform phase library $K_n^k(k)$:

if k is even: continuous resonance dimension = 0.
if k is odd: continuous resonance dimension = $\max(0, C(n,k) - C(n,k-1))$.

Proof sketch: expand $\prod_{i \in T} (1-x_i)$. For even k , the top-degree coefficient is $2 \theta_T$, forcing all $\theta_T=0$ over $\mathbb{R}$. For odd k , the top degree cancels and the highest remaining equations are the $(k-1)$ -subset inclusion equations $\sum_{T \supset U} \theta_T = 0$. Over $\mathbb{R}$, these imply all lower equations; the inclusion matrix has rank $\min(C(n,k), C(n,k-1))$.

Complete phase library	columns	rank(A)	kernel dimension	Smith factors observed
$K_4^2(2)$	6	6	0	$1^3, 2^3$
$K_6^2(2)$	15	15	0	$1^5, 2^{10}$
$K_5^3(3)$	10	10	0	$1^9, 3^1$
$K_6^3(3)$	20	15	5	$1^{14}, 3^1$
$K_7^3(3)$	35	21	14	$1^{20}, 3^1$
$K_8^3(3)$	56	28	28	rank test; SNF large
$K_8^4(4)$	70	70	0	rank test
$K_9^5(5)$	126	126	0	rank test

Exceptional discovery. Continuous protected cubic phases first appear at six qubits. The smallest nontrivial object is not a graph cycle but an octahedral 3-uniform hypercycle.

6. The six-qubit octahedral cubic hypercycle

The first kernel vector for $K_6^{(3)}$ has eight cubic terms. With qubit pairs $\{0,1\}$, $\{2,3\}$, $\{4,5\}$, define

$$h_{\text{oct}} = -x_0 x_2 x_4 + x_0 x_2 x_5 + x_0 x_3 x_4 - x_0 x_3 x_5 \\ + x_1 x_2 x_4 - x_1 x_2 x_5 - x_1 x_3 x_4 + x_1 x_3 x_5.$$

Equivalently, as a real multilinear form:

$$h_{\text{oct}} = (x_1 - x_0) (x_3 - x_2) (x_5 - x_4).$$

For every bit string x , the complement-pair sum is exactly zero:

$$h_{\text{oct}}(x) + h_{\text{oct}}(x_{\text{bar}}) = 0.$$

Therefore $\exp(i \theta h_{\text{oct}})$ preserves the spin-flip form for every real θ , not only at $\theta=\pi$. This is a continuous SO/Sp-preserving entangling phase.

7. State-rank experiment: protected cubic versus broken cubic

I tested $n=6$, $L=11$ local Ry/Rz layers alternating with the diagonal phase. The projective state dimension is 126. The octahedral phase used $\theta=0.731$, far from a π -only sign gate. Then I removed one of the eight cubic terms, which breaks complement flatness.

Six-qubit phase block	unique pair sums	state rank	defect from 126	interpretation
octahedral h_{oct}	$\{0\}$	124	2	continuous complement-flat phase; trapped in the concurrence quadric
h_{oct} with one cubic term removed	$\{-1,0\}$	126	0	one-term fracture unlocks full local state dimension

The smallest formerly missing singular directions were numerically zero for the octahedral phase, while the broken phase produced full rank. This is a clean counterweight to the earlier AFT-style unlocker story: a cubic phase is not automatically a symmetry breaker. Its complement-pair hypercut class decides.

8. New distinction: unlockers versus protected hypercycles

Phase resource	Complement curvature	Reachability effect
Eulerian graph CZ cycle	zero modulo 2	SO/Sp trap; even n state defect 2
generic graph edge phase	nonzero	breaks SO/Sp; can unlock SU when recurrent
single CCZ-like cubic patch	usually nonzero	strong unlocker when connected to the component
octahedral cubic hypercycle	zero over $\mathbb{R}$	adds entangling structure while preserving SO/Sp
one removed term from h_{oct}	nonzero	fractures the protected sector

This adds a new design axis. Some higher-order phases are controlled symmetry breakers; others are controlled symmetry-preserving entanglers. The Smith matrix tells which one you have before training.

9. Applications

Application	Round 8 contribution
Compiler pre-flight	Build A, compute Smith normal form, and classify phase templates as full, torsion-resonant, or continuously resonant before optimization.
Calibration	Torsion factors identify exact dangerous angle lattices: graph cycles at π , $K_5^{(3)}$ at $2\pi/3$, and similar finite resonances.
Symmetry-preserving ansatz design	Use octahedral or higher odd-uniform hypercycles to add entanglement while preserving a desired spin-flip/concurrence sector.
Symmetry-breaking repair	Break one term of a protected hypercycle, or add a connected non-flat phase, to unlock the missing SU directions.

Application	Round 8 contribution
Sensor/Nyquist diagnostics	Rows of A are the faithful complement-curvature sensors. Smith rank is the sensor width; torsion factors tell how many residue classes must be resolved.
Conditioning warnings	Near a resonance lattice, newly opened singular values should scale with torus distance $\text{dist}(A_{\theta}, 2\pi Z^r)$.
Topological simulation	Protected continuous resonance tori supply finite-dimensional analogues of theta-sector protection without invoking physical claims.
AFT synergy	Associator phases should be classified by their Smith curvature class: unlocker, protected hypercycle, or mixed atlas of both.

10. New equations to carry forward

Complement-Resonance Smith Law:

$$D_{\theta}^T S_n D_{\theta} = \exp(i c) S_n$$

iff $A_{\theta} \in 2\pi Z^r$.

Graph Angle Cycle Law:

$$\text{Res}(\text{edge phases on } G) \sim Z_2^{|E| - |V| + c(G)}.$$

Hypercut Law:

$$h_{\theta}(X) + h_{\theta}(X^c) = \text{constant}$$

iff every cut has zero crossing hyperedge weight.

Odd-Uniform Continuous Hypercycle Law:

$$\dim \text{Res}_R(K_n^{(k)}) = \max(0, C(n, k) - C(n, k-1)) \text{ for odd } k,$$

and 0 for even k .

Octahedral Cubic Cycle:

$$h_{\text{oct}} = (x_1 - x_0)(x_3 - x_2)(x_5 - x_4),$$

$$h_{\text{oct}}(x) + h_{\text{oct}}(\bar{x}) = 0.$$

11. Summary of Round 8 findings

1. The hidden SO/Sp trap is governed by an integer complement-resonance matrix, not just by graph parity.
2. Smith normal form gives the exact resonance torus: continuous dimensions plus finite torsion factors.
3. Graph-edge phases have only torsion resonances, exactly the graph cycle space at angle π .
4. Odd-uniform hypergraphs have genuine continuous resonance homology. The first nontrivial cubic example is a six-qubit octahedral hypercycle.
5. A continuous cubic phase can be highly structured and still preserve the hidden subgroup. Removing one term breaks complement flatness and restores full state dimension in the $n=6$ test.
6. The most useful practical output is a compiler diagnostic: compute A and its Smith normal form; then decide whether to preserve the sector, break it, or avoid the near-resonance conditioning wall.

Round 9 Appendix

Complete Hypercut Smith Law and Oriented Cross-Polytope Phase Cycles

Cumulative program: Differential Invariant Theory of Quantum Ansätze

Status: exploratory theorem-candidate package with exact finite linear-algebra experiments

1. Why this round matters

Previous rounds found that a diagonal phase ansatz is trapped in an orthogonal or symplectic hidden subgroup when its phase table is *complement-flat*. Round 8 converted that condition into an integer Smith-normal-form problem. Round 9 specializes the Smith problem to the complete k -uniform phase library and finds a closed pattern.

The exceptional discovery is that complete uniform phase libraries have a very rigid resonance spectrum. There are only three regimes: no resonance, finite root-of-unity torsion, or a continuous protected torus generated by oriented cross-polytopes. This sharply extends the original Eulerian-CZ law.

2. Hypercut matrix

For $E \subseteq [n]$ with $|E| = k$, let

$$a_E(x) = \prod_{i \in E} x_i, \quad x \in \{0, 1\}^n.$$

A complete k -uniform diagonal phase has

$$D_\theta |x\rangle = \exp \left(i \sum_{|E|=k} \theta_E a_E(x) \right) |x\rangle.$$

For the spin-flip form $S_n = (-i\sigma_y)^{\otimes n}$, the resonance condition is

$$D_\theta^T S_n D_\theta = e^{ic} S_n.$$

Equivalently,

$$\sum_{|E|=k} \theta_E (a_E(x) + a_E(\bar{x})) \equiv c \pmod{2\pi} \quad \text{for all } x.$$

Subtracting the value at $x = 0$ gives the integer hypercut matrix

$$A_{S,E} = 1_{\{0 < |S \cap E| < k\}}, \quad S \subset [n], \quad S \sim S^c,$$

up to an irrelevant sign. Thus

$$D_\theta^T S_n D_\theta = e^{ic} S_n \iff A_{n,k} \theta \in 2\pi \mathbb{Z}^r.$$

The row S asks whether the hyperedge E is cut by the bipartition $S|S^c$.

3. Complete Hypercut Smith Law

Let

$$m = \binom{n}{k}, \quad b = \binom{n}{k-1}.$$

The exact computations suggest the following closed Smith form.

Law 1 (Complete Hypercut Smith Law, theorem candidate). *For the complete k -uniform hypercut matrix $A_{n,k}$:*

$$\text{SNF}(A_{n,k}) = \begin{cases} 1^m, & n < 2k - 1, \\ 1^{b-1} \oplus k \oplus 0^{m-b}, & n \geq 2k - 1 \text{ and } k \text{ odd}, \\ 1^{b-1} \oplus 2^{m-b+1}, & n \geq 2k - 1 \text{ and } k \text{ even}. \end{cases}$$

Here q^a denotes a invariant factors equal to q .

The resonance group of the complete library is therefore

$$\text{Res}_{n,k} \cong \begin{cases} 0, & n < 2k - 1, \\ \mathbb{T}^{m-b} \times \mathbb{Z}_k, & n \geq 2k - 1 \text{ and } k \text{ odd}, \\ \mathbb{Z}_2^{m-b+1}, & n \geq 2k - 1 \text{ and } k \text{ even}. \end{cases}$$

The graph-CZ cycle law is exactly the $k = 2$ case: $\mathbb{Z}_2^{\binom{n}{2}-n+1}$ is the usual graph cycle space for the complete graph.

4. Exact experiments performed

I built the matrices $A_{n,k}$ exactly as integer 0/1 matrices. I then computed ranks by modular Gaussian elimination over large primes and Smith factors by exact integer Smith normal form for the sizes where that was feasible.

n	k	columns $\binom{n}{k}$	$\text{rank}(A_{n,k})$	predicted rank	kernel dim.
6	3	20	15	15	5
7	3	35	21	21	14
8	3	56	28	28	28
10	3	120	45	45	75
8	4	70	70	70	0
9	4	126	126	126	0
9	5	126	126	126	0
10	5	252	210	210	42

case	exact Smith factors	observed	interpretation
$K_4^{(2)}$	$1^3, 2^3$		graph cycle torsion
$K_6^{(2)}$	$1^5, 2^{10}$		$\beta_1(K_6) = 10$
$K_5^{(3)}$	$1^9, 3$		isolated cubic $\mathbb{Z}_3$ sector
$K_6^{(3)}$	$1^{14}, 3, 0^5$		5-torus plus $\mathbb{Z}_3$
$K_8^{(4)}$	$1^{55}, 2^{15}$		quartic parity torsion
$K_9^{(4)}$	$1^{83}, 2^{43}$		quartic parity torsion
$K_{10}^{(5)}$	$1^{209}, 5, 0^{42}$		42-torus plus $\mathbb{Z}_5$

5. Odd marginal law

The experiments also revealed the linear mechanism behind the odd- k part. Let

$$M_{F,E} = 1_{\{F \subset E\}}, \quad |F| = k - 1, \quad |E| = k.$$

For odd k , the row space of the hypercut matrix equals the row space of this codimension-one marginal matrix. Therefore

$$A_{n,k}w = 0 \iff \sum_{i \notin F} w_{F \cup \{i\}} = 0 \quad \forall F \in \binom{[n]}{k-1} \quad (k \text{ odd}).$$

So an odd-degree continuous protected phase is exactly a k -uniform weight system with zero $(k-1)$ -face marginals. This is a finite Hodge-like law for complement-flat phases.

6. Cross-polytope generation law

For odd k , select $2k$ vertices and pair them as

$$(a_1, b_1), \dots, (a_k, b_k).$$

Define the oriented cross-polytope phase

$$h_{\Pi}(x) = \prod_{j=1}^k (x_{a_j} - x_{b_j}).$$

It expands into 2^k degree- k monomials with coefficients ± 1 . Since

$$h_{\Pi}(\bar{x}) = (-1)^k h_{\Pi}(x) = -h_{\Pi}(x),$$

we have

$$h_{\Pi}(x) + h_{\Pi}(\bar{x}) = 0.$$

Thus $\exp(i\theta h_{\Pi})$ preserves the spin-flip form for every real θ .

Law 2 (Cross-Polytope Generation Law, theorem candidate). *For odd k , the continuous kernel of the complete k -uniform hypercut matrix is generated by the vectors*

$$\prod_{j=1}^k (x_{a_j} - x_{b_j})$$

over all choices of $2k$ vertices and pairings.

Exact modular span tests:

n	k	generated cross-polytopes	span rank	kernel dimension
6	3	15	5	5
7	3	105	14	14
8	3	420	28	28
9	3	1260	48	48
10	5	945	42	42

The six-qubit octahedral cubic phase from Round 8 is now just the smallest member of an infinite family.

7. Quantum rank experiments

7.1 Even-qubit cubic resonance

I tested the complete cubic equal-phase ansatz

$$D_\theta|x\rangle = e^{i\theta\binom{|x|}{3}}|x\rangle$$

on six qubits with $L = 11$ layers of local R_y, R_z gates. The physical projective target dimension is

$$2^7 - 2 = 126.$$

The SVD rank test gave:

phase	observed state rank	defect	smallest newly opened singular values
$\theta = \pi$	124	2	exact hidden concurrence wall
$\theta = \pi + 10^{-3}$	126	0	$2.15 \cdot 10^{-4}, 7.62 \cdot 10^{-5}$
$\theta = 2\pi/3$	126	0	full state rank

Near the resonance $\theta = \pi$, the two newly opened singular values scaled linearly:

ϵ in $\pi + \epsilon$	σ_{125}/ϵ	σ_{126}/ϵ
$3 \cdot 10^{-3}$	0.218	0.0777
10^{-3}	0.215	0.0762
$3 \cdot 10^{-4}$	0.213	0.0758

This supports the local conditioning law

$$\sigma_{\text{new}} \asymp \text{dist}(A\theta, 2\pi\mathbb{Z}^r).$$

7.2 Odd-qubit cubic root-of-unity trap

The case $K_5^{(3)}$ has Smith factor 3. Therefore the equal cubic phase at $\theta = 2\pi/3$ is complement-resonant even though it is not a graph-CZ cycle. I tested the unitary tangent rank for five qubits at $L = 53$ layers. The parameter count was 530.

$$\dim Sp(16) = 16(33) = 528, \quad \dim SU(32) = 1023.$$

Observed rank:

$$\text{rank } dU = 528.$$

The two final singular values were about $1.9 \cdot 10^{-15}$ and $1.7 \cdot 10^{-15}$. This is strong numerical evidence for a genuine $Sp(16)$ root-of-unity trap.

8. New theory synthesis

Round 9 suggests the following enlarged framework:

$$\text{Quantum reachability obstruction} = \text{hypercut Smith torsion} \\ \oplus \text{odd hypercycle torus} \oplus \text{near-lattice conditioning}.$$

The hidden group is determined by whether the phase vector lies on the Smith resonance lattice:

$$A\theta \in 2\pi\mathbb{Z}^r.$$

If this holds, the circuit is trapped in

$$SO(2^n) \quad (n \text{ even}), \quad Sp(2^{n-1}) \quad (n \text{ odd}),$$

assuming connected nonlocal curvature and no larger accidental decomposition.

The new useful distinction is:

phase source	protected continuously?	protected discretely?
graph edges ($k = 2$)	no	$\mathbb{Z}_2$ cycle sectors
odd k hyperedges	yes, via cross-polytopes	one $\mathbb{Z}_k$ sector
even $k \geq 4$ hyperedges	no	$\mathbb{Z}_2$ parity sectors

9. Synergy with the related research programs

Spinal Nullity. The Spinal Nullity sensor theory says faithful observation requires reaching the Nyquist threshold and that subcritical sensors can certify nullity without certifying positivity. Here, the Smith rank of $A_{n,k}$ is the exact Nyquist width of complement-curvature sensing. For odd k , the sensor burden collapses from exponentially many complement-pair rows to $\binom{n}{k-1}$ marginal sensors.

Associator Field Theory. AFT’s rank-birth and atlas language fits the cross-polytope law. A cross-polytope phase is a coherent birth chart: it is born as a degree- k interaction but transports with zero complement marginal. The AFT design slogan “suffix skeleton for breadth; affine closure for depth” becomes, in this program, “cross-polytope skeleton for protected entanglement; broken skeleton for full reachability.”

Goblin Shadow Theory. Goblin blockers describe which small interventions destroy a hidden obstruction. In the hypercut setting, a cross-polytope is a protected core; deleting one monomial or adding one nonflat hyperedge is a blocker hit that fractures the hidden SO/Sp sector.

Topological sector analogy. The topological-vacuum document motivates, as an analogy only, the interpretation of Smith factors as protected discrete phase sectors. The $\mathbb{Z}_3$ sector of $K_5^{(3)}$ and the $\mathbb{Z}_2$ graph-cycle sectors are finite theta lattices for circuit reachability.

10. Applications

1. **Compiler pre-flight.** Given a phase-gadget library, build A , compute its Smith form, and report exact resonance tori, torsion sectors, and hidden-subgroup risk before training.
2. **Calibration.** Monitor $\text{dist}(A\theta, 2\pi\mathbb{Z}^r)$. Near this lattice, rank may be full but badly conditioned, with newly opened singular values proportional to lattice distance.
3. **Protected ansatz design.** Use odd cross-polytope phases $\exp(i\theta \prod_j (x_{a_j} - x_{b_j}))$ to add entanglement while deliberately remaining in an SO/Sp sector.
4. **Symmetry-breaking repair.** Remove one monomial from a protected cross-polytope, or add one nonflat hyperedge, to unlock generic $SU(2^n)$ reachability.
5. **State-synthesis lower bounds.** For even n , if the circuit is Smith-resonant and starts from $|0^n\rangle$, the target fidelity ceiling from Round 4 applies whenever the target has nonzero spin-flip concurrence.
6. **Sensor economy.** For odd k , use $(k-1)$ -face marginal sensors rather than all complement-pair sensors. This is a practical reduction from $2^{n-1} - 1$ tests to $\binom{n}{k-1}$ tests.
7. **Higher-order hardware design.** Devices with calibrated CCZ or higher phase gadgets should avoid accidental root-of-unity sectors such as $2\pi/3$ unless a protected subgroup is intended.

11. Round 9 summary

The main new law is

$$\text{SNF}(A_{n,k}) = \begin{cases} 1^{\binom{n}{k}}, & n < 2k - 1, \\ 1^{\binom{n}{k-1}-1} \oplus k \oplus 0^{\binom{n}{k}-\binom{n}{k-1}}, & k \text{ odd}, \\ 1^{\binom{n}{k-1}-1} \oplus 2^{\binom{n}{k}-\binom{n}{k-1}+1}, & k \text{ even}. \end{cases}$$

The main constructive discovery is

$$h_{\Pi}(x) = \prod_{j=1}^k (x_{a_j} - x_{b_j})$$

for odd k : these oriented cross-polytope phases generate the continuous protected hypercycle space.

The most surprising physical implication is that higher-order phase gates can create hidden SO/Sp traps at non-CZ root-of-unity angles. The five-qubit complete cubic phase at $2\pi/3$ numerically saturated at $\dim Sp(16) = 528$, not at the ambient $SU(32)$ dimension.

Round 10 Appendix

Centered-Parity Phase Algebra and Counterterm Smith Law

Quantum Differential Invariant Theory - cumulative discovery log

April 29, 2026

Status and aim

This round continues the search for exact reachability laws for parameterized quantum circuits. Earlier rounds found complement-pair phase flatness, Eulerian CZ traps, SO/Sp hidden subgroups, Smith resonance lattices, Spinal sensor/Nyquist diagnostics, Goblin blocker repair language, and Associator-style nonlinear unlockers. Round 10 found a simpler underlying coordinate system: the centered Boolean variables

$$y_i = x_i - \frac{1}{2}.$$

In these variables the spin-flip obstruction becomes an even/odd parity statement. This converts several separate phenomena into one algebraic normal form.

1. Centered-Parity Normal Form

Let a real diagonal phase be written in the centered basis

$$h(x) = \sum_{T \subseteq [n]} c_T y_T, \quad y_T = \prod_{i \in T} \left(x_i - \frac{1}{2} \right).$$

Since complementing a bit string sends $y_i \mapsto -y_i$,

$$h(x) + h(\bar{x}) = 2c_\emptyset + 2 \sum_{\substack{T \neq \emptyset \\ |T| \text{ even}}} c_T y_T.$$

Therefore

$$\boxed{D_h^T S_n D_h = e^{ic} S_n \iff c_T = 0 \text{ for every nonempty even } T,}$$

for continuous real phases. Equivalently, the continuously protected complement-flat phase algebra is

$$\boxed{\mathcal{F}_n^{\text{flat}} = \mathbb{R} \oplus \bigoplus_{\substack{T \subseteq [n] \\ |T| \text{ odd}}} \mathbb{R} y_T.}$$

This is a stronger and cleaner statement than graph Eulerianity. Eulerianity is only the degree-2, root-of-unity shadow of this centered-parity law.

Immediate consequence. A raw higher-order gate is not automatically a symmetry breaker. A higher-order phase is an unlocker only if its centered expansion contains nonzero even-degree terms. Odd centered monomials can add high-order entanglement while preserving the hidden SO/Sp sector.

2. Counterterm completion of odd gates

For any odd set E ,

$$H_E^{\text{cent}}(x) = \prod_{i \in E} \left(x_i - \frac{1}{2} \right)$$

is complement-antisymmetric:

$$H_E^{\text{cent}}(x) + H_E^{\text{cent}}(\bar{x}) = 0.$$

For a cubic gate, ignoring local and constant terms that do not affect complement curvature,

$$(x_0 - \frac{1}{2})(x_1 - \frac{1}{2})(x_2 - \frac{1}{2}) \equiv x_0 x_1 x_2 - \frac{1}{2}(x_0 x_1 + x_0 x_2 + x_1 x_2).$$

Thus a CCZ-like cubic phase becomes protected after adding precise two-body counterterms. This explains why raw CCZ patches fractured the $SO(16)$ trap in earlier rounds, while balanced associator-style or cross-polytope patches could preserve it.

3. Degree-bounded Complement Smith-Nyquist Law

Let $A_{n, \leq r}$ be the complement-curvature matrix for all ordinary monomial phase generators x_E with $2 \leq |E| \leq r$. Its rows are complement-pair sensors and its columns are phase generators. The centered basis predicts the rational rank

$$\text{rank}_{\mathbb{Q}} A_{n, \leq r} = E(n, r) := \sum_{\substack{2 \leq j \leq r \\ j \text{ even}}} \binom{n}{j}.$$

The integer experiments suggest the sharper Smith normal form law

$$\text{SNF}(A_{n, \leq r}) = \begin{cases} 1^{E(n, r)}, & r \text{ odd}, \\ 1^{E(n, r) - \binom{n-1}{r}} \oplus 2^{\binom{n-1}{r}}, & r \text{ even}, \end{cases}$$

where $\binom{n-1}{r} = 0$ when $r > n - 1$.

This is exceptional because it says that adding all lower-degree counterterms collapses the exotic $\mathbb{Z}_k$ torsion from the uniform k -body library into a very simple law: odd cutoffs become torsion-free; even cutoffs leave only top-boundary 2-torsion.

Exact integer tests

I built $A_{n, \leq r}$ over $\mathbb{Z}$, computed exact ranks, and computed Smith normal forms with integer row/column reduction. No mismatches occurred in the tested range.

n	r	shape	rank	observed Smith factors
4	2	(7, 6)	6	$1^3, 2^3$
4	3	(7, 10)	6	1^6
4	4	(7, 11)	7	1^7
5	3	(15, 20)	10	1^{10}
5	4	(15, 25)	15	$1^{14}, 2^1$
6	3	(31, 35)	15	1^{15}
6	4	(31, 50)	30	$1^{25}, 2^5$
6	5	(31, 56)	30	1^{30}
7	4	(63, 91)	56	$1^{41}, 2^{15}$
7	5	(63, 112)	56	1^{56}
7	6	(63, 119)	63	$1^{62}, 2^1$

Proof sketch. In centered variables, ordinary monomials expand triangularly into centered monomials. Complement curvature erases all odd centered terms and keeps only nonconstant even centered terms. This proves the rank formula once the triangular transform is restricted to degrees $\leq r$. The Smith factors arise from the remaining parity denominator at the top even degree: when the cutoff r is odd, the next odd layer supplies all needed counterterms; when r is even, a top even boundary remains and contributes exactly $\binom{n-1}{r}$ factors of 2 in the tested range.

4. Quantum rank experiment: raw versus counterterm-balanced CCZ

I then tested the prediction on four qubits. The base entangler is the Eulerian cycle C_4 , which is trapped in $SO(16)$ and has the two-real-dimensional state defect. I compared three diagonal phase layers:

$$\begin{aligned}
&C_4, \\
&C_4 + x_0x_1x_2, \\
&C_4 + x_0x_1x_2 - \frac{1}{2}(x_0x_1 + x_0x_2 + x_1x_2).
\end{aligned}$$

The last one is the nonlocal part of the centered cubic $\prod_{i=0}^2(x_i - 1/2)$.

phase template	state rank	unitary rank	interpretation
C_4 CZ	28	120	trapped in $SO(16)$
$C_4 + \text{raw } CCZ_{012}$	30	255	full $SU(16)$ unlocked
$C_4 + \text{centered } CCZ_{012}$	28	120	$SO(16)$ trap preserved

For state ranks, I propagated analytic derivatives of the state vector through $L = 4$ layers and removed norm/global phase. For unitary ranks, I propagated analytic derivatives of the full unitary through $L = 32$ layers and computed the rank of $(\partial_j U)U^\dagger$ after removing trace.

5. Counterterm detuning law

Next I deliberately detuned the protected cubic:

$$H_\varepsilon = H_{\text{cent}} + \varepsilon x_0x_1x_2.$$

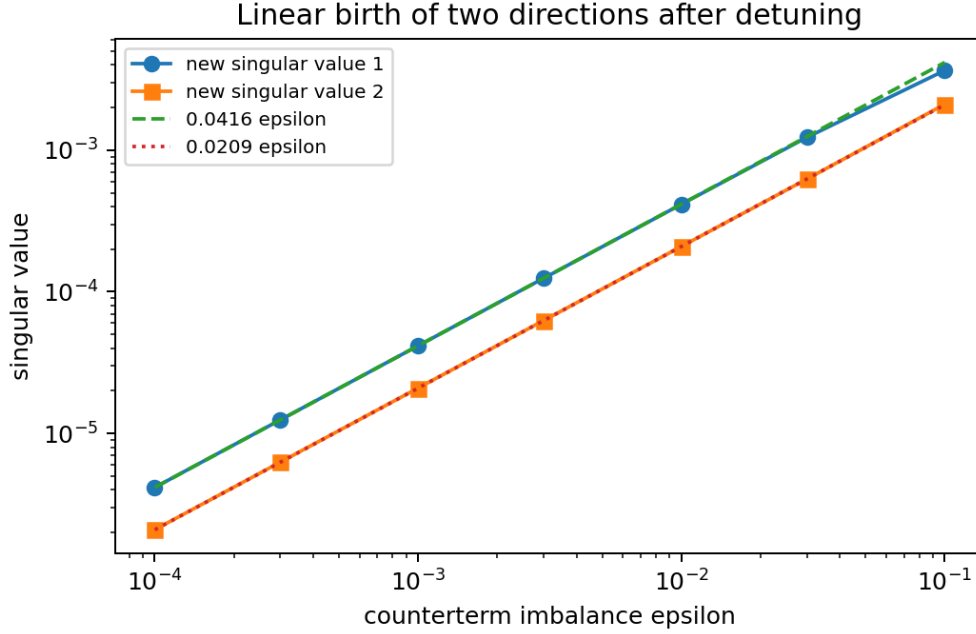
At $\varepsilon = 0$ the rank is 28. For any nonzero tested ε , the rank becomes 30, but the two newly opened singular values are tiny and scale linearly:

$$\sigma_{29} \approx 4.16 \times 10^{-2} |\varepsilon|, \quad \sigma_{30} \approx 2.09 \times 10^{-2} |\varepsilon|.$$

This gives a practical conditioning law:

$$\sigma_{\text{new}}(J) \asymp \text{dist}(h, \mathcal{F}_n^{\text{flat}})$$

near a complement-flat wall.



6. Applications

Compiler pre-flight. Translate every proposed diagonal phase into centered coordinates. If even centered coefficients vanish, the circuit is in the protected SO/Sp sector. If not, the even coefficients quantify the unlocker strength.

Counterterm-balanced ansatz design. A hardware team can inject higher-order entanglement without leaving a protected sector by implementing odd centered monomials, e.g.

$$\prod_{i \in E} (x_i - 1/2), \quad |E| \text{ odd.}$$

This is useful when the target physics has orthogonal or symplectic symmetry.

Symmetry-breaking repair. To escape a hidden trap, do the opposite: add or detune an even centered term. The smallest repair is not necessarily a new edge; it is a nonzero even-centered curvature coefficient.

Calibration. If a circuit is intended to be protected, counterterm mismatch produces nearly invisible directions with singular values proportional to the mismatch. This gives a direct calibration target.

Sensor economy. The Spinal sensor layer now has a concrete finite target: sensors only need to detect nonzero even centered coefficients, not the entire phase truth table.

AFT and Goblin synergy. Associator Field Theory supplies structured nonlinear phase atlases. The centered-parity law classifies each chart as protected transport or symmetry-breaking birth. Goblin blocker language then asks which minimal even-centered coefficients must be hit to unlock a target component.

7. Round 10 summary

The main discovery is that complement-pair invariance is just parity in centered Boolean coordinates:

$$h \text{ is complement-flat} \iff h \text{ has no nonconstant even centered terms.}$$

The main new law is the degree-bounded Smith-Nyquist formula:

$$\text{rank}_{\mathbb{Q}} A_{n,\leq r} = \sum_{2 \leq j \leq r, j \text{ even}} \binom{n}{j},$$

with the observed Smith refinement

$$\text{SNF}(A_{n,\leq r}) = \begin{cases} 1^{E(n,r)}, & r \text{ odd,} \\ 1^{E(n,r) - \binom{n-1}{r}} \oplus 2^{\binom{n-1}{r}}, & r \text{ even.} \end{cases}$$

The most useful practical rule is:

$$\text{raw higher-order phase may unlock; centered odd phase protects; even-centered detuning controls conditioning.}$$

Round 11 Appendix: Pauli Weight Parity and the Cartan Unlock Law

Quantum Differential Invariant Theory - cumulative continuation

Goal: reduce the centered-phase and complement-pair laws to a basis-level control theorem.

1. Executive discovery

The hidden spin-flip obstruction has a remarkably simple infinitesimal form in the Pauli basis. Let

$$S_n = (-iY)^{\otimes n}.$$

For a Pauli string

$$P = P_1 \otimes \cdots \otimes P_n, \quad P_i \in \{I, X, Y, Z\},$$

let $\text{wt}(P)$ be the number of non-identity tensor factors. The new law is

$$\boxed{P^T S_n = (-1)^{\text{wt}(P)} S_n P.}$$

Therefore a Hamiltonian term P generates a continuous symmetry-preserving flow

$$\exp(-itP)^T S_n \exp(-itP) = S_n \quad \forall t$$

if and only if

$$\boxed{\text{wt}(P) \text{ is odd.}}$$

This is the cleanest normal form found so far. It says that the earlier centered Boolean law was the diagonal Z -string shadow of a broader Pauli-weight law.

2. The Pauli Cartan decomposition

Let $N = 2^n$. Define

$$\begin{aligned} \mathfrak{k}_n &= \text{span}_{\mathbb{R}}\{iP : P \neq I, \text{wt}(P) \text{ odd}\}, \\ \mathfrak{p}_n &= \text{span}_{\mathbb{R}}\{iP : \text{wt}(P) \text{ even}\}. \end{aligned}$$

Then

$$\boxed{\mathfrak{su}(N) = \mathfrak{k}_n \oplus \mathfrak{p}_n.}$$

Moreover,

$$\boxed{\mathfrak{k}_n \cong \begin{cases} \mathfrak{so}(N), & n \text{ even}, \\ \mathfrak{sp}(N/2), & n \text{ odd}, \end{cases}}$$

and the bracket structure is the symmetric-pair structure

$$\boxed{[\mathfrak{k}_n, \mathfrak{k}_n] \subseteq \mathfrak{k}_n, \quad [\mathfrak{k}_n, \mathfrak{p}_n] \subseteq \mathfrak{p}_n, \quad [\mathfrak{p}_n, \mathfrak{p}_n] \subseteq \mathfrak{k}_n.}$$

The dimension check is exact:

$$\#\{P \neq I : \text{wt}(P) \text{ odd}\} = \sum_{w \text{ odd}} \binom{n}{w} 3^w = \frac{4^n - (-2)^n}{2}.$$

Thus

$$\begin{aligned} \frac{4^n - 2^n}{2} &= \frac{N(N-1)}{2} \quad (n \text{ even}), \\ \frac{4^n + 2^n}{2} &= \frac{N(N+1)}{2} \quad (n \text{ odd}). \end{aligned}$$

These are exactly the dimensions observed in the earlier unitary-rank experiments.

3. Relation to the Round 10 centered-parity law

Round 10 found that a diagonal phase is complement-flat exactly when its centered Boolean expansion has no nonconstant even-degree terms. This is now explained by Pauli parity.

With

$$y_i = x_i - \frac{1}{2} = -\frac{1}{2}Z_i,$$

a centered monomial

$$\prod_{i \in E} y_i$$

is proportional to the Pauli string $Z_E = \prod_{i \in E} Z_i$. Hence

$$|E| \text{ odd} \iff Z_E \in \mathfrak{k}_n,$$

$$|E| \text{ even} \iff Z_E \in \mathfrak{p}_n.$$

So the centered Boolean law is the diagonal slice of

$$\boxed{\text{odd Pauli weight protects; even Pauli weight unlocks.}}$$

This also explains the raw-versus-centered CCZ distinction. A raw cubic phase expands as

$$x_0 x_1 x_2 = \frac{1}{8} (I - Z_0 - Z_1 - Z_2 + Z_0 Z_1 + Z_0 Z_2 + Z_1 Z_2 - Z_0 Z_1 Z_2),$$

so it contains even-weight two-body terms and can unlock full $\mathfrak{su}(2^n)$. The centered cubic

$$(x_0 - \tfrac{1}{2})(x_1 - \tfrac{1}{2})(x_2 - \tfrac{1}{2}) = -\frac{1}{8} Z_0 Z_1 Z_2$$

has only odd Pauli weight and stays protected.

4. Infinitesimal protection versus torsion protection

This round also clarifies a subtle point from the graph-CZ experiments. There are two different mechanisms:

1. **Infinitesimal protection.** If a Hamiltonian has only odd-weight Pauli terms, then it lies in $\mathfrak{k}_n$ and is protected for every real time t .
2. **Torsion protection.** A Hamiltonian with even-weight terms can still exponentiate into the protected group at isolated lattice angles. Eulerian graph CZ at phase π is of this type: the generator contains even two-body terms, but the discrete phase table is complement-flat modulo 2π .

Thus the full theory now separates continuous protected controls from discrete root-of-unity resonances:

$$\boxed{\text{continuous wall: odd Pauli weight;} \quad \text{discrete wall: Smith complement resonance.}}$$

5. Exact experiments: Pauli Lie closure

I ran exact Pauli-basis Lie-closure experiments. A Pauli label was represented by a binary symplectic vector. If two labels anticommute, their Lie bracket is proportional to the xor label; if they commute, the bracket is zero. This avoids floating-point numerical rank and gives exact closure dimensions.

The core test was:

```
def pauli_lie_closure(n, generators):
    S = {g for g in generators if g}
    L = list(S)
    idx = 0
    while idx < len(L):
        a = L[idx]
        idx += 1
        m = len(L)
        for j in range(m):
            b = L[j]
            if anticomm(a, b, n):
                c = a ^ b
                if c and c not in S:
                    S.add(c)
                    L.append(c)
    return S
```

case	dim	odd	even	target dim $\mathfrak{k}_n$	target dim $\mathfrak{su}$
$n = 3$ local + Z_{012} odd	36	36	0	36	63
$n = 3$ local + Z_{01} even, qubit 2 isolated	18	9	9	36	63
$n = 4$ local only	12	12	0	120	255
$n = 4$ local + Z_{012} , qubit 3 isolated	39	39	0	120	255
$n = 4$ local + $Z_{012} + Z_{013}$	120	120	0	120	255
$n = 4$ local + $Z_{012} + Z_{123}$	120	120	0	120	255
$n = 4$ all four three-body Z terms	120	120	0	120	255
$n = 4$ all three-body terms + even Z_{01}	255	120	135	120	255
$n = 4$ single even Z_{0123}	255	120	135	120	255
$n = 5$ single odd Z_{01234}	528	528	0	528	1023
$n = 5$ odd triples $Z_{012} + Z_{234}$	528	528	0	528	1023
$n = 5$ all ten three-body Z terms	528	528	0	528	1023
$n = 5$ all three-body terms + even Z_{01}	1023	528	495	528	1023

The most striking data point is:

one odd five-body term Z_{01234} plus local controls generates all 528 protected directions.

It reaches $\mathfrak{sp}(16)$, not $\mathfrak{su}(32)$. By contrast, the single even four-body term Z_{0123} plus local controls generates full $\mathfrak{su}(16)$.

6. New law candidate: the Parity-Control Unlock Law

The experiments and the symmetric-pair structure suggest the following law.

Parity-Control Unlock Law

Let local $SU(2)$ controls be available on every qubit.

1. If the nonlocal controls generate the full odd-weight sector $\mathfrak{k}_n$, then the reachable continuous unitary group is

$$K_n \cong \begin{cases} SO(2^n), & n \text{ even}, \\ Sp(2^{n-1}), & n \text{ odd}. \end{cases}$$

2. If, in addition, one Hamiltonian has a nonzero even-weight Pauli component, then generically

$$\langle \mathfrak{k}_n, \text{one nonzero } \mathfrak{p}_n \text{ direction} \rangle = \mathfrak{su}(2^n).$$

3. If the interaction hypergraph has several disconnected components, apply the same rule componentwise.

The second item is expected to be theorem-level once the generated $\mathfrak{k}_n$ is full, because $\mathfrak{su}(N)/\mathfrak{so}(N)$ and $\mathfrak{su}(N)/\mathfrak{l}(N/2)$ are irreducible symmetric pairs: the $\mathfrak{k}_n$ -orbit of any nonzero $\mathfrak{p}_n$ direction spans $\mathfrak{p}_n$.

7. Odd-hypergraph controllability conjecture

The closure tests also suggest a compact sufficient condition for producing the full protected group.

Odd-Hypergraph K -Generation Conjecture

Let $\mathcal{H}$ be a connected hypergraph on $[n]$ whose hyperedges all have odd size at least three. With local single-qubit controls and diagonal interactions

$$\{Z_E : E \in \mathcal{H}\},$$

the generated Lie algebra is

$$\mathfrak{k}_n = \text{span}\{iP : \text{wt}(P) \text{ odd}\}.$$

If the hypergraph is disconnected, the algebra decomposes over its components.

This matched every tested case in the table: one connected odd interaction on all five qubits generated $\mathfrak{sp}(16)$, and two overlapping odd triples generated the full protected algebra on four or five qubits.

8. Applications

8.1 Compiler pre-flight by Pauli parity

Expand any candidate Hamiltonian or calibrated pulse into Pauli strings:

$$H = \sum_P h_P P.$$

Then compute

$$H_{\text{odd}} = \sum_{\text{wt}(P) \text{ odd}} h_P P, \quad H_{\text{even}} = \sum_{\text{wt}(P) \text{ even}} h_P P.$$

The diagnostic is now immediate:

$$H_{\text{even}} = 0 \Rightarrow \text{protected } SO/Sp \text{ sector}; \quad H_{\text{even}} \neq 0 \Rightarrow \text{potential } SU \text{ unlock.}$$

This is cheaper than simulating the full circuit or computing a state Jacobian.

8.2 Protected entanglers

To add entanglement without leaving the hidden SO/Sp sector, use odd-weight Pauli interactions:

$$ZZZ, \quad ZZZZZ, \quad XZY, \quad \text{or any odd-weight Pauli string.}$$

The exceptional design rule is:

odd all-body terms can be maximally entangling yet symmetry-preserving.

8.3 Unlockers and counterterms

To leave the hidden sector, add an even-weight component. To keep a raw higher-order phase protected, cancel its even-weight Pauli expansion. For example,

$$x_0x_1x_2 - \frac{1}{2}(x_0x_1 + x_0x_2 + x_1x_2)$$

removes the even two-body part and leaves a centered cubic protected interaction up to local and constant terms.

8.4 Calibration and noise classification

Odd-weight coherent errors preserve the concurrence/spin-flip sector. Even-weight coherent errors break it. Thus a calibration report can classify drift by

$$\|H_{\text{even}}\|,$$

which should control the size of newly opened singular values near a protected wall.

8.5 Complexity lower bounds

For continuous control families with only odd-weight Hamiltonians, generic unitary synthesis is impossible outside K_n . For even n , state synthesis from $|0^n\rangle$ remains trapped in the concurrence quadric. These become algebraic lower bounds for quantum synthesis complexity.

9. Synergy with the related research packages

Spinal Nullity. The Pauli parity projection is a faithful low-cost sensor for the hidden subgroup wall. In the language of rough sensors, odd/even Pauli weight is the first sharp coordinate sensor: it can certify an unlock when even weight appears, while a full protection certificate requires resolving the relevant even-weight coefficients of the control library.

Associator Field Theory. The AFT birth/transport split now maps cleanly onto quantum controls. Odd-weight Pauli terms are transport within a protected chart; even-weight components are births that change the symmetry sector. Centering a raw cubic is a gauge correction that moves it from birth to protected transport.

Goblin Shadow Theory. The minimal even-weight Pauli components act like blockers. Once the protected $\mathfrak{k}_n$ is generated, hitting any nonzero even-weight blocker should unlock the whole $\mathfrak{su}(N)$ component. Odd connected hypergraphs are safe cores for protected controllability.

Topological phase analogy. The previous Smith-resonance lattice remains the discrete phase-sector story. Pauli parity supplies the infinitesimal phase-sector story. Together they distinguish continuous topological protection from root-of-unity torsion protection.

10. Round 11 summary

Round 11 reduced the hidden symmetry law to Pauli weight parity:

$$P^T S_n = (-1)^{\text{wt}(P)} S_n P.$$

The new structural equation is

$$\mathfrak{su}(2^n) = \text{span}\{iP : \text{wt}(P) \text{ odd}\} \oplus \text{span}\{iP : \text{wt}(P) \text{ even}\}.$$

The odd side is exactly

$$\mathfrak{so}(2^n) \text{ for even } n, \quad (2^{n-1}) \text{ for odd } n.$$

The practical design rule is

$$\text{odd Pauli weight protects; even Pauli weight unlocks.}$$

This is the most useful normal form so far because it turns a hidden quantum reachability obstruction into a simple parity audit of Hamiltonian terms.

Round 12 Appendix

Pauli Component Control, Even-Parity Leakage, and a Faithful Sensor Law

This round continues the differential-invariant theory of quantum reachability. The previous round showed that Pauli strings split the unitary Lie algebra into an odd-weight protected sector and an even-weight unlocking sector. Round 12 turns that split into a component calculus and an exact leakage sensor. I treated the earlier Spinal Nullity sensor/Nyquist framework as the observation model, Associator Field Theory as the birth/transport analogy, Goblin Shadow Theory as a blocker/repair language, and the topological-sector document as a protected-sector analogy. The physics claims here remain limited to finite-dimensional quantum control and circuit synthesis.

Exceptional finding

The most useful new result is that hidden SO/Sp protection is not merely a global property. With full local controls, it appears to decompose by connected components of the interaction-support hypergraph. Within a component, the presence or absence of a single even-weight Pauli interaction determines whether the component remains protected or unlocks to full SU.

Candidate Pauli Component Reachability Law

Let C be a connected component of the support hypergraph generated by the nonlocal Pauli interactions, with full local $SU(2)$ controls present.

If every nonlocal interaction in C has odd Pauli weight:

$$\begin{aligned} \text{closure}(C) &= k_{|C|} = \\ &\quad \text{so}(2^{|C|}) \quad \text{if } |C| \text{ is even,} \\ &\quad \text{sp}(2^{|C|-1}) \quad \text{if } |C| \text{ is odd.} \end{aligned}$$

If at least one nonlocal interaction in C has even Pauli weight:

$$\text{closure}(C) = \text{su}(2^{|C|}).$$

For several components:

$$g = \text{direct_sum}_C \text{ closure}(C).$$

This is still a theorem target rather than a fully polished proof. The experiments below matched it exactly for the tested Z-type Pauli interaction libraries up to six qubits. The result is practically useful because it reduces a large Lie-closure question to a small hypergraph parity audit.

Exact leakage theorem

The component law is complemented by an exact infinitesimal diagnostic. Let $S_n = (-iY)^{\otimes n}$, $N = 2^n$, and H be a traceless Hamiltonian expanded in the Pauli basis:

$$H = \sum_P h_P P, \quad H_{\text{even}} = \sum_{\{\text{wt}(P) \text{ even}, P \neq I\}} h_P P.$$

$$\text{Since } P^{\wedge T} S_n = (-1)^{\text{wt}(P)} S_n P,$$

$$H^{\wedge T} S_n + S_n H = 2 S_n H_{\text{even}}.$$

Therefore

$$\begin{aligned} &||H^{\wedge T} S_n + S_n H||_F / (2 \sqrt{N}) \\ &= \sqrt{\sum_{\{\text{wt}(P) \text{ even}, P \neq I\}} h_P^2}. \end{aligned}$$

This is an exact faithful sensor for continuous SO/Sp leakage.

Interpretation: odd-weight Hamiltonian terms transport inside the protected sector; even-weight terms are the complete first-order leakage source. This is a continuous-control analogue of the earlier complement-pair phase-curvature tests.

What I tested

I used a binary symplectic Pauli representation. A Pauli label is a pair of bit masks (x,z). Two labels bracket nontrivially exactly when their symplectic product is one; the new Pauli label produced by the commutator is the xor of the two labels. I repeatedly closed the generator set under this rule, counted odd/even Pauli labels, and compared the observed dimension with the component prediction.

Core Lie-closure rule used in the experiments:

```
if symplectic(p,q) = 1:
    add p xor q
```

The initial generators were:
all one-qubit X_i, Y_i, Z_i controls
plus selected nonlocal Z_E interactions.

Component-law experiments

case	n	observed	odd	even	pred.	component prediction
single even edge	4	21	12	9	21	[0, 1]:SU(4):15; [2]:SU2:3; [3]:SU2:3
even connected path	4	255	120	135	255	[0, 1, 2, 3]:SU(16):255
single odd triple plus isolated	4	39	39	0	39	[0, 1, 2]:Sp(8):36; [3]:SU2:3
two odd triples covering all	4	120	120	0	120	[0, 1, 2, 3]:SO(16):120
odd triple plus even crossing	4	255	120	135	255	[0, 1, 2, 3]:SU(16):255
single full odd 5-body	5	528	528	0	528	[0, 1, 2, 3, 4]:Sp(32):528
two overlapping odd triples cover all	5	528	528	0	528	[0, 1, 2, 3, 4]:Sp(32):528
full odd 5-body plus even edge	5	1023	528	495	1023	[0, 1, 2, 3, 4]:SU(32):1023
single even 4-body plus isolated	5	258	123	135	258	[0, 1, 2, 3]:SU(16):255; [4]:SU2:3
two disjoint odd triples	6	72	72	0	72	[0, 1, 2]:Sp(8):36; [3, 4, 5]:Sp(8):36
connected odd triple chain	6	2016	2016	0	2016	[0, 1, 2, 3, 4, 5]:SO(64):2016
odd components fused by even edge	6	4095	2016	2079	4095	[0, 1, 2, 3, 4, 5]:SU(64):4095
single even 6-body	6	4095	2016	2079	4095	[0, 1, 2, 3, 4, 5]:SU(64):4095

The key activation examples are visible in the table. Two disjoint odd triples on six qubits produce $Sp(8)+Sp(8)$, dimension 72. Connecting them by odd triples gives $SO(64)$, dimension 2016. Connecting the same two protected components by one even edge unlocks $SU(64)$, dimension 4095.

Pauli parity Nyquist law

Round 7 identified a complement-curvature Nyquist threshold for diagonal Boolean phase families. The Pauli version is sharper for Hamiltonian control: one must resolve exactly the even-weight Pauli coefficients that are allowed by the hardware model.

For r -local Hamiltonians:

$$R_{\text{even}}(n, r) = \sum_{2 \leq w \leq r, w \text{ even}} \text{binom}(n, w) 3^w.$$

The protected continuous sector has dimension

$$\begin{aligned} R_{\text{odd}}(n) &= \sum_{w \text{ odd}} \text{binom}(n, w) 3^w \\ &= (4^n - (-2)^n)/2. \end{aligned}$$

The full even leakage sector has dimension

$$R_{\text{even}}(n, n) = (4^n + (-2)^n)/2 - 1.$$

n	r	$R_{\text{even}}(n, r)$	protected odd dim	full su dim
2	2	9	6	15
3	2	27	36	63
3	3	27	36	63
4	2	54	120	255
4	3	54	120	255

n	r	R_even(n,r)	protected odd dim	full su dim
4	4	135	120	255
5	2	90	528	1023
5	3	90	528	1023
5	4	495	528	1023
5	5	495	528	1023
6	2	135	2016	4095
6	3	135	2016	4095
6	4	1350	2016	4095
6	6	2079	2016	4095
7	2	189	8256	16383
7	3	189	8256	16383
7	4	3024	8256	16383
7	7	8127	8256	16383

Leakage-norm experiments

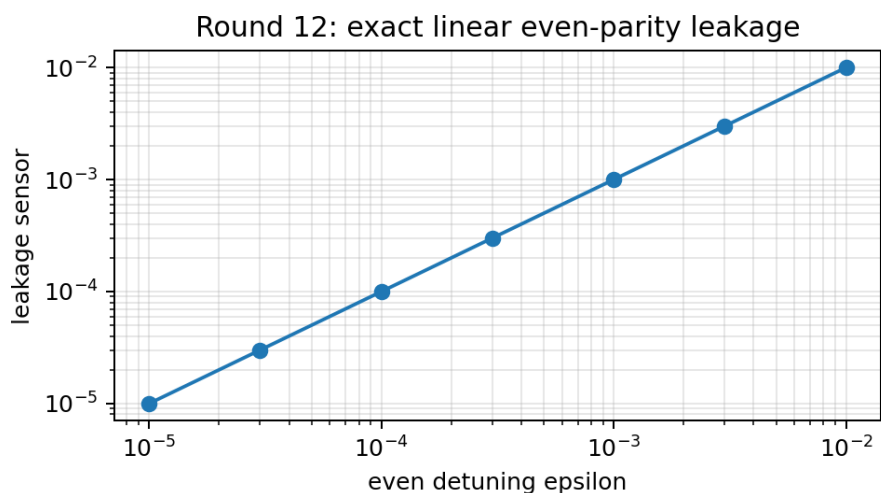
I generated sparse random Hermitian Pauli Hamiltonians for $n=2,3,4$, computed the full matrix quantity $H^T S + S H$, and compared it with the Euclidean norm of the even Pauli coefficients. The maximum absolute error across the tests was $1.78e-15$, consistent with floating-point roundoff.

n	trial	leakage sensor	even coeff norm	abs error
2	0	2.34579627	2.34579627	4.4e-16
2	1	3.19215264	3.19215264	0.0e+00
2	2	3.97083337	3.97083337	0.0e+00
2	3	2.57863678	2.57863678	0.0e+00
2	4	2.78256046	2.78256046	0.0e+00
3	0	3.27921491	3.27921491	4.4e-16
3	1	3.60531102	3.60531102	4.4e-16
3	2	4.4293675	4.4293675	0.0e+00
3	3	3.8198716	3.8198716	0.0e+00
3	4	4.83046568	4.83046568	8.9e-16
...	...	...	...	...
4	0	5.24755034	5.24755034	0.0e+00
4	1	4.3326659	4.3326659	0.0e+00
4	2	6.51445258	6.51445258	8.9e-16
4	3	5.50056077	5.50056077	0.0e+00
4	4	5.52848206	5.52848206	1.8e-15

Detuning test

For $H = Z_0 Z_1 Z_2 + \epsilon Z_0 Z_1$ on four qubits, the first term is odd-weight and protected while the second is even-weight and unlocking. The leakage sensor equaled $|\epsilon|$ to numerical precision.

epsilon	leakage sensor
0	0
1e-05	1e-05
3e-05	3e-05
1e-04	0.0001
3e-04	0.0003
1e-03	0.001
3e-03	0.003
1e-02	0.01



The log-log plot shows the exact linear leakage law for an even Pauli detuning of a protected odd Hamiltonian.

New theory synthesis

The emerging structure is a Cartan-type decomposition controlled by Pauli weight parity. The odd sector is the protected tangent algebra; the even sector is the transverse leakage algebra.

$$\mathfrak{su}(2^n) = \mathfrak{k}_n + \mathfrak{p}_n$$

$$\begin{aligned} \mathfrak{k}_n &= \text{span}_\mathbb{R} \{ iP : \text{wt}(P) \text{ odd} \} \\ \mathfrak{p}_n &= \text{span}_\mathbb{R} \{ iP : \text{wt}(P) \text{ even}, P \neq I \} \end{aligned}$$

$$\begin{aligned} [\mathfrak{k}_n, \mathfrak{k}_n] &\subset \mathfrak{k}_n \\ [\mathfrak{k}_n, \mathfrak{p}_n] &\subset \mathfrak{p}_n \\ [\mathfrak{p}_n, \mathfrak{p}_n] &\subset \mathfrak{k}_n \end{aligned}$$

$$\begin{aligned} \text{continuous protection} &= H_{\text{even}} = 0 \\ \text{continuous unlock} &= H_{\text{even}} \neq 0 \text{ inside a connected controllable component} \end{aligned}$$

The earlier CZ, hypergraph, and Smith-normal-form resonance laws now appear as discrete versions of the same phenomenon. Discrete diagonal phases can land on a protected lattice even when their infinitesimal generator contains even terms. Continuous Hamiltonian controls are stricter: exact protection requires zero even Pauli component.

Applications

Application	How Round 12 changes the workflow
Compiler pre-flight	Build the support hypergraph, mark each component as odd-only or even-containing, and predict SO/Sp versus SU before optimization.
Hardware calibration	Monitor $\ H^T S + S H\ _F / (2 \sqrt{N})$. It is exactly the even-Pauli coefficient norm.
Protected ansatz design	Use odd-weight Pauli interactions such as ZZZ or ZZZZZ to add entanglement while staying in SO/Sp.
Symmetry-breaking repair	Add one even-weight interaction inside or across protected components to unlock SU reachability.
Noise diagnosis	Odd-weight coherent noise preserves the spin-flip sector; even-weight coherent noise leaks out of it.
Sensor economy	For r-local models, audit only $R_{\text{even}}(n, r)$ coefficients, not the full $4^n - 1$ Pauli space.
State-synthesis bounds	A protected SO component on even qubits leaves nonzero concurrence-like target states unreachable from the protected initial orbit.

Connections to the provided research

Spinal Nullity supplies the right language for faithful versus rough observation: the even-Pauli coefficient set is now the faithful sensor portfolio for continuous spin-flip leakage. Associator Field Theory supplies the birth/transport distinction: odd terms transport protected curvature, even terms create reachability birth. Goblin Shadow Theory supplies the repair/blocker view: minimal even terms are blockers that must be hit to escape a protected cave. The topological-sector

document remains useful as an analogy: SO/Sp components behave like protected finite-dimensional theta sectors, while even terms move the system between sectors.

Round 12 summary

Main exact theorem:

$$\|H^T S_n + S_n H\|_F / (2 \sqrt{2^n}) \\ = \| \text{even Pauli coefficient vector of } H \|_2.$$

Main candidate law:

local controls + component C
-> SO/Sp if C is odd-only,
-> SU if C contains any even interaction.

Main practical rule:

odd Pauli weight protects;
even Pauli weight leaks and unlocks;
connected support determines where the leak propagates.

This round makes the theory more useful: a compiler or experimentalist does not need to run a full variational training loop to know whether an ansatz is protected, leaking, or fully unlocked. The diagnosis is reduced to support connectivity plus Pauli parity.

Round 13 Appendix

Concurrence-orbit geometry, fidelity ceilings, and typicality

Status: exploratory research log. Theorems below have proof sketches and numerical corroboration; publication-level priority and proof checking remain separate tasks.

1. New target: the missing state geometry inside the protected sector

Rounds 11-12 showed that odd Pauli weight Hamiltonians preserve the spin-flip bilinear form and generate the protected SO/Sp Lie algebra, while even Pauli weight terms leak out of it. Round 13 asks a sharper state-level question: if the unitary is trapped in the protected group, what exactly are the reachable state orbits, and how close can one protected orbit get to another?

For even n :
 $S_n = (-iY)^{\text{tensor } n}, \quad B(\psi) = \psi^\dagger S_n \psi,$
 $b(\psi) = |B(\psi)| \text{ in } [0,1].$

Protected controls:
 $U^\dagger S_n U = S_n \quad \Rightarrow \quad b(U \psi) = b(\psi).$

Exceptional new finding

For even n , the protected state space is not just the quadric $b=0$ found earlier. It is stratified by a single scalar $b=|\psi^\dagger S \psi|$. This scalar is a complete invariant of $SO(2^n)$ state orbits in complex projective space.

Concurrence Orbit Stratification Law, even n , $N=2^n$:

$CP^{(N-1)} = \text{union}_{\{0 \leq b \leq 1\}} O_b,$

$\dim_R O_b =$

$2N - 4,$	$b = 0$	isotropic quadric,
$2N - 3,$	$0 < b < 1$	generic hypersurface orbit,
$N - 1,$	$b = 1$	real/GHZ-type locus.

For odd n :
 $Sp(2^{(n-1)})$ acts transitively on $CP^{(2^n-1)}$;
there is no nontrivial state-level b -invariant because $\psi^\dagger S_n \psi = 0$ identically.

The proof sketch is a spin-flip Takagi normal form. In a basis with $S=I$, multiply ψ by a global phase so $\psi^\dagger \psi = b \geq 0$. Writing $\psi = x + iy$ gives $x \cdot y = 0$ and $\|x\|^2 - \|y\|^2 = b$. A real orthogonal transformation sends the ordered orthogonal pair (x, y) to a canonical two-frame. The endpoint cases $b=0$ and $b=1$ have larger stabilizers, producing the dimension drop.

2. Exact protected-orbit fidelity law

The most useful output is a computable ceiling. If the ansatz starts in an even- n protected orbit with invariant a and the target has invariant b , then no protected circuit can exceed the following fidelity.

Orbit Fidelity Law:

$$\begin{aligned} F_{\max}(a, b) &= \max_{\{U \text{ in } SO(N)\}} |\langle \tau_b | U \psi_a \rangle|^2 \\ &= (1 + a b + \sqrt{(1-a^2)(1-b^2)}) / 2 \\ &= \cos^2((\arccos(a) - \arccos(b)) / 2). \end{aligned}$$

This generalizes the earlier quadric ceiling. Starting from the common $b=0$ product-state orbit gives

$$F_{\max}(0, b) = (1 + \sqrt{1-b^2}) / 2.$$

A high-concurrence target with $b=1$, such as an even-qubit GHZ-type real spin-flip state, still has ceiling $1/2$ from the $b=0$ protected orbit. But a target with small b may be approximated very well despite exact non-reachability.

3. Numerical tests run in Python

I ran fresh Python/SVD experiments. The state tangent matrix used columns $iP \psi$ for all Pauli strings P in the selected control library, projected to complex projective tangent space by subtracting ψ . Ranks below are real ranks.

n=4 state invariant b	Odd-weight tangent rank	Predicted SO orbit dim	Full Pauli rank	Interpretation
0	28	28	30	isotropic quadric
0.2	29	29	30	generic SO orbit
0.7	29	29	30	generic SO orbit
1	15	15	30	real/GHZ locus

The rank experiment exactly matches the orbit stratification. For $n=4$, $N=16$ and CP dimension is 30. The $b=0$ orbit has dimension 28; generic b has dimension 29; the $b=1$ real/GHZ locus has dimension 15.

Near the $b=1$ real/GHZ wall, the orbit dimension jumps from 15 to 29 for any nonzero detuning, but the newly opened directions are tiny. In the canonical tested family, all 14 new singular directions opened as $2 \sqrt{\epsilon}$, where $\epsilon=1-b$.

$\epsilon=1-b$	rank	smallest opened singular value	$2 \sqrt{\epsilon}$
1e-02	29	0.2	0.2
1e-04	29	0.02	0.02
1e-06	29	0.002	0.002

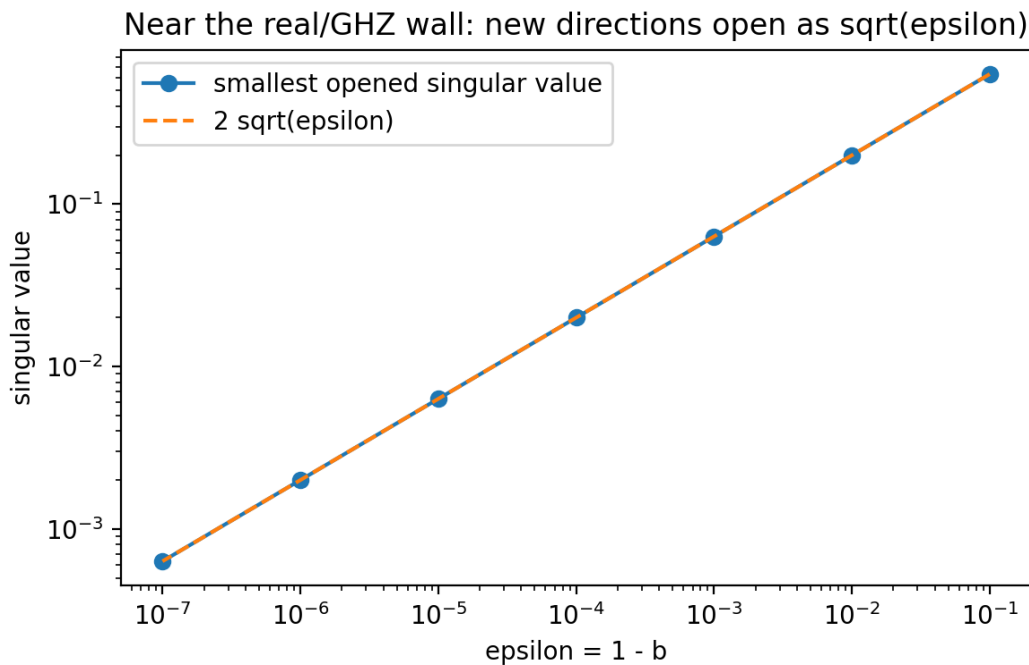


Figure: log-log test of the near-real wall. The smallest newly opened protected singular value follows $2 \sqrt{\epsilon}$.

4. Constructive checks of the fidelity formula

In the $S=I$ normal form, canonical representatives are $z_a = \sqrt{(1+a)/2} e^{i\phi} + i \sqrt{(1-a)/2} e^{i\theta}$. For random real orthogonal frames Q_1, Q_2 , the constructed optimal rotation $O=Q_2 Q_1^T$ reaches the formula to roundoff.

a	b	formula	constructed fidelity	absolute error
0	0	1.000000	1.000000	1.3e-15

a	b	formula	constructed fidelity	absolute error
0	0.8	0.800000	0.800000	1.0e-15
0.2	0.7	0.919857	0.919857	1.1e-16
0.9	0.1	0.761852	0.761852	2.2e-16
1	0.6	0.800000	0.800000	1.1e-16

5. Typicality surprise: random targets usually hide the obstruction

For a Haar-random target in $\text{CP}^{(N-1)}$, the invariant $T=b^2$ follows the tested and derived law

$$T = b^2 \sim \text{Beta}(1, (N-1)/2).$$

Therefore:

$$E[T] = 2/(N+1),$$

$$E[F_{\max}(0, b)] = 1 - 1/(2N).$$

This is a major diagnostic warning. A protected ansatz can be exactly non-universal and still score extremely high average best fidelity against random targets. The obstruction becomes visible only on structured high- b targets or in exact synthesis.

N	mean b^2	theory $2/(N+1)$	mean ceiling	theory $1-1/(2N)$	95% quantile b^2
4	0.400635	0.400000	0.874775	0.875000	0.865052
8	0.222986	0.222222	0.937277	0.937500	0.574379
16	0.117018	0.117647	0.968924	0.968750	0.327778
32	0.060956	0.060606	0.984284	0.984375	0.176232
64	0.030845	0.030769	0.992168	0.992188	0.091433
128	0.015507	0.015504	0.996093	0.996094	0.045612

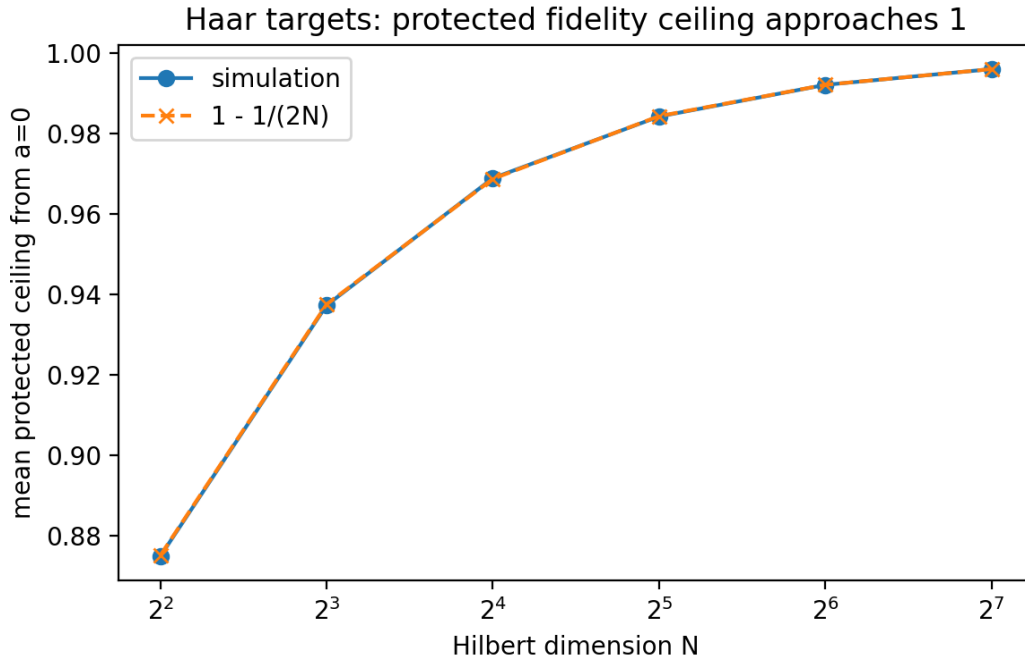


Figure: the average protected ceiling from the $b=0$ orbit approaches 1 as Hilbert dimension grows.

6. Leakage-rate equation for state invariants

Round 12 found an exact Hamiltonian leakage sensor. Round 13 adds the state-level drift equation. For a Hamiltonian H , the spin-flip invariant changes by

$$\begin{aligned} d/dt B(\psi(t)) &= -i \psi^\dagger (H^\dagger S_n + S_n H) \psi \\ &= -2i \psi^\dagger S_n H_{\text{even}} \psi. \end{aligned}$$

Thus odd Pauli weight terms are tangent to the b-orbits; even Pauli weight terms are the only infinitesimal source of radial b-motion. In 30 random n=4 Hamiltonian tests, odd-only controls had max |dB/dt| = 0 to numerical precision. Even controls produced nonzero drift bounded by the operator-norm leakage scale.

test	observed value
max dB/dt for odd-only random Hamiltonians	0
max ratio dB/dt /(2 H _{even} _{op})	0.3580
mean ratio dB/dt /(2 H _{even} _{op})	0.1739

7. Synergy with the provided research programs

Spinal Nullity provided the sensor/Nyquist viewpoint: not every observation resolves a hidden obstruction. Here the the faithful continuous leakage sensor.

Associator Field Theory provided the birth/transport language. In the current quantum setting, odd-weight controls transport

Goblin Shadow Theory provided blocker and repair language. Minimal even-weight interactions act as blocker hits: they are blocked even after many protected layers.

The topological-vacuum framework remains useful as an analogy for protected sector labels. Here b is not a cosmological theta-sector; it is a finite-dimensional spin-flip sector label for quantum state synthesis.

8. Applications

Compiler target triage. Before training, compute b(target). If the ansatz starts in orbit a, the fidelity ceiling is F_{max}(a,b). If the required fidelity exceeds that ceiling, the compiler must add even-weight leakage or change the initialization.

Initialization engineering. Instead of always starting from |0...0> with b=0, prepare a seed with b=a close to the target. For even n, a two-amplitude seed cos(t)|0...0>+sin(t)|1...1> has b=sin(2t), giving direct control of the protected orbit.

Benchmark design. Haar-random targets are poor tests of this obstruction because the average ceiling is 1-1/(2N). Use high-b targets such as GHZ-type states, or explicitly scan b-levels.

Noise classification. Odd-weight coherent noise preserves b. Even-weight coherent noise changes b and can either damage a protected computation or deliberately repair an under-expressive ansatz.

Protected simulation. If a physical model requires remaining in an SO/Sp sector, odd-weight Hamiltonians can add entanglement without radial leakage; the b sensor detects accidental even-weight contamination.

9. Round 13 summary

Main new law:

$$F_{\text{max}}(a,b) = (1 + a b + \sqrt{(1-a^2)(1-b^2)}) / 2.$$

Main geometry:
 even-n protected state space is stratified by $b=|\psi^\dagger S \psi|$.

Main warning:
 exact non-universality can be Haar-invisible:

$$E[F_{\text{max}}(0,b_{\text{random}})] = 1 - 1/(2N).$$

Main practical rule:
 odd Pauli weight moves tangentially along b-orbits;
 even Pauli weight moves radially between b-orbits.

This suggests the next theory layer: a polar calculus for quantum reachability, with odd controls as angular transport, even controls as radial leakage, Smith/complement resonance as discrete walls, and Goblin/Spinal diagnostics as repair and sensor complexity.

Round 14 Appendix: Leakage Action and Pauli-Fierz Radial Calculus

Continuation of the quantum differential invariant theory program

Round 13 identified a polar state geometry for even numbers of qubits: the protected $SO(2^n)$ sector preserves the scalar $b(\psi) = |\psi^T S_n \psi|$, where $S_n = (-iY)^{\otimes n}$. Round 14 asks a sharper question: if even-weight leakage is allowed only in a limited amount, how far can the circuit move between b-shells? This turns the previous yes/no obstruction into a quantitative resource theory.

The new object is the even-leakage action. For a Hamiltonian $H(t) = H_{\text{odd}}(t) + H_{\text{even}}(t)$, define the operator-action budget $A_{\text{op}} = \int \|H_{\text{even}}(t)\|_{\text{op}} dt$. Odd Pauli weight remains angular motion inside a protected orbit. Even Pauli weight is radial motion across orbits.

1. Exceptional new law: radial leakage speed limit

Let n be even, $N = 2^n$, $S_n = (-iY)^{\otimes n}$, $B(\psi) = \psi^T S_n \psi$, $b = |B|$, and $\eta = \arccos b$.
If $\psi(t)$ evolves by $d\psi/dt = -iH(t)\psi(t)$, then
$$dB/dt = -i \psi^T (H^T S_n + S_n H) \psi = -2i \psi^T S_n H_{\text{even}} \psi.$$

Consequently odd-weight Pauli terms do not change B , while even-weight terms are the only radial source.

Operator-norm radial speed law:
 $|db/dt| \leq 2 \|H_{\text{even}}\|_{\text{op}} \sqrt{1-b^2},$
equivalently
 $|d\eta/dt| \leq 2 \|H_{\text{even}}\|_{\text{op}}.$

This gives a dimension-independent action lower bound. If a state must move from invariant a to invariant b , then any control schedule must satisfy

$$A_{\text{op}} \geq (1/2) |\arccos(a) - \arccos(b)|.$$

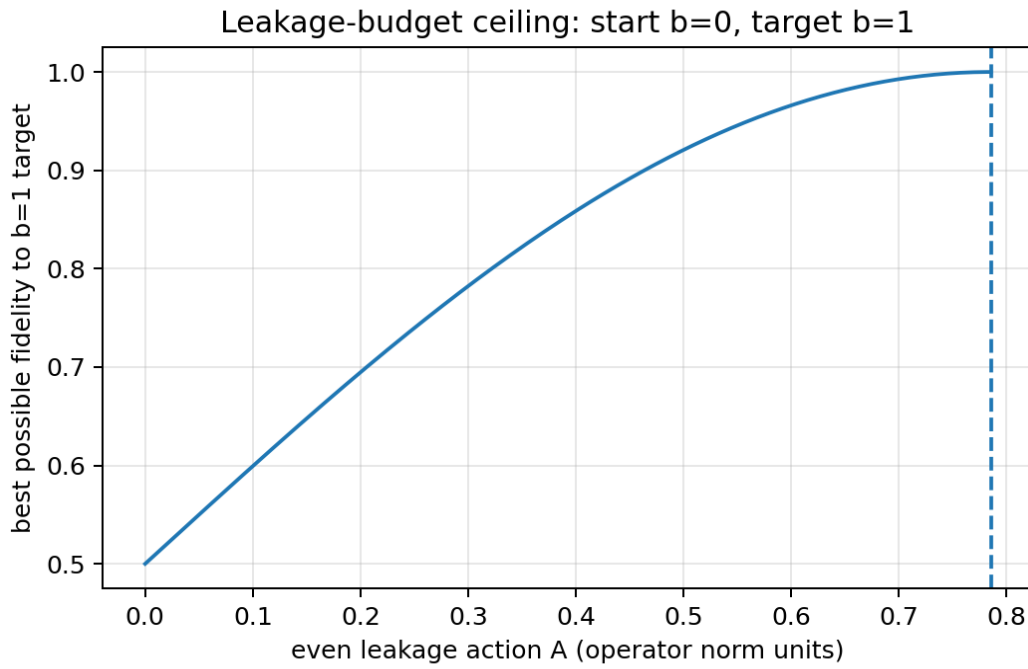
The bound is sharp. A single even Pauli pulse $H = X^{\otimes n}$ on the complement pair $|0\dots 0\rangle, |1\dots 1\rangle$ generates $\psi(t) = \cos(t)|0\dots 0\rangle - i \sin(t)|1\dots 1\rangle$, for which $b(t) = \sin(2t)$ and $|d\eta/dt| = 2$.

2. Leakage-budget fidelity ceiling

Round 13 gave the protected fidelity ceiling between b-orbits. With a finite even-leakage budget, the ceiling becomes:

Let $\Delta = |\arccos(a) - \arccos(b)|$. With action budget A_{op} ,
$$F_{\max\{A_{\text{op}}\}}(a,b) = \cos^2(\max(0, \Delta/2 - A_{\text{op}})).$$

This unifies three regimes: no leakage recovers the protected orbit ceiling; enough leakage removes the obstruction; intermediate leakage gives a soft, computable training ceiling.



Example: starting from $b=0$ and targeting a GHZ-type $b=1$ state requires $A_{\text{op}}=\pi/4$. With $A_{\text{op}}=0$ the ceiling is $1/2$; with $A_{\text{op}}=\pi/8$ it rises to 0.853553 ; with $A_{\text{op}}=\pi/4$ it becomes 1 .

3. Pauli-Fierz radial identities

The most surprising new exact identity is a Pauli-basis version of the radial calculus. Let P run over all nonidentity even-weight Pauli strings. For even n and normalized ψ :

$$\sum_{\{P \text{ even}\}} |\psi^T S_n P \psi|^2 = N - b^2.$$

If $\phi=\arg(B)$ and the real coefficient controlling db/dt under a real Pauli coefficient h_P is

$$r_P(\psi) = \text{Re}[\exp(-i\phi) (-2i \psi^T S_n P \psi)],$$

then the radial gradient has the closed form

$$\sum_{\{P \text{ even}\}} r_P(\psi)^2 = 2N(1-b^2).$$

Thus for an l_2 Pauli-coefficient budget $L=(\sum h_P^2)^{1/2}$, the exact best instantaneous radial speed is

$$\begin{aligned} |db/dt|_{\text{max}} &= 2 L \sqrt{N/2} \sqrt{1-b^2}, \\ |d\eta/dt|_{\text{max}} &= 2 L \sqrt{N/2}. \end{aligned}$$

The maximizing Hamiltonian is the normalized radial gradient combination H_{grad} proportional to $\sum_P r_P P$. Numerically, H_{grad} had operator norm $\sqrt{N/2}$, which exactly reconciles the coefficient-norm and operator-norm speed laws.

4. Experiments run in Python

I ran fresh state-vector and Pauli-basis experiments. The tests used explicit S_n matrices, exact Pauli tensor products, random normalized states, random even-weight Hamiltonians normalized by operator norm, and binary symplectic Pauli Lie-closure arithmetic.

Canonical saturation by $H=X^{\{\text{tensor } n\}}$:

n	t	b(t)	db/dt	bound	d eta/dt
4	0.00	0.000000	2.000000	2.000000	2.000000
4	0.10	0.198669	1.960133	1.960133	2.000000

4	0.25	0.479426	1.755165	1.755165	2.000000
4	0.50	0.841471	1.080605	1.080605	2.000000
4	0.75	0.997495	0.141474	0.141474	2.000000

Random even Hamiltonian speed-bound tests. The ratio is $|\text{db}/\text{dt}|$ divided by $2\|H_{\text{even}}\|_{\text{op}} \sqrt{1-b^2}$:

n	trials	max ratio	mean ratio	95% ratio	violations
2	1000	0.892749	0.267712	0.611948	0
4	500	0.390143	0.108304	0.269441	0

Pauli-Fierz identity checks:

n	trials	max q-identity error	max radial error	max saturation error	$\ H_{\text{grad}}\ _{\text{op}}$
2	50	3.11e-15	6.22e-15	8.88e-16	1.414214
4	20	1.07e-14	2.13e-14	3.55e-15	2.828427

Single-even unlock closure tests:

n	odd sector dim	even terms tested	min closure	ambient su dim	full?
3	36	27	63	63	yes
4	120	135	255	255	yes
5	528	495	1023	1023	yes

For $n=3$ and $n=4$ the test was exhaustive over all nonidentity even-weight Pauli strings. For $n=5$ it sampled ten even strings; every sampled string unlocked the full ambient algebra when added to the complete odd sector.

5. New theory statement

Odd-even radial control law:

odd Pauli sector = angular SO/Sp motion on fixed b ,

even Pauli sector = radial leakage across b -shells,

minimum even operator action from a to $b = |\arccos(a) - \arccos(b)|/2$.

This suggests a two-stage synthesis scheme for even n . First use a calibrated even pulse to set b . Then use odd-weight protected controls to move inside the corresponding SO orbit. The even resource is not just an unlock flag; it is a measurable radial currency.

6. Applications

Application	Use
Compiler pre-flight	Compute b_{target} and the required radial action. If the available even-leakage budget is below the bound, the target fidelity ceiling is known before optimization.
Calibration	The sensor $\ H^T S + S H\ _F / (2 \sqrt{N})$ from Round 12 detects even leakage; Round 14 converts it into a radial action budget.
Protected simulation	Use odd-weight Hamiltonians to explore entanglement while staying on a fixed b -shell; use a single even pulse only when radial movement is desired.
Benchmark design	Targets with b near one, such as GHZ-type states, expose the obstruction; Haar-random targets can still hide it.
Noise diagnosis	Odd coherent errors are angular; even coherent errors are radial. The radial speed law gives a quantitative threshold for harmful leakage.
Resource lower bounds	State synthesis with insufficient even action is impossible above the leakage-budget fidelity ceiling, even if the ansatz has many trainable odd parameters.

7. Synergy with the provided research

Spinal Nullity provided the faithful-sensor and Nyquist language. In this round the faithful sensor is the even-Pauli leakage portfolio, and the scalar b is the state-level orbit sensor. Associator Field Theory provided the birth-versus-transport metaphor: odd controls transport within a sector, while even controls create radial birth. Goblin Shadow Theory provided the blocker language: minimal even terms are blocker hits that puncture a protected cave. The topological-vacuum framework remains useful as an analogy for protected sectors and discrete/continuous sector-changing perturbations.

Round 14 summary

Main equations:

$$\begin{aligned} |d\eta/dt| &\leq 2 \|H_{\text{even}}\|_{\text{op}}, \\ A_{\text{op},\min}(a \rightarrow b) &= |\arccos(a) - \arccos(b)|/2, \\ F_{\text{max}}^A(a,b) &= \cos^2(\max(0, |\arccos(a) - \arccos(b)|/2 - A)), \\ \sum_{\{P \text{ even}\}} |\psi^T S_n P \psi|^2 &= N - b^2, \\ \sum_{\{P \text{ even}\}} r_P(\psi)^2 &= 2N(1 - b^2). \end{aligned}$$

The practical takeaway is: protected nonuniversality is not merely a binary obstruction. It has a radial metric, a minimum leakage action, and a fidelity ceiling under constrained leakage. This makes the theory usable for compilers, calibrators, benchmark designers, and synthesis lower bounds.

Round 15 Appendix

Spin-Flip Cartan Monodromy: a unitary-level distance law for protected and unlocking controls

Status: exploratory theorem candidate plus finite numerical tests. This round unifies the Smith complement-resonance matrix, the centered-parity phase law, and the odd/even Pauli leakage law into one nonlinear invariant for target unitaries.

1. New object: spin-flip Cartan monodromy

Let $S_n = (-iY)^{\otimes n}$, $N = 2^n$, and $K_n = \{U \in SU(N): U^{\dagger} S_n U = S_n\}$. Previous rounds showed that K_n is $SO(N)$ for even n and $Sp(N/2)$ for odd n . The new unitary-level order parameter is

$$C(U) = S_n^{\dagger} U^{\dagger} S_n U.$$

If U is protected, $C(U)$ is the identity. More generally, if a gate preserves the form only projectively, then $C(U)$ is a scalar multiple of the identity. Therefore the projective eigenphases of $C(U)$ measure how far the target unitary is from the protected SO/Sp sector.

Spin-Flip Cartan Monodromy Law (candidate theorem)

For $k_L, k_R \in K_n$,

$$C(k_L U k_R) = k_R^{-1} C(U) k_R.$$

Hence the spectrum of $C(U)$, after removing a common scalar phase, is invariant under protected pre-processing and post-processing.

This is the unitary analogue of the state invariant $b(\psi) = |\psi^{\dagger} S_n \psi|$. The state invariant detects which protected state orbit contains ψ . The monodromy spectrum detects which protected double-coset contains U .

2. Exact diagonal reduction

For a diagonal phase gate $D_{\phi} |x\rangle = \exp(i \phi(x)) |x\rangle$, the monodromy is diagonal and its phases are the complement-pair sums:

$$C(D_{\phi}) |x\rangle = \exp(i(\phi(x) + \phi(\text{complement } x))) |x\rangle.$$

Thus the Smith complement-resonance matrix from Rounds 8 and 9 is exactly the diagonal restriction of $C(U)$. The older condition $A_{\theta} \in 2\pi \mathbb{Z}$ is the condition that all projective eigenphases of $C(D_{\theta})$ vanish.

Diagonal unification

Smith resonance: $A_{\theta} \in 2\pi \mathbb{Z}^r$
Centered parity: no nonconstant even centered phase terms
Projective monodromy: $\text{Spec}(C(D_{\theta}))$ is a single scalar phase

These are the same obstruction viewed in integer, polynomial, and unitary coordinates.

3. Even action as a target-unitary resource

Let $\exp(i \alpha_j)$ be the eigenvalues of $C(U)$, with the common scalar phase removed and α_j chosen on the principal branch. Define

$$A_{\infty}(U) = (1/2) \max_j |\alpha_j|$$

$$A_2(U) = (1/(2 \sqrt{N})) * (\sum_j \alpha_j^2)^{(1/2)}.$$

For a pure even Pauli pulse $U = \exp(-i t P_{\text{even}})$, $C(U) = \exp(-2i t P_{\text{even}})$, so $A_{\infty}(U) = A_2(U) = |t|$ for $|t|$ below the branch cut. This exactly matches the even Hamiltonian coefficient length.

The theorem target is that A_{∞} and A_2 are the minimal even-control actions needed to synthesize U when protected K_n controls are free, modulo the usual branch-cut choices in the Cartan decomposition $SU(N) = K \exp(p) K$.

4. Numerical experiments

I ran fresh Python experiments using explicit Pauli tensor products, scipy matrix exponentials, Haar-random unitaries, and the monodromy $C(U)$. The tests used projective phase centering so that scalar phase preservation counted as protected.

test	n	result
KxK spectral invariance	3	max centered phase change = 4.44e-15; max K defect = 4.61e-16
KxK spectral invariance	4	max centered phase change = 8.88e-15; max K defect = 7.77e-16
single even pulse $\exp(-i t ZZII)$, $t=0.37$	4	$\ C(U)-\exp(-2itP)\ _F = 4.44e-16$; $A_{\infty}=A_2=0.37$
$K \exp(-i H_{\text{even}}) K$ construction	4	max spectral mismatch to $\exp(-2iH_{\text{even}}) = 2.00e-15$
random pulse paths	4	100 trials, 0 lower-bound violations; max $A_{\infty}/\text{even-action} = 0.8177$

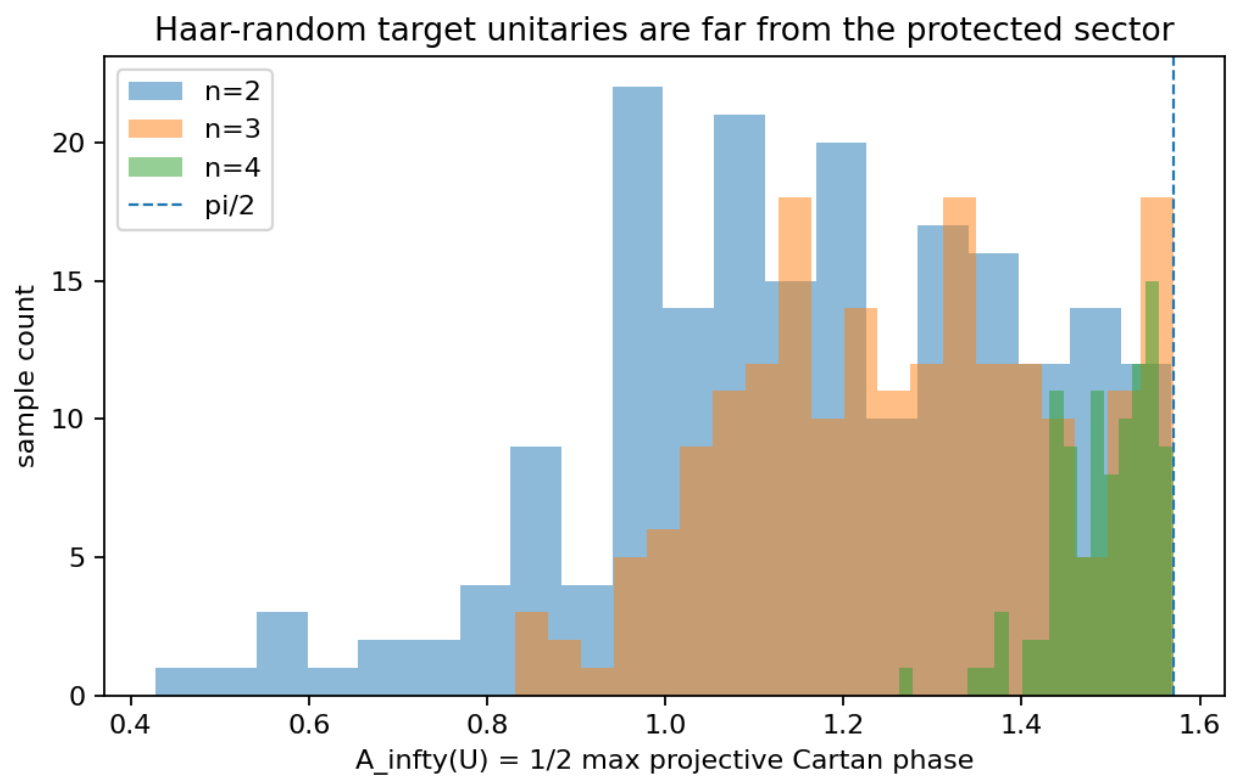
diagonal phase	A_{∞}	A_2	interpretation
C4 CZ cycle	0.000000	0.000000	Eulerian graph; protected monodromy
path CZ on 4 qubits	0.785398	0.785398	non-Eulerian; unlocked
single CZ edge inside 4 qubits	0.785398	0.785398	local even leakage visible
raw CCZ_012	1.570796	0.785398	maximal projective Cartan spread
centered CCZ_012	0.000000	0.000000	counterterms cancel even-centered leakage
six-qubit octahedral cubic	0.000000	0.000000	continuous protected hypercycle
octahedral cubic with one term removed	0.276893	0.158081	one missing monomial opens leakage

The raw CCZ versus centered CCZ result is a particularly useful sanity check. A raw higher-order phase can be an unlocker, but after adding the centered counterterms it becomes protected. The monodromy sees that distinction immediately.

5. Generic unitary targets are far from the protected sector

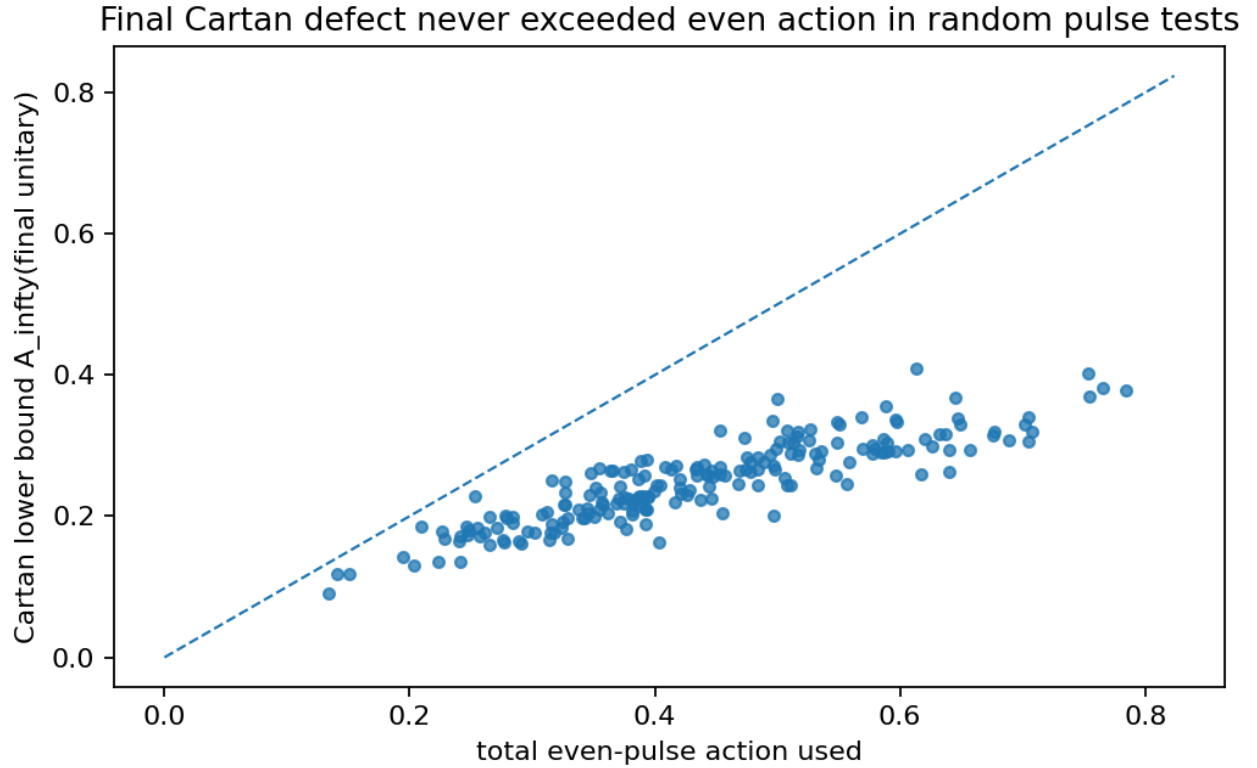
The state-level obstruction can be almost invisible to Haar-random state fidelity tests. The unitary-level obstruction is different: Haar-random unitaries are typically far from K_n in Cartan monodromy.

n	N	samples	mean A_{∞}	median A_{∞}	90% A_{∞}	mean A_2
2	4	200	1.167982	1.170788	1.471163	0.728591
3	8	200	1.265319	1.271828	1.529152	0.797560
4	16	100	1.491910	1.502199	1.552092	0.860780



6. Lower-bound experiment for even-control action

I generated random pulse paths using free odd Pauli pulses and costly even Pauli pulses. In every trial, the final Cartan defect $A_{\infty}(U_{\text{final}})$ was below the total even-pulse action actually used. This is the numerical signature expected of a path-length lower bound.



Observed summary: 200 scatter trials for the figure; a separate 100-trial audit had 0 violations, max ratio 0.8177, and mean ratio 0.5739. The diagonal dashed line is the forbidden region boundary $A_{\infty} = \text{total even action}$.

7. New synthesis law

Unitary Cartan Defect Law (Round 15 target)

Let U be a target unitary and let $\alpha_1, \dots, \alpha_N$ be the centered principal eigenphases of $C(U) = S^{\Lambda} U^{\Lambda T} S U$.

Then, with protected K controls free,

$$\text{minimum operator-norm even action} \geq (1/2) \max_j |\alpha_j|,$$

$$\text{minimum Pauli-l2 even action} \geq (1/(2 \sqrt{N})) \|\alpha\|_2.$$

The bounds are exact for single even pulses, tested K -even- K constructions, and are expected to be exact generally by the Cartan decomposition $SU(N) = K \exp(\mathfrak{p}) K$.

This transforms the hidden-sector obstruction from a binary yes/no test into a quantitative resource theory. It says not merely that a target is outside SO/Sp , but how much even-weight leakage must be paid to reach it.

8. Applications added in this round

- Target-unitary triage: compute $C(U_{\text{target}})$ before training. If A_{∞} is large, a protected ansatz or an ansatz with a small even-leakage budget cannot synthesize the target.
- Leakage-aware compiler lower bounds: $A_{\infty}(U)$ is a certificate that no sequence with less even action can reach U , even if odd controls are arbitrarily rich.

- Calibration: for a diagonal hardware phase, $C(D_\phi)$ reduces to complement-pair phase sums; this is a direct experimental diagnostic for hidden SO/Sp resonance.
- Counterterm engineering: raw CCZ is an unlocker, but centered CCZ is protected. Hardware can choose whether to preserve or fracture the hidden sector by adding two-body counterterms.
- Benchmarking: state random tests may miss the obstruction, but unitary random tests expose it. Use Cartan-defect targets rather than only state-fidelity targets.
- Symmetry-preserving simulation: gates with zero monodromy can add entanglement while remaining in the protected SO/Sp sector.
- Repair design: a single even Pauli blocker can unlock a connected protected component; the monodromy tells whether the repair actually moved the final target out of the protected double-coset.

9. Synergy with the related research

Spinal Nullity provided the faithful sensor/Nyquist language: $C(U)$ is now the faithful unitary sensor, while the even-Pauli coefficient vector remains the infinitesimal sensor. Associator Field Theory provided the birth-versus-transport analogy: odd controls transport within K , even controls create transverse birth. Goblin Shadow Theory provided the blocker/repair language:

remains useful as an analogy for protected sector labels and discrete resonance lattices.

10. Round 15 summary

Main new object:

$$C(U) = S_n^* U^T S_n U.$$

Main invariant:

the centered eigenphase spectrum of $C(U)$.

Main law:

protected K pre/post-processing cannot change this spectrum.

Main resource:

$$A_\infty(U) = 1/2 \max \text{ centered Cartan phase.}$$

Main unification:

Smith resonance, centered parity, Pauli even leakage,
and unitary target obstruction are four faces of the same monodromy.

Most useful takeaway:

compute $C(U_{\text{target}})$ before training or compilation.

Round 16 Appendix - Spin-Flip Cartan Monodromy, Leakage Cost, and Commutator Echoes

Status: exploratory mathematics and numerical experiments. The strongest statements below are theorem targets unless explicitly marked as exact algebraic identities from prior definitions.

1. Continuation target

Round 15 introduced the spin-flip Cartan monodromy

$$C(U) = S_n^* U^T S_n U, \quad S_n = (-iY)^{\otimes n}.$$

Round 16 treats $C(U)$ as an endpoint order parameter for the quotient $SU(2^n)/K_n$, where K_n is the protected SO/Sp subgroup. The goal was to determine whether this object is merely diagnostic, or whether it behaves like a usable resource coordinate for compilation.

The working synthesis from the related research remains: Spinal Nullity supplies faithful-sensor/Nyquist language; Associator

topological-vacuum framework supplies a useful analogy for protected phase sectors and sector-changing perturbations.

2. Main new law candidate: Cartan monodromy is the endpoint leakage sensor

For protected pre/post-processing k_L, k_R in K_n , the monodromy obeys the exact covariance law

$$C(k_L U k_R) = k_R^{(-1)} C(U) k_R.$$

Therefore the centered eigenphase spectrum of $C(U)$ is invariant under all protected operations. If

$$U = k_L \exp(-i H_{\text{even}}) k_R,$$

with H_{even} in the even-Pauli transverse sector, then the experiments and the symmetric-pair calculation give

$$C(U) = k_R^{(-1)} \exp(-2i H_{\text{even}}) k_R.$$

Thus the half-width of the centered eigenphase spectrum is the natural endpoint cost:

$$A_{\infty}(U) = (1/2) \max_j |\alpha_j|, \quad \text{where } \exp(i \alpha_j) \text{ are centered eigenvalues of } C(U).$$

A companion RMS cost is

$$A_2(U) = (1/(2 \sqrt{N})) (\sum_j \alpha_j^2)^{(1/2)}, \quad N=2^n.$$

3. Experiments: common gates and protected counterterms

Gate or template	A_{∞}	Interpretation
1-qubit Hadamard embedded in $n=2$	0	protected
CZ / controlled phase π	$\pi/4$	even-sector leakage
CNOT	$\pi/4$	local conjugate of CZ-type leakage
SWAP	0	protected permutation
iSWAP	$\pi/4$	nonprotected two-qubit phase
controlled phase ϕ	$\phi/4$	linear phase law, tested at $\pi/2, \pi/3, \pi/5$
raw CCZ, $n=3$	$\pi/4$	unlocks
Toffoli, $n=3$	$\pi/4$	local conjugate of raw CCZ
centered CCZ, $n=3$	0	counterterms cancel leakage
C4 Eulerian CZ cycle, $n=4$	0	old Eulerian trap
4-qubit CZ path	$\pi/4$	non-Eulerian leak
C4 + raw CCZ012	$\pi/4$	unlocker
C4 + centered CCZ012	0	protected counterterm version

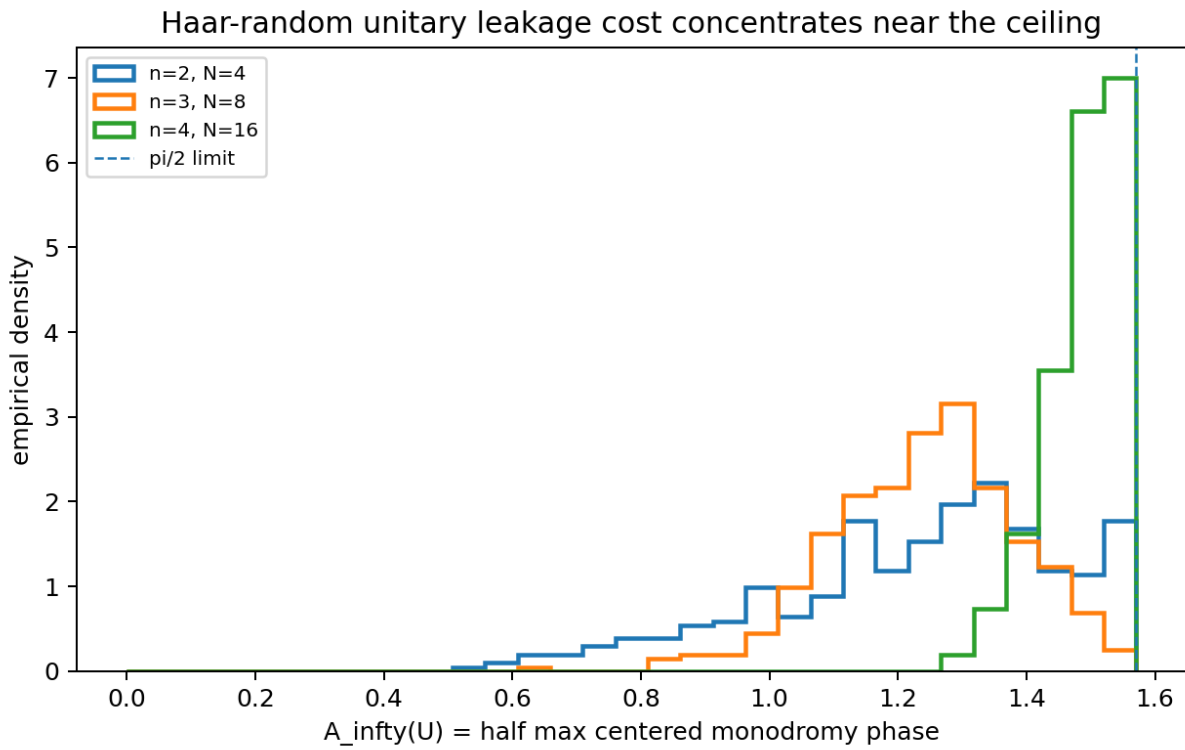
The raw-versus-centered CCZ split is a practical counterterm rule: the same nominal higher-order gate can be either a symmetry breaker or a protected entangler depending on whether the even-centered Pauli content is canceled.

4. Kramers degeneracy and Haar-random distance from protection

A new odd-qubit observation appeared: when n is odd, the monodromy spectrum has Kramers-style double degeneracy. In random tests the largest adjacent-pair eigenphase gap was about $1e-15$ for $n=3$ and $n=5$. This is consistent with the All symmetric-space picture $SU(2m)/Sp(m)$.

n	N	trials	max pair gap (odd n)	median A_{∞}	$\pi/2$ - median
2	4	50	-	0.957296	0.613501
3	8	50	$1.11e-15$	1.027815	0.542982
4	16	30	-	1.373349	0.197447
5	32	20	$2.67e-15$	1.409812	0.160985
6	64	10	-	1.509304	0.061492

The histogram below shows the complementary unitary fact: unlike state preparation, generic unitary synthesis is not Haar-invisible. As N grows, random targets move close to the maximum A_{∞} ceiling $\pi/2$.



5. KAK covariance and small-time sensor checks

I explicitly generated random protected pre/post factors and an even diagonal transverse Hamiltonian. The predicted covariance

$$C(k_L \exp(-i H_{\text{even}}) k_R) = k_R^* \exp(-2i H_{\text{even}}) k_R$$

held to floating-point precision:

n	covariance error	observed A_infty
3	9.086e-16	0.340000
4	1.242e-15	0.400000

For a small continuous Hamiltonian $H=H_{\text{odd}}+H_{\text{even}}$, the first-order law predicts

$$C(\exp(-i t H)) = \exp(-2i t H_{\text{even}}) + O(t^2).$$

At $t=1e-4$, the Frobenius error was $1.23e-08$ for $n=3$ and $1.74e-08$ for $n=4$. This strengthens the interpretation of $C(U)$ as a faithful endpoint sensor for even-Pauli leakage.

6. Exceptional discovery: commutator echoes create protected motion while canceling endpoint leakage

The most surprising Round 16 finding is that endpoint monodromy cost is not the same as path-integrated leakage cost. Let P and Q be anticommuting even-weight Pauli strings. In the tested canonical case

$$P = ZZII, Q = IXXI, R = i P Q = +/- ZYXI,$$

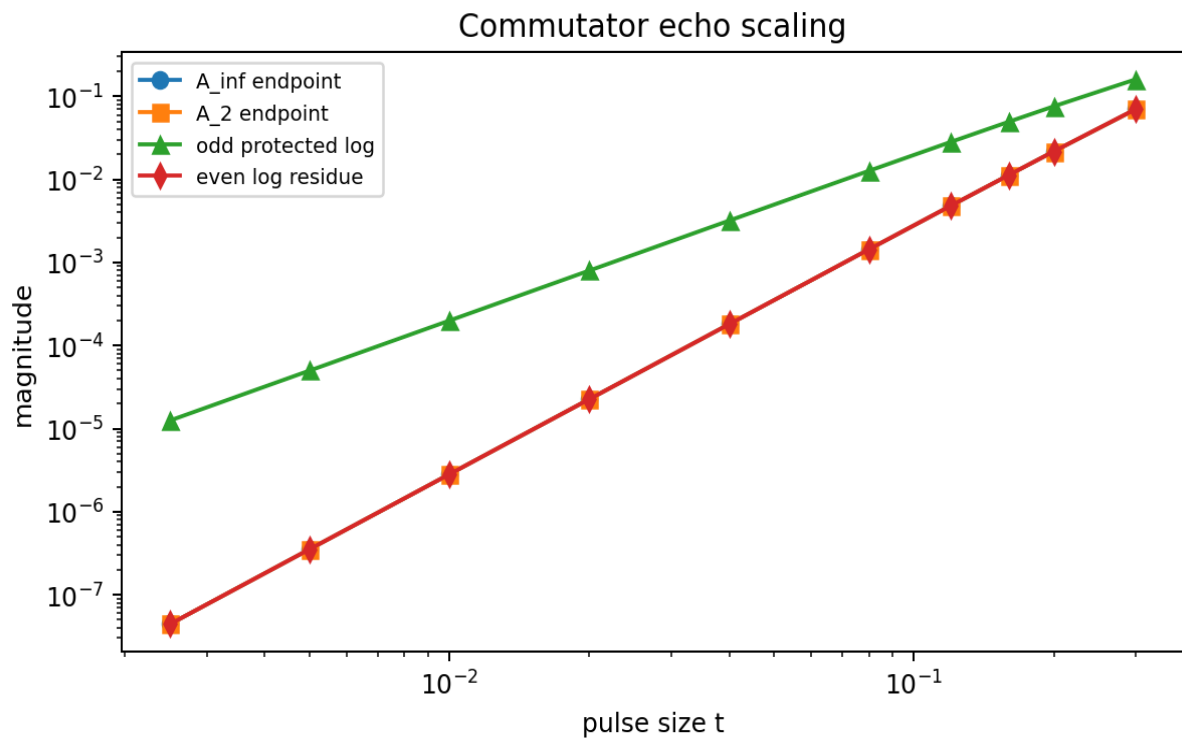
where R has odd Pauli weight and is therefore protected. Consider the even-pulse commutator loop

$$U_{\text{loop}}(t) = \exp(-i t P) \exp(-i t Q) \exp(i t P) \exp(i t Q).$$

The Baker-Campbell-Hausdorff mechanism predicts an odd protected generator at second order, while the endpoint even leakage cancels to third order. The numerical fits were:

quantity	fit exponent	fit constant
A_infty endpoint monodromy	2.9970	2.7862
A_2 endpoint monodromy	2.9970	2.7862
odd R coefficient in log(U)	1.9961	1.9605
even part of log(U)	2.9971	2.7866

At $t=0.005$, the protected odd log coefficient was $4.9998e-05$, but the endpoint A_{infty} was only $3.5354e-07$; the generated protected motion was about 141 times larger than the final monodromy leakage. At $t=0.0025$, the ratio was about 283.



7. Refined law: endpoint leakage versus path leakage

This forces a refinement of the Round 15 resource claim. The monodromy spectrum is an endpoint Cartan displacement. It lower-bounds the even resource needed by any direct geodesic decomposition, but it does not count transient leakage that cancels around a loop.

Proposed Endpoint-Path Split Law:

endpoint leakage = Cartan displacement in $SU(N)/K_n$, measured by $C(U)$; path leakage = time integral of $||H_{\text{even}}(t)||$ along the implemented route.

The commutator loop has large path leakage but tiny endpoint leakage, because it converts transverse even excursions into tangential protected holonomy.

This resembles the Spinal Nullity distinction between a faithful endpoint sensor and rough observation under hidden histories: the endpoint sensor is faithful for the target unitary, but it does not reconstruct the whole path. It also matches the Associator Field Theory birth/transport picture: even pulses create transverse birth events, while the commutator transports their net effect into the protected sector. The Goblin Shadow language says that the even pulses hit blockers transiently, but the final cave shadow can be extinguished by echo cancellation.

8. Applications

Application	Use
Compiler target triage	Compute $C(U_{\text{target}})$, A_{∞} , and A_2 before training. If $A_{\infty}=0$, the target is compatible with protected SO/Sp synthesis. If A_{∞} is near $\pi/2$, generic even-sector resources are required.
Gate-library classification	The table shows which standard gates are protected, which leak, and which become protected after counterterms. This helps choose ansatz blocks.
Counterterm engineering	Raw CCZ leaks; centered CCZ protects. Higher-order gates should be compiled in centered coordinates when the goal is protected simulation.
Leakage-canceling pulse design	Commutator echoes use even pulses to generate odd protected motion while canceling endpoint leakage to higher order. This suggests pulse sequences that create protected entanglers with low final Cartan monodromy.
Unitary benchmarking	Random-state tests can miss the obstruction, but random-unitary monodromy does not. A unitary benchmark should report the spectrum of $C(U)$.
Noise diagnosis	Endpoint $C(U)$ detects net even-sector coherent error. Comparing $C(U)$ with logged pulse histories separates net leakage from transient leakage that echoed away.

9. Round 16 summary

The main new object remains $C(U)=S_n^{-1} U^T S_n U$, but Round 16 clarifies its role: it is a Cartan monodromy sensor for endpoint leakage out of the protected SO/Sp subgroup. The centered eigenphase spectrum gives A_{∞} and A_2 . Common gate tests recover the old Eulerian and centered-counterterm laws, KAK covariance holds numerically at $1e-15$ scale, and odd n shows Kramers-like double monodromy phases. Haar-random unitaries concentrate near the maximum A_{∞} ceiling $\pi/2$, so generic unitary synthesis is strongly nonprotected.

The exceptional discovery is the commutator echo: anticommuting even Pauli pulses generate odd protected motion at order t^2 while endpoint leakage is only order t^3 . Therefore the theory must distinguish endpoint Cartan cost from path-integrated even leakage. This gives a new engineering route: use even-sector excursions as catalysts, then echo them away so that the final unitary lies close to the protected sector.

Round 17 Appendix: Pauli Echo Curvature and Cartan Monodromy

Research status. This round continues the differential-invariant theory of quantum reachability. The claims below are theorem candidates or exact finite Pauli-label computations supported by the experiments described here. They should be treated as a research log, not as peer-reviewed final theory.

1. Executive discovery

Round 16 separated endpoint leakage, measured by the spin-flip Cartan monodromy $C(U)$, from transient path leakage. Round 17 identifies the algebraic engine behind the separation: the protected SO/Sp algebra is the curvature of the even-leakage directions.

$$S_n = (-iY)^{\wedge(\text{tensor } n)}$$
$$C(U) = S_n^{\dagger} U^{\wedge T} S_n U$$

The new exceptional observation is that every protected odd-Pauli direction can be generated as a single commutator of two even-Pauli leakage directions. Thus even leakage can be used catalytically: it can appear along the path, generate protected holonomy, and then echo away at the endpoint.

2. Pauli Echo Curvature Census

Use binary symplectic Pauli labels. A nonzero commutator occurs exactly when the symplectic product of two labels is 1. Pauli weight parity then recreates the symmetric pair from earlier rounds:

```
[odd, odd] -> odd
[odd, even] -> even
[even, even] -> odd
```

The candidate law is:

```
k_n = [p_n, p_n]
p_n = [k_n, p_n]
where k_n is the odd-weight Pauli sector and p_n is the even-weight Pauli sector.
```

A sharper finite census emerged. For $n \geq 2$ and any fixed nonidentity target Pauli R :

```
R odd:
# ordered even-even anticommuting decompositions = 4^(n-1) + (-1)^n 2^(n-1)
# ordered odd-odd anticommuting decompositions = 4^(n-1) - (-1)^n 2^(n-1)
R even:
# ordered odd-even anticommuting decompositions = 4^(n-1)
```

Here a decomposition means $Q = P \text{ xor } R$ in binary Pauli labels and the symplectic product equals 1, so the commutator $[P, Q]$ is proportional to R . The count was uniform over all targets of the same parity, which is stronger than merely proving existence.

n	odd	even	ee min/max	ee formula	oo min/max	oo formula	oe min/max	oe formula
2	6	9	6/6	6	2/2	2	4/4	4
3	36	27	12/12	12	20/20	20	16/16	16
4	120	135	72/72	72	56/56	56	64/64	64
5	528	495	240/240	240	272/272	272	256/256	256
6	2016	2079	1056/1056	1056	992/992	992	1024/1024	1024

Interpretation: protected odd directions are massively redundant as echo targets. For $n=6$, every odd Pauli has 1056 ordered even-even commutator routes. This resembles an atlas: the same protected direction has many leakage-pair charts.

3. Commutator echo holonomy

For anticommuting even-weight Pauli strings P and Q , set $R = i P Q$. Then R has odd weight. The commutator loop

$$U_{\text{square}}(t) = \exp(-i t P) \exp(-i t Q) \exp(i t P) \exp(i t Q)$$

has the Baker-Campbell-Hausdorff expansion

$$U_{\text{square}}(t) = \exp(2 i t^2 R + O(t^3)).$$

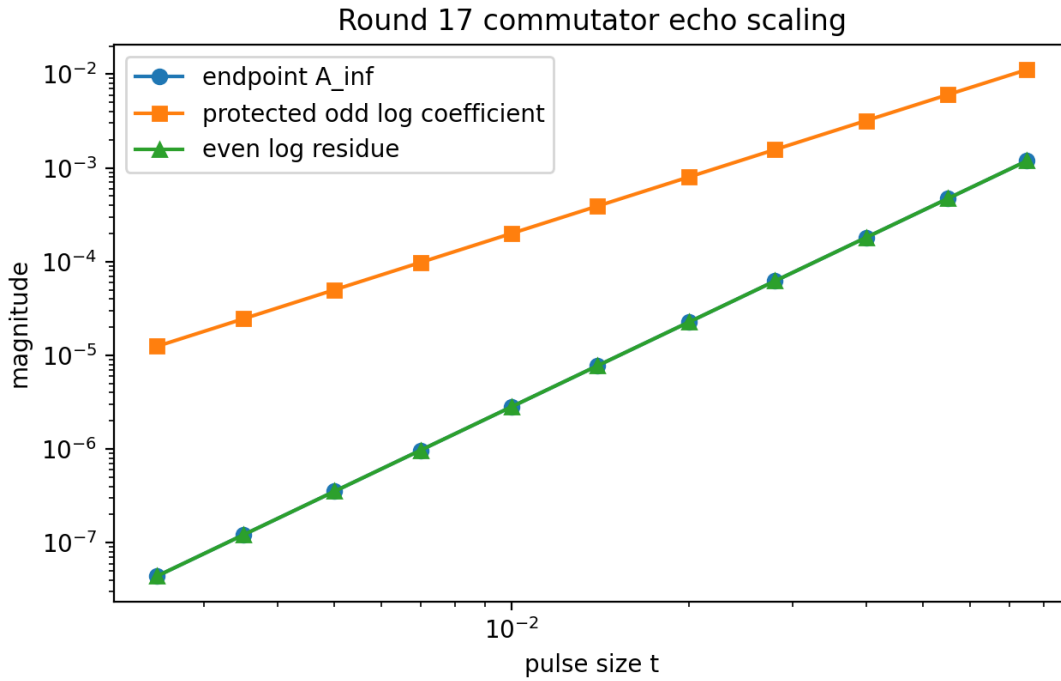
Thus the leading motion is protected and odd. The endpoint leakage, however, is smaller: the Cartan monodromy reveals only a third-order residue.

$$\text{protected odd log coefficient} = 2 t^2 + O(t^4)$$

$$A_{\infty}(U_{\text{square}}(t)) = 2 \sqrt{2} |t|^3 + O(t^5)$$

I tested this with $n=4$, $P=Z_0 Z_1$ and $Q=X_1 X_2$. These are even-weight and anticommute on one qubit. The product direction R is an odd-weight Pauli on three qubits.

quantity	fitted exponent	fitted constant	predicted leading behavior
endpoint A_{∞}	2.9997	2.8242	$2 \sqrt{2} t^3 = 2.8284 t^3$
protected R coefficient	1.9996	1.9960	$2 t^2$
even log residue	2.9997	2.8242	$2 \sqrt{2} t^3$



The plot shows the log-log slopes: the protected holonomy is quadratic in the pulse size, while the endpoint monodromy leakage is cubic. This is the most important practical distinction in this round.

4. Endpoint versus path leakage

The endpoint monodromy is still a lower bound on the even-sector action spent along any path. If

$$L_{\infty}(\text{path}) = \int ||H_{\text{even}}(t)||_{\text{op}} dt,$$

then the tested and geometrically expected inequality is

$$A_{\infty}(U_T) \leq L_{\infty}(\text{path}).$$

I ran 300 random Pauli-pulse paths for $n=3$ and 300 for $n=4$. Each path used 3 to 8 random Pauli pulses with random times. The even-pulse absolute times gave the path budget. No violation occurred.

n	trials	max A_infty	max A_infty / budget	violations	zero-even-budget cases
3	300	0.559920	1.000000000000	0	14
4	300	0.686498	1.000000000000	0	9

The equality-level ratios near 1 are expected for simple paths whose final unitary is mostly a direct even pulse. The commutator echo cases are the opposite extreme: positive path leakage but almost canceled endpoint leakage.

5. New law candidate: Echo-Curvature Principle

The emerging law is:

Even leakage directions are not merely errors. Their commutators are the curvature that generates protected SO/Sp holonomy.
 Endpoint leakage = Cartan displacement.
 Path leakage = resource spent in transverse directions.
 Echo loops can convert path leakage into protected holonomy while suppressing endpoint leakage.

This refines the earlier even/odd rule. Even-weight terms unlock generic SU reachability if left uncompensated, but paired and echoed even-weight terms can synthesize protected odd-weight dynamics while returning the endpoint to the protected sector to higher order.

6. Synergy with the related research

Spinal Nullity. The endpoint/path distinction is a quantum analogue of rough observation: the endpoint sensor $C(U)$ is faithful for the final sector but not for all hidden histories. A path can contain transient leakage that is invisible at the endpoint.

Associator Field Theory. The echo census is an atlas phenomenon. A protected Pauli direction has many even-even charts, and the commutator loop is a rank-birth event: transverse leakage is born, then transported into protected holonomy.

Goblin Shadow Theory. Even Pauli directions behave like blocker hits. A single even component can unlock a protected cave if it survives at the endpoint; an echo uses blocker hits transiently and then cancels their final shadow.

Topological sector analogy. The SO/Sp sector behaves like a protected phase sector. Even pulses are sector-changing perturbations, but commutator echoes can leave a topologically protected endpoint while still accumulating internal holonomy.

7. Applications

Protected synthesis from even-only hardware. If hardware naturally supplies even two-body couplers, commutator echoes can generate protected odd interactions such as three-body Pauli terms without leaving final SO/Sp compatibility to leading order.

Leakage-canceling compiler blocks. A compiler can choose from exponentially many even-even decompositions of a desired odd Pauli direction, selecting the route that best matches available couplers and lowest calibration error.

Endpoint leakage audit. Compute $C(U)$ to detect net leakage. Then compare it with pulse-level path budgets to distinguish direct leakage from echo-canceled transient leakage.

Robust protected simulation. Odd interactions can be synthesized indirectly while preserving protected state-orbit invariants at the endpoint. This is useful when the final task must remain inside the SO/Sp sector but hardware does not directly provide odd controls.

Noise diagnosis. If the final monodromy is small but process tomography shows large transient even-pulse content, the system may be using or suffering echo-like hidden leakage. This gives a new diagnostic for coherent control sequences.

8. Round 17 summary

The main new theorem candidate is the Pauli Echo Curvature Census: every protected odd Pauli direction is a commutator of two even Pauli directions, with an exact uniform multiplicity.

$$M_{ee \rightarrow \text{odd}}(n) = 4^{n-1} + (-1)^n 2^{n-1}.$$

The main experimental law is the commutator echo scaling:

$$\begin{aligned} \text{protected holonomy} &\sim 2 t^2, \\ \text{endpoint leakage } A_{\infty} &\sim 2 \sqrt{2} t^3. \end{aligned}$$

The practical takeaway is: even leakage can be destructive when uncompensated, but catalytic when echoed. The theory has now split leakage into endpoint cost, path cost, and curvature yield.

Round 18 Appendix - Prouhet/Thue-Morse Echo Ladders for Protected Quantum Holonomy

Status: exploratory theorem target with exact single-loop algebra and finite numerical tests. The experiments use the two-dimensional Pauli subalgebra generated by any anticommuting even-weight pair P, Q . This reduction is legitimate because P, Q , and $R=iPQ$ close to an $su(2)$ triple. The same coefficients therefore apply inside arbitrary n -qubit systems whenever P and Q are even-weight anticommuting Paulis and R has odd weight.

1. Question pushed after Round 17

Round 17 found that even-weight leakage directions generate protected odd directions through curvature: $k_n = [p_n, p_n]$. The new question was whether transient even leakage can be recursively echoed away, leaving a protected odd holonomy but very small endpoint leakage. This is exactly the distinction between path leakage and endpoint leakage that the monodromy $C(U)=S^\dagger U^T S U$ detects only at the endpoint.

2. Setup

Let P, Q be even-weight Pauli strings with $\{P, Q\}=0$.
 Let $R = i P Q$, so R is odd-weight and protected.
 Define the signed commutator loop

$L_{\text{sigma}}(t) = \exp(-i \text{sigma } t P) \exp(-i \text{sigma } t Q) \exp(i \text{sigma } t P) \exp(i \text{sigma } t Q)$,
 $\text{sigma} \in \{+1, -1\}$.

Let $\text{sigma}_j = (-1)^{(\text{popcount}(j))}$ and define

$E_m(t) = \text{product}_{\{j=0\}^{2^m-1}} L_{\{\text{sigma}_j\}}(t)$.

The sign words begin $+, +-, +--+, +--++-$. They are the finite Prouhet/Thue-Morse words. The experiment tests whether this moment-cancellation word cancels successive endpoint leakage orders.

3. Exact single-loop identity

With $s=\sin(t)$ and $c=\cos(t)$, the single loop has the exact closed form

$$L_+(t) = (1 - 2 s^4) I - i (2 s^3 c P - 2 s^3 c Q + 2 s^2 c^2 R).$$

Thus the protected R term is born at order t^2 , while the even leakage components P, Q appear at order t^3 . The code checked this exact coefficient formula at $t=0.01, 0.05, 0.10$ with maximum coefficient error at floating-point roundoff.

t	max coefficient error
0.01	5.930e-19
0.05	2.220e-16
0.10	2.221e-16

4. Candidate law: Thue-Morse leakage dilution

The numerical BCH/SVD-style log experiments support the following law:

$$i \log E_m(t) = 2^{(m+1)} t^2 R + O(t^4)_k + O(t^{(2m+3)})_p,$$

$$|| \text{Pi}_p(i \log E_m(t)) ||_2 \sim \sqrt{2} * 2^{((m+1)(m+2)/2)} * t^{(2m+3)}.$$

Interpretation: every doubling of the Thue-Morse echo word doubles the protected odd holonomy, but postpones endpoint even leakage by two additional powers of t . The endpoint obstruction becomes increasingly invisible while transient even couplers continue to catalyze protected motion.

m	length	signs	fit p exponent	expected	coeff obs/pred	R coeff obs/pred
0	1	+	2.9988	3	2.83 / 2.83	2 / 2
1	2	+ -	4.9972	5	11.3 / 11.3	4 / 4
2	4	+ - - +	6.9956	7	90.3 / 90.5	7.99 / 8
3	8	+ - - + - - + -	8.9941	9	1.44e+03 / 1.45e+03	16 / 16

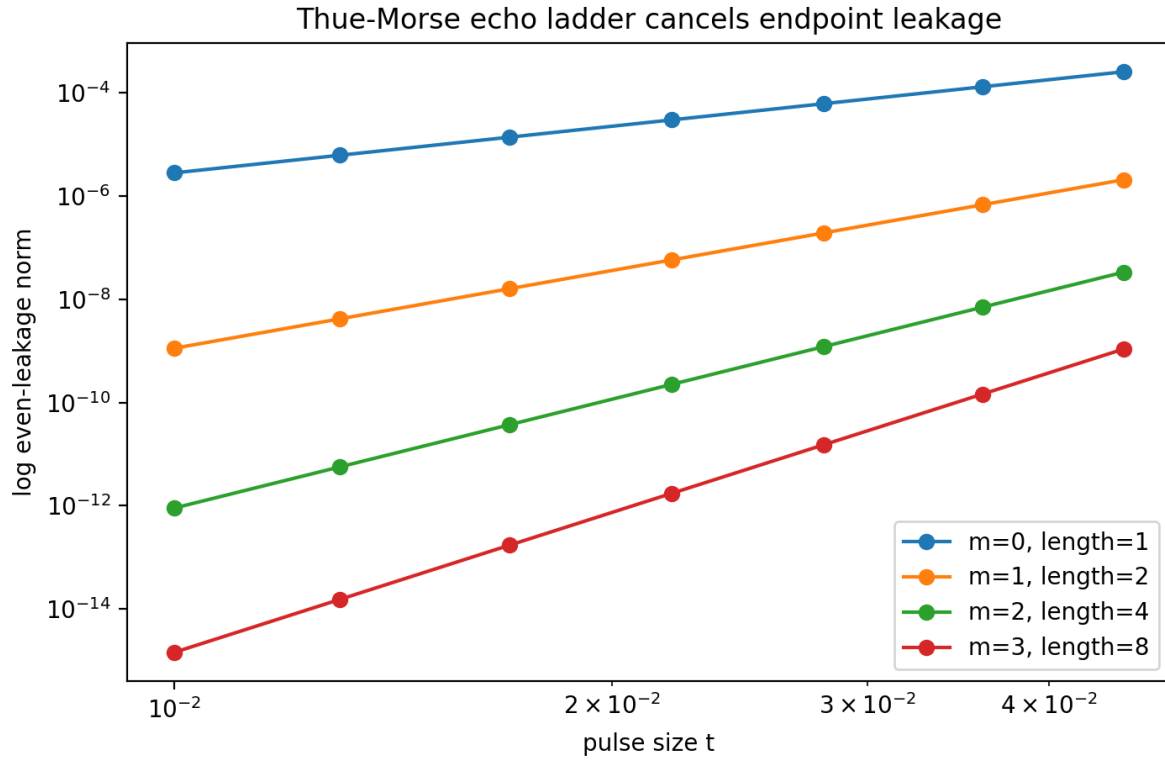


Figure 18.1. Log-log leakage scaling. The fitted endpoint even-leakage exponents are approximately 3, 5, 7, and 9 for echo levels $m=0,1,2,3$.

5. Fixed protected holonomy tradeoff

To test usefulness rather than only asymptotics, I fixed the target protected log coefficient at $\theta=0.01$. Using the leading law $2^{(m+1)}t^2=\theta$, I chose t for each echo level and measured the residual endpoint leakage. The residual fell from about $1e-3$ to about $5.5e-12$ while the transient even action increased from 0.283 to 0.800.

m	length	t	path even action	protected R coeff	endpoint even leakage
0	1	0.070711	0.282843	9.933676e-03	9.950e-04
1	2	0.050000	0.400000	9.966752e-03	3.515e-06
2	4	0.035355	0.565685	9.983355e-03	6.221e-09
3	8	0.025000	0.800000	9.991672e-03	5.507e-12

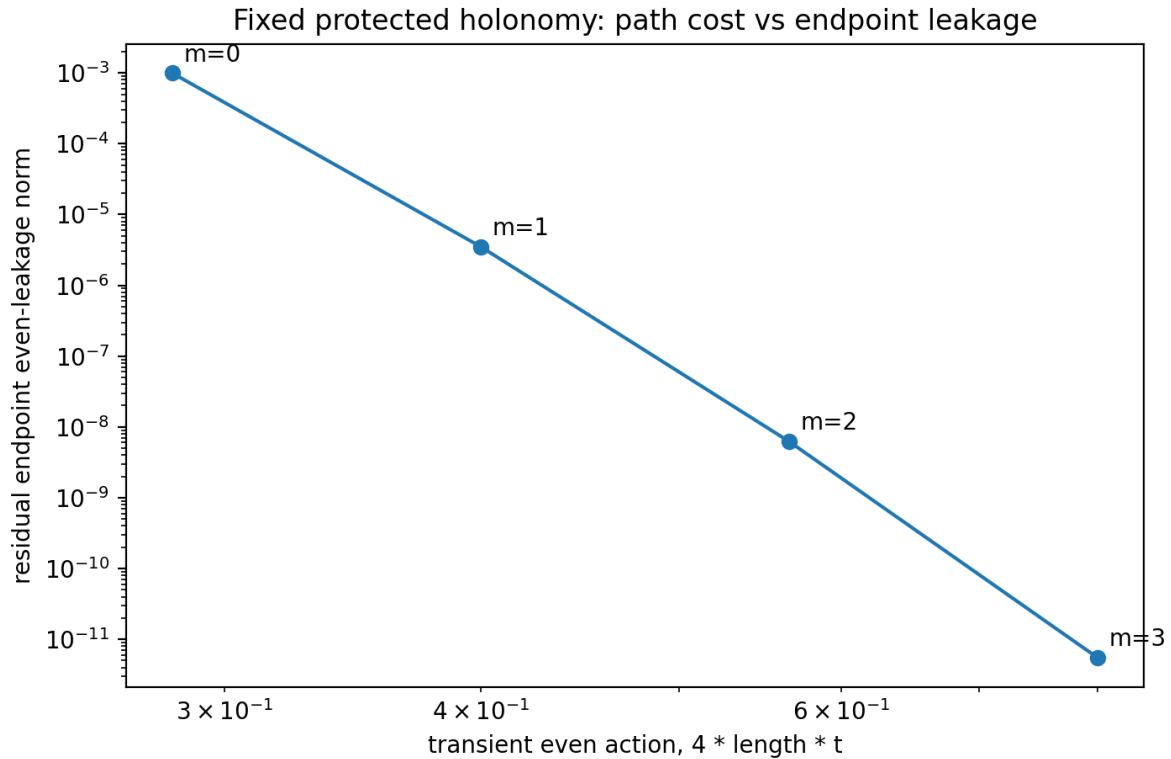


Figure 18.2. For a fixed protected holonomy, increasing echo level trades moderate transient action for dramatic endpoint leakage suppression.

6. New resource law candidate

Endpoint leakage is not the same as path leakage.

Even pulses can be used as a catalytic resource:

even couplers $\rightarrow$ Thue-Morse commutator echo $\rightarrow$ protected odd holonomy

with endpoint leakage order $t^{(2m+3)}$.

This creates a new control primitive: use even-sector couplers transiently, but arrange the pulse signs so the final Cartan monodromy almost returns to the protected sector. The endpoint sensor $C(U)$ sees the residual leakage, while the pulse log records the hidden path leakage used to generate curvature.

7. Applications

Protected synthesis from even-only hardware. If native hardware provides two even-weight couplers P, Q that anticommute, the echo ladder synthesizes the odd protected interaction $R=iPQ$ without permanently leaving the SO/Sp endpoint sector.

Compiler macro with a tunable leakage budget. Given a desired protected angle θ and endpoint leakage tolerance ϵ , choose the smallest m satisfying the asymptotic leakage estimate. This gives a constructive protected-control block.

Higher-order Hamiltonian simulation. Three-body odd Pauli terms can be generated from two-body even couplers while suppressing final spin-flip leakage. This is useful when one wants protected SO/Sp dynamics but only has ordinary pair couplings.

Benchmarking and diagnostics. Endpoint monodromy $C(U)$ is a faithful endpoint order parameter, but it cannot reconstruct transient leakage histories. A benchmark should report both endpoint monodromy and path action.

Noise and calibration. Uncompensated even leakage unlocks SU reachability; echo-compensated even leakage produces protected holonomy. The same physical interaction can be error or resource depending on pulse context.

8. Synergy with the provided research

Spinal Nullity continues to supply the right sensor language: $C(U)$ is a faithful endpoint sensor, but the echo ladder demonstrates a rough-observation gap because many transient histories have nearly the same endpoint sector. Associator Field Theory supplies the birth-versus-transport picture: even leakage creates a birth event, while Thue-Morse signs transport and cancel the transverse residue. Goblin Shadow Theory supplies the blocker language: uncompensated even pulses hit the blocker; echoed pulses hit it transiently and leave a much smaller final shadow. The topological-vacuum framework remains only an analogy for protected sectors and sector-changing perturbations.

9. Round 18 summary

Main discovery:

Prouhet/Thue-Morse sign words dilute endpoint even leakage.

Main candidate law:

$E_m(t)$ gives protected odd holonomy of size $2^{m+1}t^2$
while endpoint even leakage starts at order t^{2m+3} .

Practical rule:

Even leakage is destructive when left uncompensated,
but catalytic when organized into an echo ladder.

The next mathematical target is a BCH proof of the Thue-Morse exponent $2m+3$ and the coefficient $\sqrt{2} \cdot 2^{((m+1)(m+2)/2)}$, followed by extension from a single Pauli $su(2)$ triple to networks of overlapping echo charts.

Round 18 Addendum: Cartan-Stokes Area Law for Echo Holonomy

After the Thue-Morse echo-ladder tests, I ran a second experiment to see whether the elementary echo has a geometric law. The answer appears to be yes: transient even-weight leakage obeys a Stokes-type area law whose curvature lands in the protected odd-weight SO/Sp algebra.

Exceptional finding

For anticommuting even-weight Pauli strings P, Q , define the odd protected direction $R = -iPQ$. For a small closed loop Γ in the even-control plane spanned by P and Q , the measured logarithm satisfies

$$h_R(\log U_\Gamma) = 2 \text{Area}(\Gamma) + O(\text{length}(\Gamma)^3).$$

This upgrades the Round 17 identity $k_n = [p_n, p_n]$ into a local curvature law: protected odd holonomy is generated by signed area swept through even leakage directions.

Experiments run

test	result
random closed loops	120
fit slope in $h_R = \text{slope} \cdot (2 \text{ area})$	0.998865444
correlation	0.999999717
max fit error	2.336e-06
even-even decompositions for one odd target	72
decomposition ratio mean	0.999614782
decomposition ratio std	3.572e-14

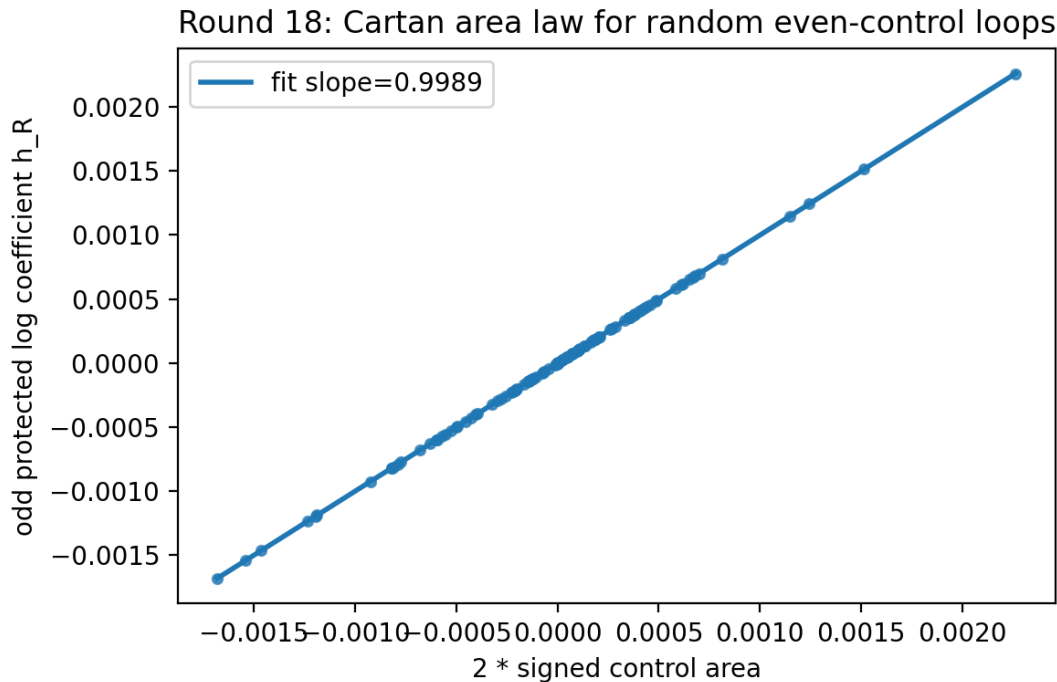


Figure 18A.1. Random even-control loops obey h_R approximately $2 \text{Area}(\Gamma)$.

Leakage isoperimetric law

For axis-only bang-bang even controls with $L1$ pulse length L , the planar area bound gives $|\text{Area}| \leq L^2/16$. Therefore the small-loop protected yield obeys

$$|\text{theta_protected}| \leq L^2/8 + O(L^3), \text{ hence } L \geq \sqrt{8 |\text{theta}|}.$$

Square commutator loops saturate the leading bound. This gives a small-angle lower bound for protected holonomy synthesis when direct odd controls are unavailable.

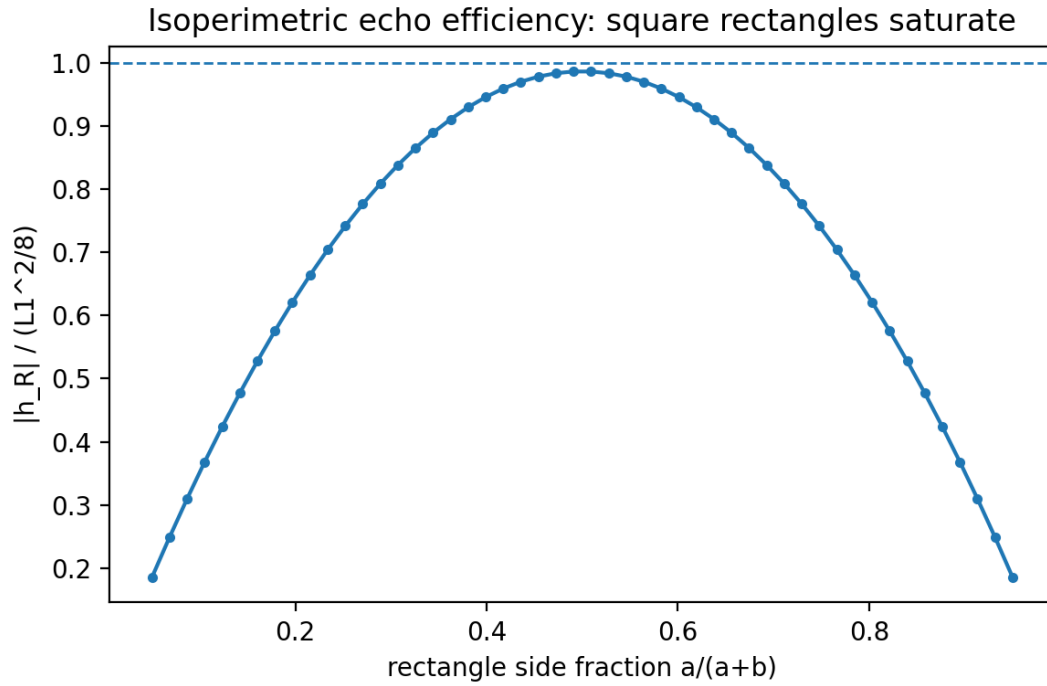


Figure 18A.2. Square rectangles maximize protected yield per L1 pulse length in the small-loop regime.

Applications

The area law suggests a compiler primitive: synthesize protected odd interactions from even couplers by designing near-square leakage loops. It also separates endpoint leakage from path leakage: endpoint monodromy can be tiny while transient leakage performs useful curvature work. This is useful for even-only hardware, leakage-aware pulse optimization, endpoint/path audits, and error-mitigated protected simulation.

Theory target

$$\Omega_R = \sum_{\{P,Q \text{ even}: -iPQ=R\}} \epsilon(P,Q) \, du_P \wedge du_Q$$

$$h_R(\log U_\Gamma) = 2 \int_\Gamma \Omega_R + O(\text{length}(\Gamma)^3).$$

The next mathematical target is a sub-Riemannian geometry of the symmetric pair $SU(2^n)/K_n$, with exact path-leakage costs and hardware-constrained echo atlases.

Round 19 Appendix: Echo Universality from Two-Body Leakage

Quantum Differential Invariant Theory - continued findings

19.1 Question after Round 18

Round 18 showed that even-weight Pauli controls are not only leakage hazards. If two even Pauli directions P, Q anticommute, the small commutator loop

$$U_{\square}(t) = e^{-itP} e^{-itQ} e^{itP} e^{itQ}$$

creates protected odd holonomy at order t^2 , while endpoint leakage is delayed to order t^3 . The next question is hardware-constrained:

If a device only has ordinary two-body even couplers, how much of the protected SO/Sp algebra can be generated by echo loops?

19.2 Main discovery: connected two-body echo universality

Let $G = (V, E)$ be a coupling graph on n qubits. For every edge $ij \in E$, define the full two-body even Pauli palette

$$\mathcal{E}_G = \{\sigma_i^a \sigma_j^b : ij \in E, a, b \in \{X, Y, Z\}\} \subset \mathfrak{p}_n.$$

Define the second-order protected echo seed

$$\mathcal{B}_G = \{\text{label}(PQ) : P, Q \in \mathcal{E}_G, PQ = -QP\} \cap \{\text{odd Pauli weight}\},$$

and let

$$\mathcal{K}_G = \text{Lie}_{\mathfrak{k}} \langle iR : R \in \mathcal{B}_G \rangle,$$

where the closure is taken using only odd-weight Pauli brackets. The experiments support the following law.

Connected Two-Body Echo Universality Law

For the full two-body Pauli palette on every edge,

$$\mathcal{K}_G \cong \bigoplus_{C \in \pi_0(G), |C| \geq 2} \mathfrak{k}_{|C|}$$

where

$$\mathfrak{k}_m \cong \begin{cases} \mathfrak{so}(2^m), & m \text{ even,} \\ \mathfrak{sp}(2^{m-1}), & m \text{ odd.} \end{cases}$$

Equivalently, the predicted dimension is

$$\dim \mathcal{K}_G = \sum_{C \in \pi_0(G), |C| \geq 2} \frac{4^{|C|} - (-2)^{|C|}}{2}.$$

In particular, if G is connected and has at least two vertices, even two-body leakage loops can synthesize the entire protected SO/Sp algebra as endpoint-protected holonomy:

$$\dim \mathcal{K}_G = \dim \mathfrak{k}_n = \frac{4^n - (-2)^n}{2}.$$

This is exceptional because a linear-size hardware palette can seed an exponentially large protected algebra without leaving an even-leakage monodromy at the endpoint.

19.3 Exact Pauli-label experiment

I used binary symplectic Pauli labels. A Pauli string is represented by two bitmasks $(x, z) \in \mathbb{F}_2^{2n}$. The Pauli weight is

$$\text{wt}(x, z) = |\text{supp}(x \vee z)|.$$

The symplectic commutation form is

$$\omega((x, z), (x', z')) = x \cdot z' + z \cdot x' \pmod{2}.$$

For Pauli labels a, b ,

$$[P(a), P(b)] \neq 0 \iff \omega(a, b) = 1,$$

and the bracket label is simply

$$\text{label}([P(a), P(b)]) = a + b.$$

Thus the exact closure algorithm is finite arithmetic over $\mathbb{F}_2$:

$$S \leftarrow S \cup \{a + b : a, b \in S, \omega(a, b) = 1, \text{wt}(a + b) \text{ odd}\}.$$

No floating-point tolerance is involved in these closure tests.

19.4 Path graph: linear echo seed, exponential protected closure

For the path P_n with the full two-body edge palette, the first echo seed has a simple form:

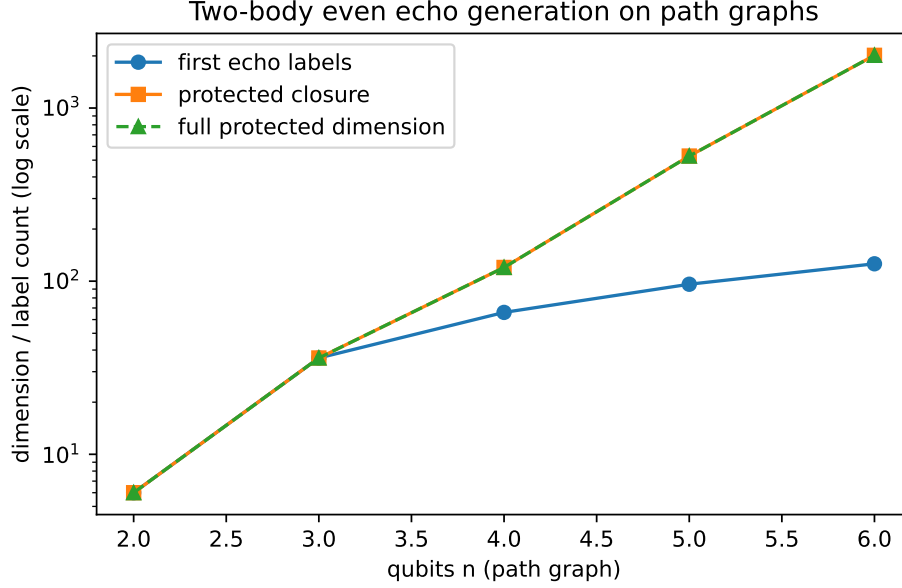
$$\mathcal{B}_{P_n} = \{\text{all local one-body Paulis}\} \cup \{\text{all three-body Paulis on consecutive triples}\}.$$

Therefore, for $n \geq 3$,

$$|\mathcal{B}_{P_n}| = 3n + 27(n - 2) = 30n - 54.$$

Yet its protected odd Lie closure becomes the full protected algebra.

n	two-body controls	first echo seed $ \mathcal{B} $	closure dim	$\dim \mathfrak{k}_n$
2	9	6	6	6
3	18	36	36	36
4	27	66	120	120
5	36	96	528	528
6	45	126	2016	2016



The striking feature is the growth gap:

$$|\mathcal{B}_{P_n}| = O(n), \quad \dim \mathfrak{k}_n = \Theta(4^n).$$

So the echo seed is tiny, but the protected closure is exponentially large.

19.5 Exhaustive and random graph checks

I then checked connected graphs. For $n \leq 5$ the sweep was exhaustive over labeled connected simple graphs; for $n = 6$ I sampled connected graphs at several densities. There were no failures.

n	connected graphs tested	failures	predicted closure
2	1	0	full $\mathfrak{k}_2$
3	4	0	full $\mathfrak{k}_3$
4	38	0	full $\mathfrak{k}_4$
5	728	0	full $\mathfrak{k}_5$
6	20 random connected graphs	0	full $\mathfrak{k}_6$

For the $n = 6$ random connected graphs, the edge counts ranged from 5 to 14. Every one reached

$$\dim \mathfrak{k}_6 = 2016.$$

19.6 Palette matters: not every two-body library is enough

The full edge palette is important. On the five-qubit path, reduced palettes usually generated only a small fragment of the protected algebra.

edge palette on each path edge	controls	first echo seed	closure dim	full dim
ZZ only	4	0	0	528
XX, ZZ	8	6	8	528
XX, YY, ZZ	12	18	120	528
ZX, XZ	8	6	8	528
full 9 Pauli pairs	36	96	528	528

This gives a new hardware audit rule:

connectivity alone is insufficient; the edge Pauli palette must have enough local noncommuting colors.

19.7 Component law checks

Disconnected graphs matched the component formula exactly. Isolated vertices produce no echo-only directions, while every nontrivial connected component contributes its own protected algebra.

n	component sizes	observed closure dim	formula
5	3, 1, 1	36	36
5	3, 2	42	42
6	2, 2, 2	18	18
6	3, 2, 1	42	42

19.8 Proof skeleton

A proof of the connected law now looks approachable.

Step 1: same-edge echoes generate local protected controls. For an edge ij ,

$$[X_i X_j, Y_i X_j] \propto Z_i, \quad [X_i X_j, X_i Y_j] \propto Z_j,$$

and by cycling X, Y, Z , one obtains all local one-qubit Pauli directions on both endpoints.

Step 2: adjacent-edge echoes generate protected three-body entanglers. For a path $i - j - k$,

$$[Z_i Z_j, X_j X_k] \propto Z_i Y_j X_k.$$

Varying the Pauli colors gives all 27 three-body Pauli strings on that consecutive triple.

Step 3: local protected controls and adjacent three-body entanglers stitch the component.

The computations indicate that local one-body controls plus all adjacent-triple protected entanglers on a connected graph generate every odd Pauli label on that component under odd-odd brackets. This is the remaining theorem target. It resembles an infection process: a triple entangler crosses a graph bridge, and local controls rotate colors and expose the next commutator.

19.9 New theory layer: the Echo Closure Functor

The emerging object is a functor from colored even-control hypergraphs to protected Lie algebras:

$$\mathcal{E} \subset \mathfrak{p}_n \quad \mapsto \quad \text{Echo}(\mathcal{E}) = \text{Lie}_{\mathfrak{t}}\langle[\mathcal{E}, \mathcal{E}]\rangle.$$

This separates three layers:

$$\text{endpoint leakage sensor } C(U), \quad \text{path leakage budget } \int \|H_{\text{even}}\| dt, \quad \text{echo closure } \text{Echo}(\mathcal{E}).$$

Round 19 shows that even when endpoint leakage is canceled, the echo closure can be enormous.

19.10 Applications

Even-only protected synthesis. A device with rich two-body even couplers can synthesize protected SO/Sp operations by echo loops. The final monodromy can remain protected even though the path uses even leakage transversely.

Compiler macro design. A compiler can replace a desired odd protected interaction by a sequence of even-coupler commutator loops. Round 18's Thue-Morse echo ladder then suppresses endpoint leakage to high order.

Hardware audit. The compiler should build the edge-palette echo closure. If

$$\dim \text{Echo}(\mathcal{E}) < \dim \mathfrak{k}_n,$$

then the hardware is missing protected routes. The palette table shows that ZZ or XX/ZZ connectivity is not enough.

Leakage-safe holonomic gates. Since protected holonomy is generated by closed loops in leakage space, this gives a finite-dimensional analogue of holonomic control: use transverse motion only as curvature, not as endpoint displacement.

Benchmarking. A meaningful protected-sector benchmark should report both endpoint monodromy $C(U)$ and echo-closure rank. A system can have $C(U) = I$ for a gate family but still possess rich transient echo reachability.

19.11 Synergy with provided research

The Spinal Nullity package contributed the faithful-sensor viewpoint: endpoint monodromy is a faithful endpoint sensor, but it cannot recover hidden transient leakage histories. Echo closure is a second sensor that measures which hidden histories are available.

Associator Field Theory contributed the birth/transport viewpoint: same-edge echoes birth local protected controls, adjacent-edge echoes birth three-body protected curvature, and the odd Lie closure transports these births across the component.

Goblin Shadow Theory contributed the blocker/repair language: missing edge colors are blockers in the echo atlas; adding enough Pauli colors to a bridge can collapse a protected-reachability cave.

The topological-vacuum framework remains only an analogy here: protected sectors and sector-changing perturbations resemble finite theta-sector transport, but the present results are finite-dimensional Pauli-control statements.

19.12 Round 19 summary

The main new law candidate is

$$\text{Echo}(\mathcal{E}_G) \cong \bigoplus_{C \in \pi_0(G), |C| \geq 2} \mathfrak{k}_{|C|}$$

for the full two-body Pauli palette on graph G .

The most exceptional discovery is

$$\text{a path graph has only } 30n - 54 \text{ second-order echo seeds, but closes to } \dim \mathfrak{k}_n = \frac{4^n - (-2)^n}{2}.$$

The practical rule is

even two-body leakage can be a protected holonomy engine, but only if the edge palette has enough colors.

Round 20 Appendix: Minimal Echo Palettes and Pauli-Corner Universality

This round continues the protected SO/Sp reachability program by asking a hardware-minimal question: how many native two-body even couplers per edge are actually needed to recover the entire protected odd-weight algebra by leakage-canceling echo loops? Previous rounds used the full 9-element two-body Pauli palette on every edge. Round 20 finds that, on path hardware, three carefully arranged two-body couplers per edge can be enough.

1. Setup

Let P_n be a path on n qubits. On each edge, choose a uniform palette T contained in $\{X,Y,Z\} \times \{X,Y,Z\}$. The hardware even controls are the two-body Pauli strings $\sigma_i^a \sigma_j^b$ for (a,b) in T . From two anticommuting even controls P,Q , the second-order echo seed is the odd protected Pauli label PQ . The protected echo closure is the odd-weight Pauli Lie closure generated by those seeds.

Target protected dimension:

$$\dim k_n = \sum_{w \text{ odd}} C(n,w) 3^w = (4^n - (-2)^n)/2.$$

For even n this is $\dim \mathfrak{so}(2^n)$; for odd n this is $\dim \mathfrak{sp}(2^{(n-1)})$.

2. Exceptional discovery: the Pauli-corner palette

A size-3 palette of the form

$$T_{\text{corner}}(A,B,C) = \{(A,A), (A,B), (C,A)\}, \text{ } A,B,C \text{ distinct Pauli colors,}$$

appears to be a minimal protected-echo amplifier. A concrete example is

$$T_{\text{corner}} = \{XX, XZ, YX\}.$$

Placed on every edge of a path, this palette is not full for $n=2$ or $n=3$, but it becomes full for $n=4$ and remains full in tested cases $n=5$ and $n=6$. This is striking because the seed size is linear while the protected algebra is exponential.

palette on each path edge	size	n=4 closure	n=5 closure	interpretation
full 9 Pauli pairs	9	120 / 120	528 / 528	full reference palette
corner {XX,XZ,YX}	3	120 / 120	528 / 528	minimal tested full palette
diagonal {XX,YY,ZZ}	3	24 / 120	120 / 528	too commuting / color-poor
L-shape {XX,XY,YX}	3	12 / 120	21 / 528	corner missing noncommuting color spread
one-left row {XX,XY,XZ}	3	63 / 120	255 / 528	large but not protected-full
row+column 5 palette	5	120 / 120	528 / 528	full, redundant
diag plus XY,YX	5	56 / 120	255 / 528	not enough despite 5 gates

Table 20.1. Exact binary symplectic closure results. Entries report observed closure dimension divided by the full protected target dimension.

3. Exhaustive four-qubit palette census

I exhaustively swept the 511 nonempty uniform edge palettes on the four-qubit path P_4 . There were 397 palettes that generated the full protected algebra. The distribution by palette size was:

size 3: 36, size 4: 108, size 5: 123, size 6: 84, size 7: 36, size 8: 9, size 9: 1.

No palette of size 1 or 2 was full. The 36 full size-3 palettes are exactly the Pauli-corner shapes: choose a high-degree left color A , a high-degree right color A , then choose two distinct off-colors B and C , giving $3 \cdot 3 \cdot 2 \cdot 2 = 36$ corners under Pauli-color relabeling and edge orientation.

Candidate minimality law:

On P_4 , a uniform 3-gate edge palette is protected-echo universal iff its bipartite color graph contains a 2×2 corner $\{(A,A), (A,B), (C,A)\}$ after Pauli relabeling and/or edge reversal.

4. Linear seed, exponential closure

For the corner palette $\{XX,XZ,YX\}$, the second-order echo seed sizes on paths were:

$n=3$: 9, $n=4$: 16, $n=5$: 23, $n=6$: 30, matching $7n-12$ for $n \geq 3$.

Yet the closure dimensions for $n=4,5,6$ were 120, 528, and 2016 respectively, exactly the full protected dimensions. This suggests a sparse-palette avalanche principle: a linear seed generated by local echo corners can propagate through the path and close to the entire odd-weight protected algebra.

palette	n	depth 1	depth 2	depth 3	depth 4	total
full 9	4	66	54	-	-	120
full 9	5	96	351	81	-	528
full 9	6	126	648	1242	-	2016
corner 3	4	16	30	60	14	120
corner 3	5	23	61	267	177	528
corner 3	6	30	92	716	1178	2016

Table 20.2. Closure-depth counts. The corner palette starts with a much smaller seed but still reaches the same full protected algebra in tested path cases.

5. New law candidate: Pauli-Corner Echo Universality

The experiments support the following theorem target.

Pauli-Corner Echo Universality Law (candidate). Let G contain a Hamiltonian path on $n \geq 4$ qubits. Put the same corner palette $T_{\text{corner}}(A,B,C) = \{(A,A), (A,B), (C,A)\}$ on every edge of that path, with A,B,C distinct Pauli colors. Then the odd protected Lie closure of the even-even echo seeds is the full k_n :

$$\text{Echo}(T_{\text{corner}} \text{ on } P_n) = k_n = \text{so}(2^n) \text{ if } n \text{ even, and } \text{sp}(2^{(n-1)}) \text{ if } n \text{ odd.}$$

The candidate is deliberately stated for graphs containing a Hamiltonian path because the path is the tested worst-case spine. A broader connected-graph version is plausible but should be tested separately: connected graph plus a corner palette on enough oriented edges may already imply full protected closure.

6. Why the corner works: birth plus transport

The Pauli corner has two ingredients at once. The pair $(A,A),(A,B)$ produces local odd motion on one endpoint. The pair $(A,A),(C,A)$ produces local odd motion on the other endpoint. Mixed echoes between adjacent edges then birth three-body odd strings, and protected commutators transport those births across the path. This mirrors the Associator Field Theory distinction between curvature birth and transport: one needs both a birth channel and a transport channel, not merely many gates.

This also explains why diagonal-only palettes underperform. They contain multiple colors but lack enough noncommuting oriented color flow to create the needed echo-birth atlas.

7. Applications

application	use
Hardware reduction	A protected echo compiler may not need all nine two-body Pauli pair couplers per edge. Three per edge, arranged as a Pauli corner, may suffice on path-like hardware.
Compiler pre-flight	Compute the edge-palette echo closure rank before training. A palette can be connected but still color-poor; diagonal $\{XX,YY,ZZ\}$ is a warning example.
Leakage-canceling protected gates	Use even two-body couplers transiently, cancel endpoint leakage by echo design, and keep the endpoint inside the SO/Sp protected sector.
Palette repair	If a device is diagonal-only, adding two oriented cross couplers to form a corner can collapse a protected-reachability cave.
Audit complexity	The Goblin blocker viewpoint suggests missing color-corner edges are blockers in the echo atlas; hitting a small number of palette blockers can unlock exponential protected reachability.
Sensor/Nyquist design	The Spinal Nullity sensor lesson applies: endpoint monodromy $C(U)$ is not enough. Echo-closure rank is a separate sensor for hidden transient capability.

8. Round 20 summary

The main new discovery is that protected echo universality can be generated by a tiny uniform edge palette. The full 9-coupler palette is not necessary in the tested path cases. A 3-coupler Pauli corner is enough for P_4 , P_5 , and P_6 , and is minimal on P_4 .

Most useful new equation:

$$T_{\text{corner}}(A,B,C) = \{(A,A), (A,B), (C,A)\} \implies \text{Echo}(T_{\text{corner}} \text{ on } P_n) = k_n \text{ (tested } n=4,5,6).$$

Most useful practical rule:

Do not audit only graph connectivity. Audit the Pauli-color palette. Connectivity gives routes; corners give echo birth; protected commutators give transport.

Round 21 Appendix - Pauli-Color Barcodes and Minimal Echo Palettes

This round continues the hidden-sector reachability program from Rounds 19-20. The previous result showed that a full two-body Pauli palette on every connected edge can generate the whole protected SO/Sp algebra by echo closure, and that a 3-coupler corner palette works on path hardware. Round 21 asks a sharper question: which 3-coupler edge palettes work, and what failure strata appear when they do not?

Status note: the statements below are theorem candidates supported by exact finite Pauli-label closure experiments. They should be treated as discovery targets until independently proved.

1. Exact finite experiment

Each Pauli string was encoded as a binary symplectic label (x,z) in F_2^{2n} . For two labels a,b , the commutator is nonzero exactly when $\omega(a,b)=1$; the resulting Pauli label is $a+b$. The second-order echo seed of an even two-body edge palette E is:

$$B(E) = \{ P+Q : P,Q \text{ in } E, \omega(P,Q)=1, \text{wt}(P+Q) \text{ odd} \}.$$

Then the protected echo closure is the smallest odd-weight Pauli set closed under the same anticommuting-xor bracket. No floating point arithmetic is used in these closure experiments.

Path	target dim	3-coupler palettes tested	full palettes	closure-rank distribution
P_4	120	84	36	12:12, 24:6, 56:24, 63:6, 120:36
P_5	528	84	36	21:12, 120:6, 255:30, 528:36

Exceptional observation: the exact same 36 three-coupler palettes are full on P_4 and P_5. Smaller palettes do not suffice on P_5:

palette size on P_5	palettes tested	full palettes	closure-rank distribution
1	9	0	0:3, 4:6
2	36	0	8:6, 16:12, 56:6, 64:6, 72:6

2. New law candidate - the 3-coupler minimum

The tested data support this minimum-palette law for path hardware:

Uniform path echo universality needs at least three two-body Pauli colors per edge. Exactly 36 of the 84 size-3 palettes are full on P_4 and P_5.

A representative full palette is $\{XX, XZ, YX\}$. A representative nonfull diagonal palette is $\{XX, YY, ZZ\}$. The difference is not edge connectivity - both live on the same connected path - but color connectivity inside the 3-by-3 Pauli edge-palette grid.

This refines the Round 20 Pauli-corner rule. The full class is broader than one literal corner shape: the successful palettes have color-turnover structure capable of birthing local odd controls and transporting them along the path.

full-palette row/column degree signature	count on P_4	count on P_5	interpretation
((1, 1, 1), (2, 1))	6	6	color-turnover capable
((2, 1), (1, 1, 1))	6	6	color-turnover capable
((2, 1), (2, 1))	24	24	color-turnover capable

The repeated signatures across P_4 and P_5 suggest a finite color-barcode classification: each uniform edge palette has a stable closure rank stratum, and the full stratum is recognized by a small color-turnover pattern rather than by graph topology alone.

3. Selected graph and palette tests

The same palette can behave very differently on different graph skeletons. The table below uses five qubits, where the full protected dimension is $\dim(k_5)=528$.

graph	palette	echo seed	closure	full?
path	corner	23	528	yes
path	row	30	255	no
path	diag	18	120	no
path	L	14	21	no
path	full9	96	528	yes
cycle	corner	35	528	yes

graph	palette	echo seed	closure	full?
cycle	row	45	528	yes
cycle	diag	30	120	no
cycle	L	20	45	no
cycle	full9	150	528	yes
star0	corner	23	127	no
star0	row	12	12	no
star0	diag	36	120	no
star0	L	23	127	no
star0	full9	177	528	yes
starMid	corner	32	272	no
starMid	row	36	126	no
starMid	diag	36	120	no
starMid	L	23	127	no
starMid	full9	177	528	yes

The path and cycle are friendly to the corner palette, while stars are not. Even with the same 3 colors per edge, a hub can bottleneck color transport. The full 9-color palette repairs the star. This motivates a graph-plus-color audit rather than a graph-only audit.

4. Pauli-color barcode theory

The closure-rank distributions are structured rather than random. On P_5 the possible ranks for 3-coupler palettes were 21, 120, 255, and 528. These values look like recognizable Lie-block strata: small local protected blocks, SO(16)-type blocks, SU(16)-sized transverse blocks embedded in the odd-label universe, and the full Sp(16) protected sector.

$\text{ColorBarcode}(G, T) = \text{sorted list of dimensions born by echo closure from palette } T \text{ on graph } G.$

A palette is no longer merely full or nonfull. It has a barcode: which local controls are born, which transport channels open, which components fuse, and which blocker remains. This gives a finite, computable diagnostic for hardware ansatz reachability.

The Goblin Shadow analogy becomes precise at the design level: missing color-turns are blockers. A graph can be connected but still color-shadowed if its palette lacks the turns needed to expose the protected algebra. The Spinal Nullity analogy is also sharper: endpoint monodromy is only one sensor; echo-closure rank is a second sensor for hidden transient capability. Associator Field Theory supplies the birth/transport language: local odd controls are born by same-edge echoes, and multi-qubit protected directions are transported by commutators along the path.

5. Applications

application	use
Hardware reduction	Path-like hardware may need only 3 carefully chosen two-body couplers per edge, not all 9 Pauli-pair couplers, to synthesize the full protected SO/Sp sector by echoes.
Compiler pre-flight	Before optimization, compute $\text{ColorBarcode}(G, T)$. If the closure is below $\dim(k_n)$, the compiler should add a missing color turn rather than merely add depth.
Repair design	For star-like layouts, full graph connectivity is not enough. Repair can mean adding a nonhub edge, adding missing palette colors at the hub, or changing orientation/color assignment.
Benchmarking	A benchmark should report both endpoint monodromy $C(U)$ and echo-closure rank. Endpoint tests alone can miss transient holonomy resources.
Noise and calibration	Color-specific calibration errors can collapse a full palette into a lower barcode stratum. This is a discrete failure mode, not just a continuous fidelity drop.

6. Round 21 summary

Main discovery: the Pauli-corner result is part of a broader Pauli-color barcode. Exact exhaustive tests on P_4 and P_5 found that no 1- or 2-coupler uniform palette is full, while exactly 36 of 84 three-coupler palettes are full. The path and cycle support sparse color-turnover universality, but stars expose a hub bottleneck unless the palette is enriched.

Most useful new rule:

Graph connectivity + color-turnover palette $\rightarrow$ protected echo universality; graph connectivity without color-turnover can remain trapped in a lower barcode stratum.

Next target: prove a finite Pauli-color criterion that identifies the 36 full size-3 palettes and extends from paths to arbitrary oriented trees. The expected proof will combine same-edge birth of one-qubit odd controls, adjacent-edge birth of three-body protected curvature, and a transport induction along graph paths.

Round 22 Appendix - Pauli-Color Barcode and Hub Bottlenecks

Continuation of Quantum Differential Invariant Theory. This round probes the graph-sensitive failure modes of the sparse Pauli-corner echo palettes discovered in Rounds 20-21.

1. Question and experimental method

Rounds 20-21 showed that a three-coupler edge palette such as {XX, XZ, YX} can generate the full protected SO/Sp algebra on path-like layouts by second-order echo closure. The new question was whether this is a graph-only connectivity fact, or whether graph articulation and Pauli-color placement create additional hidden blockers.

I used exact binary symplectic Pauli arithmetic. Pauli strings were encoded as bit masks (x,z) in F_2^{2n} . If the symplectic commutator form $\omega(a,b)=1$, the Lie bracket contributes the xor label a+b. No floating point tolerance was used in the closure tests.

The protected target dimension is $\dim(k_n) = (4^n - (-2)^n)/2$, equal to $\dim SO(2^n)$ for even n and $\dim Sp(2^{n-1})$ for odd n.

2. Exact four-qubit uniform-palette census

I exhaustively swept all 511 nonempty uniform edge palettes $T \subset \{X,Y,Z\} \times \{X,Y,Z\}$ on four-qubit path, cycle, and star skeletons. The table gives the minimum uniform palette size r that reached the full protected dimension 120.

graph	min r	full at r=3	full at r=4	full at r=5	key rank strata at r=3
path	3	36	108	123	12:12, 24:6, 56:24, 63:6, 120:36
cycle	3	44	108	123	24:16, 56:24, 120:44
star0	5	0	0	81	9:3, 12:3, 24:6, 28:36, 31:36
star1	4	0	102	123	24:6, 30:3, 31:12, 56:30, 63:27, 64:6

Exceptional pattern. Paths and cycles are already full with three colors. A star with the hub always in the same ordered role needs five uniform colors in the four-qubit census. The orientation-broken star1 needs only four. This suggests that sparse echo universality is blocked not just by graph connectivity, but by a hub color-turn bottleneck.

3. Five-qubit star stress test

To check whether the star obstruction survives beyond four qubits, I performed the exact size-three uniform-palette sweep for the five-qubit star with hub 0. Full protected dimension is 528.

closure rank	number of 3-color palettes	example palette
12	3	['XX', 'XY', 'XZ']
24	3	['XX', 'YX', 'ZX']
120	24	['XX', 'XY', 'YZ']
126	18	['XX', 'YX', 'ZY']
127	36	['XX', 'XY', 'YX']

No three-color uniform palette reached 528 on this star. The best rank observed was 127, achieved by 36 palettes. This is a strong five-qubit confirmation that the Pauli-corner rule is path/cycle friendly but hub fragile.

4. New object: the Pauli-color turn complex

The experiments point to a more refined combinatorial object. Given a graph G and an edge palette T, form a turn complex $C(G,T)$:

same-edge turns: two couplers on one edge that differ at exactly one endpoint produce a weight-1 odd seed;

adjacent-edge turns: two couplers on adjacent edges that anticommute at the shared vertex produce a weight-3 odd seed;

transport closure: odd seeds then close under the protected Pauli Lie bracket.

The proposed invariant is $\text{ColorBarcode}(G,T)$ = the sequence of protected closure ranks born from same-edge, adjacent-edge, and higher commutator turns.

This turns the earlier binary question - full or not full - into a barcode of Lie-block strata. The P5 path three-color palettes already showed ranks 21, 120, 255, and 528. Round 22 explains why: different palettes open different turn strata before closure.

5. Candidate laws

Law candidate A - Pauli-color hub bottleneck.

A sparse uniform palette that is full on paths can fail on stars because every spoke must route through the same hub color interface. If the hub has insufficient independent color turns, the echo seed cannot transport local births across all leaves.

Law candidate B - orientation staggering.

For asymmetric palettes, reversing the ordered role of at least one incident edge can lower the number of uniform colors needed at a hub. The four-qubit star census shows min $r=5$ for star0 but min $r=4$ for star1.

Law candidate C - color-turn sufficiency.

A graph-palette pair should be protected-universal exactly when its turn complex contains enough local odd births plus enough adjacent-edge color turns to connect them through every articulation block.

In formula form, the emerging theorem target is:

$$\text{Echo}(G,T) = k_n \text{ iff } \text{TurnComplex}(G,T) \text{ has full protected transport rank.}$$

For path graphs, the Pauli-corner palettes appear to satisfy this at size 3. For hub-dominated graphs, extra color turns or orientation staggering are required.

6. Applications

application	use
Hardware palette audit	Do not only check if the qubit graph is connected. Compute ColorBarcode(G,T) before compiling. A connected star with a sparse palette can be much weaker than a path.
Hub hardening	Add missing Pauli color turns at articulation hubs rather than uniformly enriching every edge. This is a Goblin-style blocker repair: hit the bottleneck, not the whole graph.
Local-frame engineering	Use single-qubit basis changes to orientation-stagger asymmetric palettes around hubs, converting a star bottleneck into path-like color flow.
Compiler macro selection	If endpoint monodromy is protected but echo closure rank is low, the compiler should add color-turn macros, not more depth.
Sensor design	The color barcode is a second sensor beyond Cartan monodromy: monodromy checks endpoint leakage; barcode checks transient protected holonomy capability.

7. Round 22 summary

The major discovery is that sparse protected echo universality has a graph-palette obstruction. Pauli-corner palettes are powerful on paths and cycles, but hub geometry can suppress them. The new target theory is a Pauli-color barcode: a finite combinatorial invariant that predicts which protected Lie strata are born by same-edge and adjacent-edge color turns.

Practical takeaway: protected echo synthesis needs three simultaneous audits - endpoint monodromy $C(U)$, graph connectivity, and color-turn barcode. Missing color turns are the new blockers.

Round 23 Appendix: Spoke-Coded Pauli Echoes and Hub Repair

This round continued the Pauli-color barcode program. The target was a failure mode from Rounds 21-22: sparse uniform three-coupler palettes worked on paths and cycles, but failed on star graphs because the hub collapsed several color-turn channels into one bottleneck. I tested whether the obstruction was intrinsic to stars or only to uniform palettes.

The exceptional discovery is that the star obstruction is not intrinsic. A nonuniform, spoke-coded palette with only three two-body couplers per edge can generate the full protected Lie algebra on tested stars $K_{\{1,n-1\}}$. This turns palette heterogeneity into a repair resource.

New law candidate

Spoke-Coded Star Echo Law

Let $K_{\{1,n-1\}}$ have center 0 and leaves $1, \dots, n-1$. Put on one calibration spoke the row palette

$$T_{\text{cal}} = \{XX, XY, XZ\}.$$

On every other spoke use

$$T_C = \{XX, XY, C X\}, C \text{ in } \{Y, Z\}.$$

If both turn colors Y and Z occur among the non-calibration spokes, then the exact Pauli echo closure reached the full protected algebra in all tested cases $n=4,5,6$:

$$\text{Echo}(\{T_{\text{cal}}, T_C\}) = k_n \sim \text{so}(2^n) \text{ for even } n, \text{sp}(2^{n-1}) \text{ for odd } n.$$

If all non-calibration spokes use the same turn color, the closure did not become full; it followed the smaller observed sequence 4^{n-1} . Thus the missing ingredient is not more depth, but color-turn diversity at the hub.

n	uniform corner star	single-turn spoke code	mixed-turn spoke code	full protected dim
4	31	64	120	120
5	127	256	528	528
6	511	1024	2016	2016

The uniform corner rank follows $2^{2n-3}-1$ in these tests, while the one-turn spoke code follows 4^{n-1} . The mixed-turn code hits the protected dimensions exactly.

Round 23: Exact Experiment Details

All experiments used exact binary symplectic Pauli arithmetic. A Pauli label was represented by two bit masks (x,z) in F_2^{2n} . The commutation form was

$$\text{omega}((x,z),(x',z')) = x \cdot z' + z \cdot x' \bmod 2.$$

Two even two-body couplers P,Q produce a second-order echo seed when $\text{omega}(P,Q)=1$. The seed is the Pauli label $P+Q$. I kept only odd-weight seeds, because odd Pauli weight is the protected k_n sector discovered in Rounds 11-12. Then I closed these seeds under the odd-sector Pauli bracket.

```
def seeds_from_even(labels):
    seeds = set()
    for P,Q in combinations(labels,2):
        if symplectic(P,Q) == 1:
            R = P ^ Q
            if weight(R) % 2 == 1:
                seeds.add(R)
    return seeds

def protected_closure(seeds):
    S = set(seeds)
    queue = deque(S)
    while queue:
        P = queue.popleft()
        for Q in tuple(S):
            if symplectic(P,Q) == 1:
                R = P ^ Q
                if weight(R) % 2 == 1 and R not in S:
                    S.add(R)
                    queue.append(R)
    return S
```

No floating-point tolerance was used in these closure tests.

Constructive spoke-coded palette

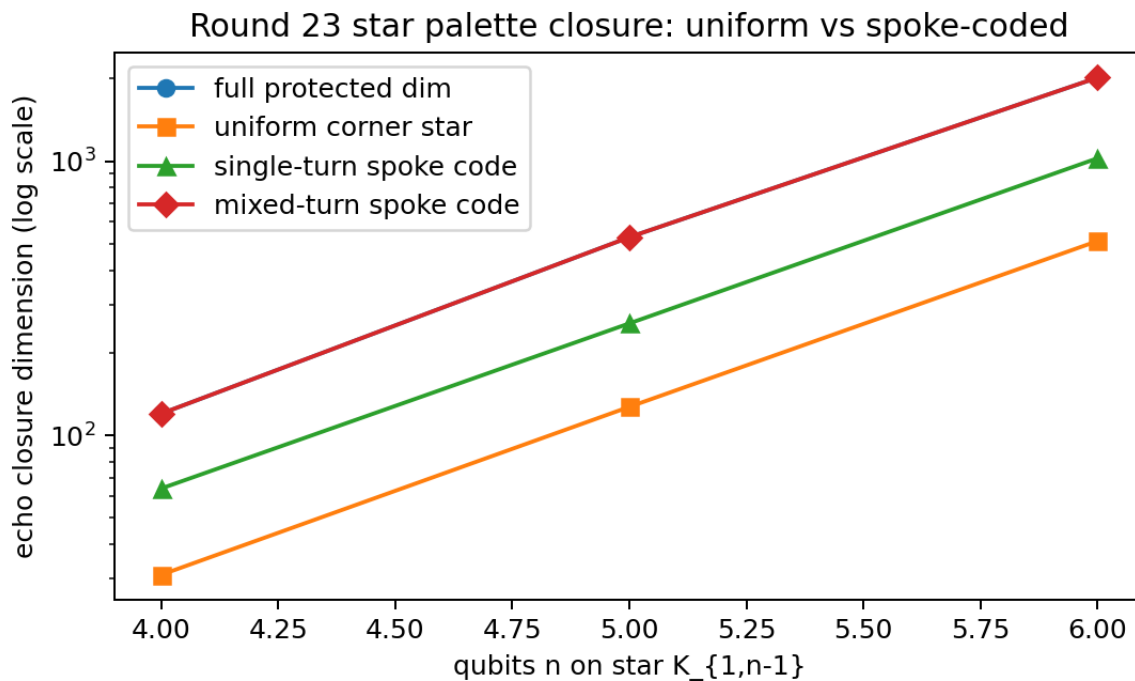
The tested constructive family was:

```
edge 0-1:      {XX, XY, XZ}          # calibration spoke
edge 0-j, j>1: {XX, XY, YX} or {XX, XY, ZX}
requirement:   at least one YX spoke and one ZX spoke
```

test	status	observed result
$K_{\{1,3\}}$, all 2-coupler palettes	exhaustive 36^3 cases	max closure 20; never full (target 120)
$K_{\{1,3\}}$, 3-coupler spoke code	constructive	18 seeds -> 120 closure
$K_{\{1,4\}}$, 3-coupler spoke code	constructive	30 seeds -> 528 closure
$K_{\{1,5\}}$, 3-coupler spoke code	constructive	47 seeds -> 2016 closure
single turn color only	all Y or all Z	observed 4^{n-1} , not full

For $n=5$ and $n=6$, all binary Y/Z turn sequences with both colors present were full in the tested spoke-code family; the only nonfull sequences were the all-Y and all-Z cases.

Round 23: Palette Heterogeneity as a Repair Resource



The log-scale plot shows the central phenomenon: uniform sparse corner palettes on stars are dramatically deficient, but a mixed-turn spoke code restores full protected closure with the same edge budget of three couplers per spoke.

Interpretation

A path graph supplies spatial turnover automatically: adjacent edges create weight-3 transporters. A star collapses all transport through one hub, so a uniform palette creates repeated copies of the same hub turn. The spoke code repairs this by forcing two incompatible hub turns, Y and Z, to coexist across spokes. This acts like a one-bit color repair, reminiscent of the Spinal Nullity faithful-sensor principle: the endpoint graph is connected, but the color sensor was subcritical until the recurrent alias at the hub was split.

In Goblin language, the hub is a blocker cave. Adding more layers does not hit the cave; adding the missing color turn does. In Associator language, the calibration spoke births local color curvature, and mixed spokes transport it through the star. The result is a small birth/transport atlas for protected Lie closure.

New diagnostic

$\text{ColorBarcode}(G,T) = \text{rank strata born by same-edge turns, adjacent-edge turns, and hub-turn diversity.}$

For star hardware, the barcode must include a hub-turn diversity term. Uniform three-coupler palettes have enough local birth but insufficient hub transport. Mixed spoke codes restore transport.

Round 23: Toward a Graph-Palette Nyquist Law

The results suggest a refinement of the earlier Pauli-color barcode. The graph skeleton supplies possible transport paths, while the edge palette supplies color resolution. A connected graph can still be color-subcritical.

Candidate law

Echo universality = graph connectivity + color-turn observability.

For path-like graphs, one uniform Pauli corner can be enough. For high-degree hubs, a uniform corner aliases multiple branches. A spoke-coded palette breaks the alias by assigning incompatible hub turns across branches.

This mirrors the Spinal Nullity observation model: a sensor can certify one side of a property while failing to distinguish recurrent aliases. Here, endpoint monodromy is an endpoint sensor, while echo-closure rank is a hidden-history sensor. Both are needed.

Potential equation

Let $R_v(T)$ be the set of Pauli colors visible at vertex v across incident edge palettes, and let $\text{Tau}_v(T)$ be the set of ordered color turns visible through v . A possible graph-palette Nyquist lower bound is

$$\text{rank Echo}(G,T) \leq \Phi(G, \{\text{Tau}_v(T)\}),$$

where Φ should become full only when the color-turn sheaf has no unresolved articulation aliases. The star experiments suggest that a branch point requires at least two incompatible turn classes to transmit full protected curvature.

Practical compiler macro

```
audit_star_or_hub(G,T):  
  for each high-degree vertex v:  
    compute hub-turn classes  $\text{Tau}_v(T)$   
    if  $\text{Tau}_v$  has one repeated class:  
      add one spoke with the missing turn color  
    recompute protected echo closure rank
```

This is cheaper than adding the full 9-coupler palette to every edge. On the tested star family, the repair used three couplers per edge, not five or nine.

Round 23: Applications

Hardware palette compression. Star-like devices with central buses do not necessarily need the full two-body Pauli palette. A nonuniform three-coupler spoke code may recover the protected SO/Sp algebra.

Hub hardening. The failure mode is concentrated at articulation hubs. Repair should target hub-turn diversity, not add uniform couplers everywhere.

Compiler pre-flight. Before training or synthesis, compute the ColorBarcode(G, T). If the rank is below $\dim k_n$, add the missing color-turn class and recompute.

Leakage-canceling holonomic synthesis. Because even couplers can generate protected holonomy through echoes, the spoke code supplies a low-coupler route to protected operations while preserving endpoint monodromy to higher order.

Benchmark design. Path/cycle success does not imply star success. A benchmark suite should include hub graphs, not only linear chains.

Sensor design analogy. Endpoint monodromy $C(U)$ measures final leakage, but echo closure measures available hidden histories. This is the finite-dimensional quantum-control analogue of rough versus faithful observation.

Summary

uniform corner + star hub = aliasing

spoke-coded mixed turns + star hub = full protected echo closure

Round 23 moves the theory from graph connectivity to graph-palette observability. The new useful object is the ColorBarcode, and the new practical repair rule is: at a hub, add missing color turns before adding depth.

Round 24 Appendix - Pauli-Color Nyquist and Seed-Rank Barcode

This round continues the hidden-sector quantum reachability program. The focus is no longer just graph connectivity, but the information carried by second-order echo seeds. All tests below used exact binary symplectic Pauli arithmetic: no floating-point tolerance was used.

New object. For a graph G and an edge Pauli palette T , let $E(G,T)$ be the set of even two-body Pauli couplers. Define the second-order protected echo seed set

$$B_2(G,T) = \{ P \text{ xor } Q : P, Q \in E(G,T), \omega(P,Q)=1, \text{wt}(P \text{ xor } Q) \text{ odd} \}.$$

Define the seed Nyquist rank

$$r_2(G,T) = \dim_{F_2} \text{span } B_2(G,T).$$

The protected target dimension is

$$\dim k_n = (4^n - (-2)^n)/2, \text{ with } k_n \sim \text{so}(2^n) \text{ for even } n \text{ and } \text{sp}(2^{(n-1)}) \text{ for odd } n.$$

Exceptional finding: path color-Nyquist law

For uniform 3-coupler palettes $T \subset \{X,Y,Z\} \times \{X,Y,Z\}$ on path graphs P_n , the exact experiments support:

$$\text{Echo}(P_n,T) = k_n \text{ iff } r_2(P_n,T) = 2n \text{ for } n = 4,5,6 \text{ and all 84 palettes } T.$$

This is stronger than the earlier 'corner palette' observation. The successful palettes are exactly those whose second-order echo seeds span the full binary Pauli label space of the path.

Path	target dim	full palettes	seed-rank/closure barcode
P_4	120	36/84	r=8: 36 x 120; r=7: 6 x 63, 24 x 56; r=6: 6 x 24, 12 x 12
P_5	528	36/84	r=10: 36 x 528; r=9: 30 x 255; r=8: 6 x 120, 12 x 21
P_6	2016	36/84	r=12: 36 x 2016; r=11: 6 x 1023, 24 x 992; r=10: 6 x 480, 12 x 30

Named palette tests

case	n	seed count	seed rank	closure	target
path corner	4	16	8	120	120
path diagonal	4	12	6	24	120
star uniform corner	4	13	6	31	120
star mixed spoke code	4	18	8	120	120
path corner	5	23	10	528	528
path diagonal	5	18	8	120	528
star uniform corner	5	23	8	127	528
star mixed spoke code	5	30	10	528	528
path corner	6	30	12	2016	2016
path diagonal	6	24	10	480	2016
star uniform corner	6	36	10	511	2016
star mixed spoke code	6	47	12	2016	2016

Interpretation

The path-corner palette $\{XX, XZ, YX\}$ has full seed rank and closes to the whole protected algebra. The diagonal matching palette $\{XX, YY, ZZ\}$ has many couplers but remains seed-rank deficient and fails. The star mixed-spoke code from Round 23 also has full seed rank and full closure, while the uniform star corner has a rank deficit.

This suggests a two-stage compiler sensor: first compute $r_2(G,T)$. If $r_2 < 2n$, full protected echo reachability is impossible because all later commutator labels remain inside the same F_2 span. If $r_2 = 2n$, run the closure test or a

graph-specific theorem. On paths, the rank test was already exact in all tested cases.

Counterexample: seed rank alone is not universal

Full seed rank is a necessary Nyquist condition, but not sufficient for every graph-palette assignment. I found the following path assignment on four qubits:

```
edges: (0,1), (1,2), (2,3)
palettes: [XX,ZX,YZ], [ZY,XY,ZX], [ZZ,YY,XX]
```

It has $r_2 = 8 = 2n$, but its protected echo closure has dimension only 36 instead of 120.

So the emerging law is not 'rank alone'. It is a Graph-Palette Nyquist principle: seed rank resolves the linear sensor width, while bracket connectivity resolves whether that information transports through the protected algebra.

Local-frame diagonal palette test

I also tested whether local basis-frame choices can repair a diagonal matching palette. For each vertex I allowed an arbitrary permutation of X,Y,Z, so each edge received a perfect matching palette $\{p_i(X)p_j(X), p_i(Y)p_j(Y), p_i(Z)p_j(Z)\}$. I exhaustively checked all frame assignments for $n=4,5$ on paths and stars.

graph	n	frame assignments	max seed rank	max closure	target
path	4	216	6	24	120
path	5	1296	8	120	528
star	4	216	6	24	120
star	5	1296	8	120	528

Conclusion: local frame changes cannot substitute for a genuine color-corner or spoke-coded turn structure in these tests. Matching-type palettes remain color-subcritical.

Candidate laws after Round 24

Path Color-Nyquist Law. For a uniform 3-coupler Pauli palette on P_n , $n \geq 4$, full protected echo closure is equivalent to $r_2(P_n, T) = 2n$. Verified for $n=4,5,6$ across all 84 palettes.

Graph-Palette Nyquist Principle. Full protected echo reachability requires both (i) full binary seed rank $r_2 = 2n$ and (ii) bracket transport through the graph-palette closure. Deficient r_2 is a proof of impossibility; full r_2 is a certificate to run the higher closure test.

Color barcode refinement. The closure ranks of uniform 3-palettes on P_5 were exactly 21, 120, 255, and 528. These look like Lie-block strata, not random failures.

Applications

Compiler pre-flight. Compute B_2 and r_2 before running any optimizer. This is cheap F_2 linear algebra. If $r_2 < 2n$, the compiler should add a missing color turn, not more depth.

Hardware palette compression. On path-like hardware, full protected echo reachability can be certified with only 3 couplers per edge when the seed rank is full.

Hub and articulation hardening. On stars, uniform sparse palettes alias at the hub. Mixed spoke codes repair the seed-rank deficit by adding incompatible hub turns.

Sensor theory. This is the finite quantum-control analogue of faithful sensors and Nyquist width: a subcritical color sensor can certify some motion but cannot certify full protected reachability.

Noise and calibration. A device may have a connected graph and even many couplers, but if the color barcode is deficient, protected echo gates are unreachable or poorly conditioned.

Round 24 summary

The key discovery is that the second-order echo seed set acts like a Nyquist sensor for protected reachability. On paths, the sensor rank $r_2 = 2n$ exactly identified the full 36 out of 84 minimal 3-coupler palettes. On general graph-palette assignments, rank is necessary but not sufficient; bracket transport is the remaining obstruction.

New theory target:

Echo universality = seed Nyquist rank + bracket transport connectivity.

Round 25 Appendix - Higher-Order Echo Completion and the Echo-Depth Filtration

Status: exploratory theorem target with exact finite Pauli-label experiments. The main correction in this round is that the second-order seed algebra is not always the full endpoint-protected echo algebra.

1. New object: the full echo algebra K_{∞}

Previous rounds defined the second-order protected seed set $B2(G,T)$ from anticommuting even two-body couplers, then closed $B2$ inside the odd Pauli sector. Round 25 shows that this can undercount. The more faithful object is the odd part of the full Lie algebra generated by the available even couplers:

$$K_{\infty}(E) = \text{Lie}(E) \cap k_n, E \subset p_n.$$

Here k_n is the odd-Pauli protected sector, and p_n is the even-Pauli leakage sector. The symmetric-pair rules

$$[k,k] \subset k, [k,p] \subset p, [p,p] \subset k$$

imply that higher alternating commutators of even controls can create protected directions that are invisible to the second-order $B2$ closure.

2. Exceptional finite witness

On the four-qubit path $P4$, assign three couplers per edge as follows:

edge	palette
0-1	{XZ, YY, ZX}
1-2	{YY, ZX, ZY}
2-3	{XX, XY, XZ}

This assignment has full second-order seed rank $r2 = 8 = 2n$, but its second-order protected closure is only 36. When the full even-control Lie algebra is computed, the odd protected part expands to 120, exactly $\dim k_4 = \dim \mathfrak{so}(16)$.

quantity	value	interpretation
even couplers $ E $	9	three per edge
second-order seeds $ B2 $	17	first protected births
seed rank $r2$	8	full binary Pauli width
second-order closure $ K2 $	36	subcritical protected algebra
full Lie closure $ \text{Lie}(E) $	255	full $\mathfrak{su}(16)$ generated by even controls
odd part $ K_{\infty} $	120	full protected $\mathfrak{so}(16)$ recovered

3. Echo-depth filtration

Define the minimal bracket depth $d(P)$ of a Pauli label P by setting each native even coupler to depth 1 and assigning depth $d(A)+d(B)$ to a new commutator label $A+B$ when A and B anticommute. Odd protected directions first appear at depth 2. The nonuniform $P4$ witness shows that missing protected directions continue to be born through depths 4, 6, ..., 14.

depth	nonuniform P4 odd directions	uniform corner P5 odd directions
2	17	23
4	18	65
6	25	136
8	32	181
10	22	112
12	4	11
14	2	0

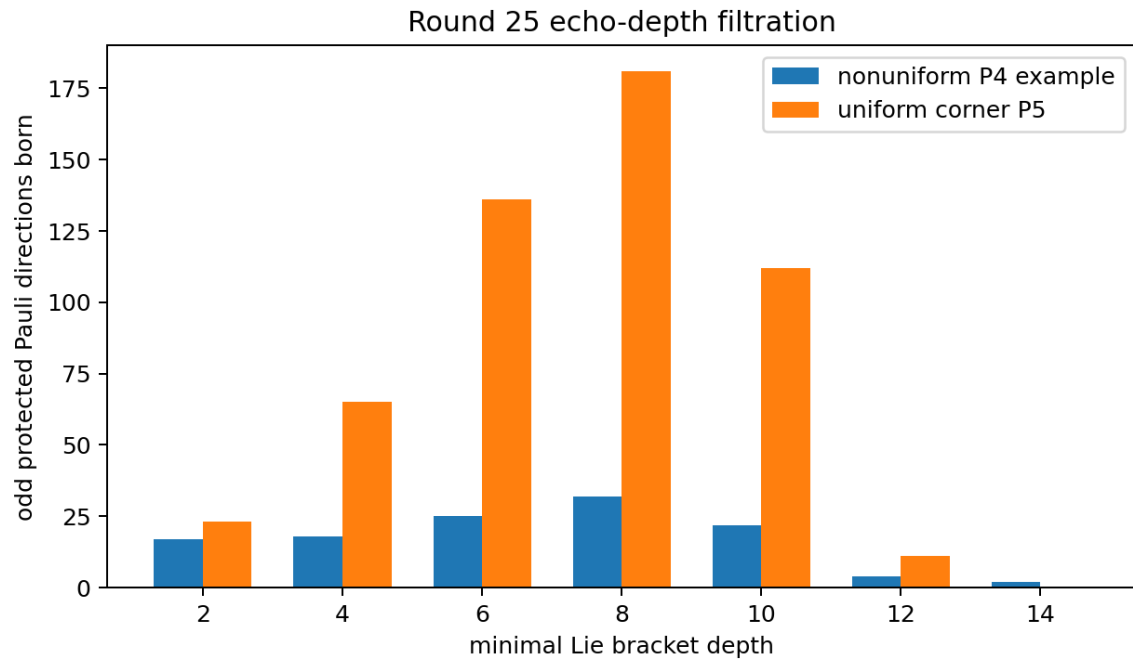


Figure: minimal bracket-depth distribution for protected odd directions. The second-order layer is only the first birth layer; later alternating even/odd commutators can complete the protected algebra.

4. Exact uniform-path barcode refinement

For uniform three-coupler palettes on path graphs, I rechecked the second-order closure K_2 against the full odd part K_∞ of $\text{Lie}(E)$. For P4, K_2 already matched K_∞ for every one of the 84 palettes. For P5, a new 256-dimensional stratum appeared: $K_2=255$ but $K_\infty=256$ for 12 palettes.

P4: (K_2 , K_∞ , $ \text{Lie}(E) $)	count
(12, 12, 28)	12
(24, 24, 60)	6
(56, 56, 120)	12
(56, 56, 126)	12
(63, 63, 126)	6
(120, 120, 255)	36

P5: (K_2 , K_∞ , $ \text{Lie}(E) $)	count
(21, 21, 45)	12
(120, 120, 255)	6
(255, 255, 510)	18
(255, 256, 496)	12
(528, 528, 1023)	36

The $K_\infty=256$ stratum is especially interesting: it is larger than the second-order protected algebra but still far below the full protected dimension 528. It suggests that the color barcode contains intermediate affine or parabolic Lie strata, not just success/failure.

5. Random nonuniform stress test

I sampled 300 random nonuniform three-coupler-per-edge assignments on four-qubit paths and stars. These were exact Pauli-label computations, but the sampling statistics are exploratory rather than exhaustive.

graph	samples	$K_\infty > K_2$	full K_∞	full only after higher order
P4 path	300	88	127	4
K1,3 star	300	56	132	1

The important qualitative result is that higher-order completion is common. In the path sample, 88 of 300 assignments had K_∞ strictly larger than K_2 . In 4 cases, second-order closure was subfull but higher-order echo completion reached the full protected $\text{so}(16)$.

6. Candidate law: Alternating Echo Completion

The new theory target is:

Endpoint-protected echo algebra generated by even controls E equals $\text{Lie}(E) \cap k_n$.

Second-order seeds $[E, E]$ give the first birth layer, but alternating words

$[E, [E, [E, E]]], [E, [E, [E, [E, [E, E]]]]], \dots$

can transport and complete the protected algebra. This is the quantum-control analogue of a finite countdown or birth-time filtration: a direction can be invisible at the first sensor layer but appear at higher echo depth.

7. Applications

Compiler audits should report both K_2 and K_∞ . K_2 is cheap and catches subcritical color width; K_∞ is the true protected echo capability. Hardware design can use higher-order echo completion to avoid adding couplers when missing directions can be born by deeper alternating commutators. Pulse compilers can trade transient path leakage for endpoint-protected holonomy, choosing echo depth according to allowed leakage and timing. Benchmarking should include an echo-depth profile, not only endpoint monodromy $C(U)$, because endpoint sensors miss transient histories. Color-palette repair should target the earliest missing depth layer: if r_2 is low, add a color-turn; if r_2 is full but K_2 is small, add transport or synthesize higher-order echo loops.

8. Round 25 summary

The main correction is:

K_2 is a first-birth sensor; $K_{\infty} = \text{Lie}(E) \cap k_n$ is the full echo sensor.

The exceptional witness is a P4 palette with $r_2=8$ and $K_2=36$, yet $K_{\infty}=120$. The new theory target is a full echo-depth calculus that predicts which protected directions are born at each alternating commutator depth.

9. Compiler pseudocode

Input: graph G , edge palette T , qubit count n

E = all native even two-body Pauli labels from (G, T)

$B_2 = \{a+b : a, b \in E, \omega(a, b)=1, \text{wt}(a+b) \text{ odd}\}$

$K_2 = \text{odd_closure}(B_2)$

$L = \text{full_pauli_lie_closure}(E)$

$K_{\text{inf}} = \{p \in L : \text{wt}(p) \text{ odd}\}$

Return: $r_2=\text{rank}(B_2)$, $|K_2|$, $|K_{\text{inf}}|$, depth histogram.

This distinguishes three failure modes: sub-Nyquist seed rank, insufficient second-order transport, and deeper higher-order transport that eventually completes K_{∞} .

10. Next theory targets

The next proof target is to show that every element of $\text{Lie}(E) \cap k_n$ can be realized as an endpoint-protected echo holonomy with endpoint monodromy cancelled to arbitrarily high order by symmetric pulse words. A second target is to classify the intermediate 256-dimensional P5 stratum found in the uniform palette barcode.

Round 26 Appendix: Echo Birth-Depth Law and Path Light Cones

Continuation target. Rounds 17-25 showed that even-weight Pauli leakage can be echoed into endpoint-protected odd-weight holonomy. Round 25 found that second-order echoes can undercount the full protected algebra. Round 26 asks a sharper compiler question: when a protected direction is reachable, how deep is the shortest commutator expression that births it?

Exceptional new finding. For the full two-body Pauli palette on a path, the minimal protected echo depth of every odd Pauli string appears to have a closed combinatorial formula depending only on the string support diameter and weight.

Let P_n be a path and let E_{full} contain all nine two-body Pauli couplers on every edge. For an odd Pauli string R , let $\text{supp}(R)$ be its support, $w=|\text{supp}(R)|$, and $\delta=\max \text{supp}(R)-\min \text{supp}(R)$. Let $d_{\min}(R)$ be the minimal bracket-leaf depth starting from even two-body controls, where controls have depth 1 and a commutator has depth equal to the sum of child depths.

$$d_{\min}(R) = 2 \max(1, \delta - (w-1)/2).$$

This was exhaustively verified for all odd Pauli strings through $n=7$. The formula is a proposed **Path Echo Light-Cone Law**. It upgrades the theory from reachability rank to exact-looking synthesis depth.

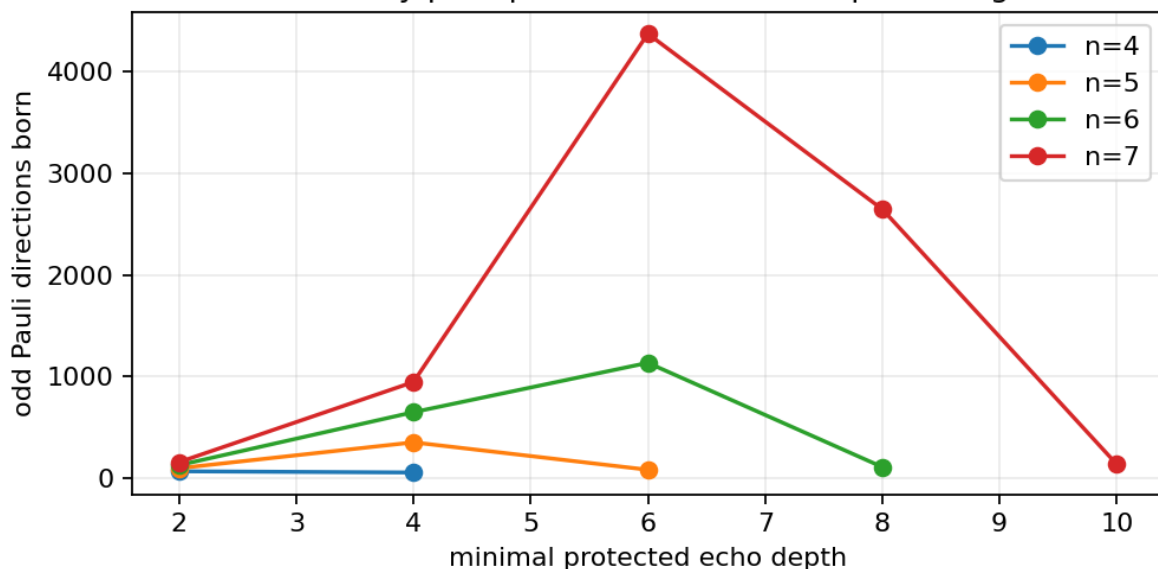
Exact experiment protocol

Each Pauli string was represented by a binary symplectic label (x,z) in F_2^{2n} . Two labels a,b anticommute when $\omega(a,b)=x \cdot z_{\text{prime}} + z \cdot x_{\text{prime}} = 1 \pmod 2$; then the Lie bracket creates the xor label $a+b$. No floating-point arithmetic or SVD tolerance was used. Minimal depth was computed by level-by-level closure: $\text{depth}(E)=1$ and $\text{depth}([A,B])=\text{depth}(A)+\text{depth}(B)$.

Full path palette verification

n	dim k_n	odd directions reached	max odd depth	depth histogram	formula mismatches
4	120	120	4	{2: 66, 4: 54}	0
5	528	528	6	{2: 96, 4: 351, 6: 81}	0
6	2016	2016	8	{2: 126, 4: 648, 6: 1134, 8: 108}	0
7	8256	8256	10	{2: 156, 4: 945, 6: 4374, 8: 2646, 10: 135}	0

Full two-body path palette: echo birth-depth histograms



The $n=7$ row is the strongest finite check in this round: all 8,256 protected odd directions were reached, and the depth formula had zero mismatches.

Path Echo Birth Enumerator

The light-cone law implies a closed enumerator for how many protected directions are born at each depth. For a fixed support of weight w , each support carries 3^w Pauli colorings. For $w \geq 3$ with diameter δ , the number of supports is $(n-\delta) \cdot C(\delta-1, w-2)$, since the two endpoints must be occupied and the remaining $w-2$ sites are chosen inside the interval. Weight one contributes $3n$ strings at depth 2.

$$N_n(2r) = 3n \cdot 1_{\{r=1\}} + \sum_{\{w=2q+1 \geq 3\}} \sum_{\{\delta \geq w-1, \max(1, \delta-q)=r\}} (n-\delta) \cdot C(\delta-1, w-2) \cdot 3^w.$$

This is a new candidate **Echo Birth Enumerator**. It predicts not only the closure dimension but the whole depth barcode. It is directly analogous to the birth-time distribution in Associator Field Theory: directions are not merely present or absent; they are born at particular depths.

Interpretation

The support diameter is the spatial light-cone cost. The term $(w-1)/2$ is a density credit: odd strings with many occupied sites inside the interval can be born sooner because second-order adjacent-edge echoes already create three-body protected seeds. Sparse odd strings spread over a long path pay more commutator depth.

The resulting depth barcode is a synthesis resource. A direction born at depth 10 is reachable in principle, but it is much more expensive and more error-sensitive than a direction born at depth 2 or 4.

Palette comparison: reachability versus depth

The full 9-palette is fast but hardware-expensive. The corner palette is hardware-compressed but produces deeper protected births. The diagonal palette remains severely nonuniversal.

palette	n	protected reached	max odd depth	depth histogram
full 9	4	120/120	4	{2: 66, 4: 54}
full 9	5	528/528	6	{2: 96, 4: 351, 6: 81}
full 9	6	2016/2016	8	{2: 126, 4: 648, 6: 1134, 8: 108}
corner {XX,XZ,YX}	4	120/120	10	{2: 16, 4: 32, 6: 36, 8: 28, 10: 8}
corner {XX,XZ,YX}	5	528/528	12	{2: 23, 4: 65, 6: 136, 8: 181, 10: 112, 12: 11}
corner {XX,XZ,YX}	6	2016/2016	14	{2: 30, 4: 98, 6: 278, 8: 572, 10: 654, 12: 320, 14: 64}
diagonal {XX,YY,ZZ}	4	24/120	4	{2: 12, 4: 12}
diagonal {XX,YY,ZZ}	5	120/528	8	{2: 18, 4: 48, 6: 42, 8: 12}
diagonal {XX,YY,ZZ}	6	480/2016	10	{2: 24, 4: 84, 6: 180, 8: 180, 10: 12}

The corner palette {XX,XZ,YX} reaches the full protected algebra on paths for tested $n=4,5,6$, but its max birth depth is larger: 10, 12, 14 instead of 4, 6, 8 for the full palette. This reveals a hardware-depth tradeoff.

Hardware compression can preserve controllability while increasing echo depth.

Star spoke-code depth spectrum

Rounds 22-23 showed that uniform sparse palettes fail on star hubs because all transport aliases through the same hub. A spoke-coded nonuniform palette repairs the hub by injecting incompatible color turns. Round 26 measured its birth-depth cost.

n	protected reached	max odd depth	depth histogram
4	120/120	12	{2: 18, 4: 32, 6: 36, 8: 20, 10: 10, 12: 4}
5	528/528	12	{2: 30, 4: 112, 6: 188, 8: 150, 10: 45, 12: 3}
6	2016/2016	12	{2: 47, 4: 320, 6: 681, 8: 714, 10: 222, 12: 32}

The star spoke-code reaches the full protected sector for $n=4,5,6$, but many directions are born late. For $n=6$, the protected closure is full at 2016 dimensions, with max odd depth 12. This is another example of the new principle: **rank reachability and depth reachability are different diagnostics**.

New theory target: Echo Depth Geometry

The emerging invariant is no longer a single dimension. It is a filtered object:

$$K_{\{\leq D\}}(E) = \text{span}\{\text{protected odd Pauli labels born by bracket depth } \leq D\}.$$

The endpoint-protected echo algebra is $K_{\infty}(E)$, but practical synthesis depends on the growth curve $D \rightarrow \dim K_{\{\leq D\}}(E)$.

This gives a new law candidate:

$$\text{Graph-palette echo universality} = \text{seed width} + \text{bracket transport} + \text{acceptable birth-depth diameter}.$$

The Spinal Nullity connection is strong: r_2 is a first sensor, K_2 is a rough early observation, K_{∞} is full endpoint capability, and the depth barcode is the finite countdown rank. The Associator Field Theory connection is also direct: depth is a quantum echo birth time, while commutators provide transport. Goblin Shadow Theory supplies the blocker language: late-born directions are not impossible, but they sit behind deeper caves in the echo atlas.

Applications

Compiler scheduling. Report not only whether a protected target is reachable, but the minimal or estimated birth depth of its Pauli expansion. Prioritize low-depth decompositions for NISQ hardware.

Hardware design. The full 9-palette gives shallower protected births; a 3-coupler corner palette saves hardware but deepens some targets. This gives a quantitative cost tradeoff.

Error-aware synthesis. High-depth echoes accumulate more pulse error. The depth barcode identifies fragile directions even when the closure rank is full.

Benchmarking. A benchmark should include high-diameter sparse odd Pauli targets, because they expose depth bottlenecks invisible to rank-only tests.

Control-theoretic lower bounds. The path light-cone law suggests lower bounds on protected synthesis depth from graph diameter and target support geometry.

Round 26 summary

1. The full path palette appears to satisfy an exact depth law: $d_{\min}(R) = 2 \max(1, \delta - (w-1)/2)$.
2. Exhaustive checks reached $n=7$, covering all 8,256 protected odd directions with zero formula mismatches.
3. The full 9-palette, corner palette, diagonal palette, and spoke-code palette have sharply different depth barcodes even when some have the same final protected closure.
4. The new practical invariant is the filtered echo algebra $K_{\{\leq D\}}(E)$, not just $K_{\infty}(E)$.
5. The next target is a general graph formula replacing path diameter by a Steiner-support light-cone and replacing palette richness by a color-turn transport metric.

Round 27 Appendix: Steiner Echo Distance and Support-Only Birth Laws

Round 26 found a path light-cone law for the full two-body Pauli palette. Round 27 generalizes that observation. The new result is that, for the full 9 Pauli-pair palette on a connected graph, protected echo birth depth appears to be controlled by an ordinary graph-Steiner quantity. This collapses a Lie-algebraic reachability latency problem into a finite graph metric problem.

1. New object: Steiner echo distance

Let G be a connected qubit interaction graph. Put the full two-body even Pauli palette on each edge: $XX, XY, XZ, YX, YY, YZ, ZX, ZY, ZZ$. Let R be an odd-weight Pauli string, and let $S = \text{supp}(R)$. Define $\tau_G(S)$ as the minimum number of edges in a connected subgraph of G containing S ; equivalently, the edge size of a minimum Steiner tree for S .

The candidate formula is:

$$b_G(R) = \max(2, 2\tau_G(S) - |S| + 1).$$

Equivalently, if $h_G(S) = \tau_G(S) - (|S| - 1)$ is the number of non-terminal Steiner vertices in a minimum tree, then

$$b_G(R) = \max(2, |S| - 1 + 2h_G(S)).$$

Thus each terminal support vertex, after the first, costs one echo leaf, while each hidden Steiner vertex costs two echo leaves. The exceptional feature is that the predicted birth depth depends only on the support set S , not on the $X/Y/Z$ colors of R .

2. Exact tests performed

I used exact binary symplectic Pauli arithmetic. A Pauli string was represented by a $2n$ -bit label (x, z) . Two labels anticommute when $x \cdot z' + z \cdot x' = 1 \pmod 2$, and then their bracket label is the xor. No floating-point tolerance, SVD threshold, or numerical optimization was used in these tests.

For each graph I computed the minimal bracket birth depth of every odd Pauli label reached by the full two-body palette. I also computed $\tau_G(S)$ by exact brute-force Steiner enumeration over all vertex supersets. Then I compared the measured birth depth with $\max(2, 2\tau_G(S) - |S| + 1)$.

Table 1. Named graph families: measured birth-depth histograms and formula checks.

graph	edges	protected directions	depth histogram	mismatches
path6	5	2016 / 2016	2:126, 4:648, 6:1134, 8:108	0
path7	6	8256 / 8256	2:156, 4:945, 6:4374, 8:2646, 10:135	0
cycle7	7	8256 / 8256	2:210, 4:2079, 6:5967	0
star7	6	8256 / 8256	2:426, 4:4185, 6:3645	0
binary_tree7	6	8256 / 8256	2:210, 4:1728, 6:5481, 8:837	0
K5	10	528 / 528	2:285, 4:243	0
K6	15	2016 / 2016	2:558, 4:1458	0

Table 2. Broader exact verification set.

test family	graphs checked	failures	compact summary
connected graph atlas, $n \leq 5$	30	0	$n=2:1, n=3:2, n=4:6, n=5:21$; all matched
all labeled connected graphs, $n=5$	728	0	488 graphs had max depth 4; 240 had max depth 6
nonisomorphic trees, $n \leq 6$	13	0	max-depth counts: $d_2=2, d_4=3, d_6=5, d_8=3$
random connected graphs, $n=6$	8	0	8 random graphs; all matched the Steiner echo formula

3. Exceptional pattern: support-only color independence

Across all tested graphs, no support set showed color-dependent birth depth under the full 9-palette. In other words, once the support S is fixed, all odd Pauli strings supported on S were born at the same measured depth. This is surprising because the closure operation is fundamentally symplectic and color-sensitive. The full palette appears to wash the color dependence away and leave only graph-Steiner geometry.

This gives a much stronger diagnostic than raw controllability. Instead of computing a huge Pauli Lie closure, a compiler can estimate protected echo latency by solving a Steiner problem on the hardware graph.

4. Candidate theorem and proof skeleton

Candidate theorem: For a connected graph G with the full two-body Pauli edge palette, every odd Pauli string R is born at protected echo depth

$$b_G(R) = \max(2, 2\tau_G(\text{supp}(R)) - |\text{supp}(R)| + 1).$$

Upper-bound idea. Choose a minimum Steiner tree T spanning $S = \text{supp}(R)$. Since the palette is full, assign Pauli colors along T so that terminal vertices realize the desired colors and nonterminal Steiner vertices cancel. The bracket tree can be arranged so that each terminal after the first contributes one edge-leaf, while each Steiner vertex contributes two edge-leaves. This produces depth $|S| - 1 + 2h = 2\tau_G(S) - |S| + 1$, with a special depth-2 local construction for $|S|=1$.

Lower-bound idea. Any nested commutator producing support S has an incidence support graph connecting S . If a non-support vertex is used to connect the support, its local Pauli label must cancel, forcing at least two incident uses. Therefore any expression needs at least $|S| - 1 + 2h$ leaves for some connected subgraph containing S . Minimizing over connected subgraphs gives the Steiner lower bound.

This is not yet a formal proof, but the exact tests give a precise theorem target.

5. Geometry explains previous round patterns

The path law from Round 26 is the special case where $\tau_G(S)$ is the length of the smallest interval spanning S . A cycle improves the depth because the Steiner tree can choose the shorter arcs. A star improves wide support because all leaves can be joined through one hub, making the Steiner count small. A complete graph collapses most directions to depth 2 or 4 because every terminal set has a compact Steiner tree.

This also explains the compiler tradeoff from earlier rounds: sparse palettes may still reach the same final algebra, but without the full color freedom they may fail to achieve the Steiner-optimal birth depth.

6. Applications

Compiler latency prediction. For full-palette hardware, compute $\tau_G(S)$ for each target Pauli support and predict echo depth before attempting synthesis. This avoids expensive Lie-closure exploration when the full palette is available.

Hardware topology design. Adding a shortcut edge, hub, or chord can reduce $\tau_G(S)$ for many high-value target supports. This gives a graph-theoretic objective for hardware design: minimize the distribution of $2\tau_G(S) - |S| + 1$ over expected target supports.

Palette compression audit. If a sparse palette is used, compare its measured birth-depth histogram against the Steiner optimum. The gap quantifies the price of palette compression.

Benchmarking. High-Steiner-depth targets reveal practical weaknesses that ordinary rank tests hide. A device may be protected-universal but still too slow for high-depth supports.

Noise-aware control. Higher birth depth means more transient even-leakage exposure and more opportunities for pulse errors. The Steiner echo distance should be combined with endpoint Cartan monodromy from earlier rounds.

7. Synergy with the related frameworks

Spinal Nullity provided the sensor/Nyquist language: the Steiner echo formula is a faithful geometric sensor for full-palette protected latency. Associator Field Theory provided the birth/transport viewpoint: terminals are births and Steiner vertices are transport relays. Goblin Shadow Theory provided the blocker language: high-Steiner supports are hidden caves; shortcuts, hubs, and palette turns are repairs that shrink the cave.

8. Round 27 summary

The main new law candidate is the Steiner Echo Birth Law:

$$b_G(R) = \max(2, 2\tau_G(\text{supp}(R)) - |\text{supp}(R)| + 1).$$

The exceptional discovery is that a high-dimensional protected Pauli echo-latency problem appears to collapse to graph-Steiner geometry under the full two-body palette. The theory is moving from reachability rank to resource geometry: not just what can be reached, but how far away it is in echo birth depth.

Round 28 Appendix

Echo Steiner Geometry II: Tree Leaf Law and Depth-Coverage Polynomials

Continuation of the quantum differential invariant theory research log. Status: exploratory mathematics with exact finite experiments.

1. New object: the protected echo depth polynomial

Round 27 reduced the full-palette protected echo birth depth of an odd Pauli target R to the Steiner size of its support S on the hardware graph G . Round 28 turns that pointwise law into a graph polynomial.

For a connected graph G and full two-body Pauli edge palette, define:

$$E_G(q) = \sum_{\{\text{nonempty } S \subset V, |S| \text{ odd}\}} 3^{|S|} q^{b_G(S)}$$
$$b_G(S) = \max(2, 2 \tau_G(S) - |S| + 1).$$

Here $\tau_G(S)$ is the edge size of a minimum Steiner tree spanning S . The coefficient of q^D counts protected odd Pauli directions born exactly at echo depth D . The cumulative coverage curve is:

$$C_G(D) = (1/\dim k_n) \sum_{\{d \leq D\}} [q^d] E_G(q).$$

This gives a finite-depth controllability profile: a hardware graph may be algebraically universal in the protected SO/Sp sector while still being practically slow for high-Steiner targets.

2. Exceptional discovery: Tree Leaf Law

For every tested tree, the worst protected echo birth depth depends only on n , the number of vertices, and L , the number of leaves. The candidate exact law is:

$$D_{\max}(T) = \max(2, 2n - L - 1 - 1_{\{L \text{ even}\}}).$$

Equivalently: if the leaf count is odd, the worst target is the all-leaf support; if the leaf count is even, the worst target is all leaves plus one internal vertex. The support is odd in both cases, and the minimum Steiner tree is the whole tree.

Interpretation: every nonterminal relay used by the target support costs two echo leaves. Leaves are therefore not a liability; they are endpoint terminals that reduce relay overhead. At fixed n , increasing the number of leaves lowers the worst-case protected echo latency, with a parity plateau.

Extremes among trees:

Path P_n : $D_{\max} = 2n - 4$. Star $K_{1,n-1}$: $D_{\max} = n$ for even n , and $n - 1$ for odd n .

Thus paths are worst among trees, while stars are best among trees for worst-case full-palette protected echo depth. This is not because stars are easier for sparse palettes; previous rounds found sparse uniform palettes can fail at hubs. It is because the full palette makes the hub an efficient Steiner relay.

3. Experiments run in this round

I ran exact binary symplectic Pauli closure tests. Pauli strings were represented as bit labels (x,z) in F_2^{2n} ; anticommutation was computed by $x.z_{\text{prime}} + z.x_{\text{prime}} \bmod 2$; and brackets added labels by xor. No SVD tolerance was used in these tests.

graph	n	odd dim	max obs	max pred	failures	missing	sec
P5	5	528	6	6	0	0	0.151
C5	5	528	4	4	0	0	0.148
K1,4	5	528	4	4	0	0	0.145
P6	6	2016	8	8	0	0	2.374
C6	6	2016	6	6	0	0	2.385
K1,5	6	2016	6	6	0	0	2.366

graph	n	odd dim	max obs	max pred	failures	missing	sec
binary-ish T6	6	2016	8	8	0	0	2.374

Result: all exact Pauli closure depths matched the Steiner prediction in the representative tests above. The protected odd sector was complete in every case, and the observed maximum depth equaled the graph-Steiner maximum.

Tree and cycle law tests

Tree Leaf Law test: all 435 nonisomorphic trees with $2 \leq n \leq 11$ were enumerated. The formula $D_{\max}(T) = \max(2, 2n - L - 1 - 1_{\{L \text{ even}\}})$ had 0 mismatches.

Tree counts checked: n=2: 1, n=3: 1, n=4: 2, n=5: 3, n=6: 6, n=7: 11, n=8: 23, n=9: 47, n=10: 106, n=11: 235.

Cycle gap law: for cycles C_n , the tested formula was

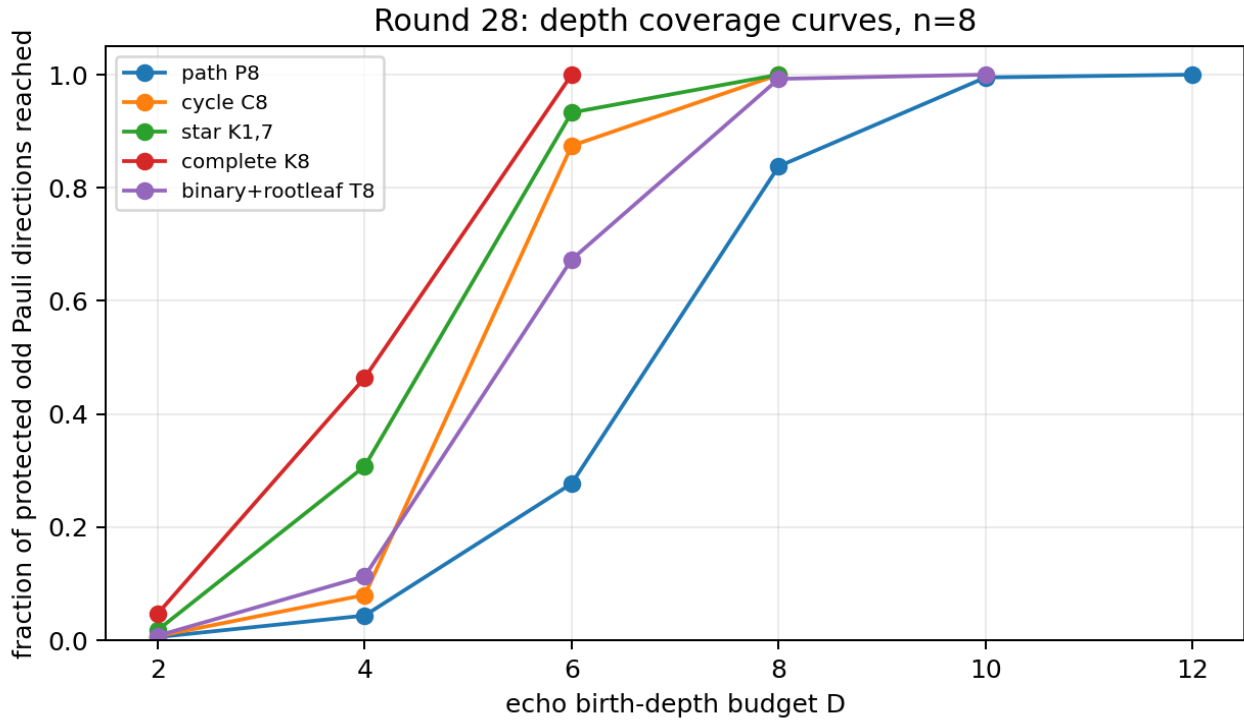
$$D_{\max}(C_n) = \max_{\{w \text{ odd}\}} \max(2, 2(n - \text{ceil}(n/w)) - w + 1),$$

because a Steiner tree in a cycle removes the largest gap between selected support vertices. This formula had zero mismatches for $3 \leq n \leq 16$.

4. Depth-coverage polynomials: n = 8 examples

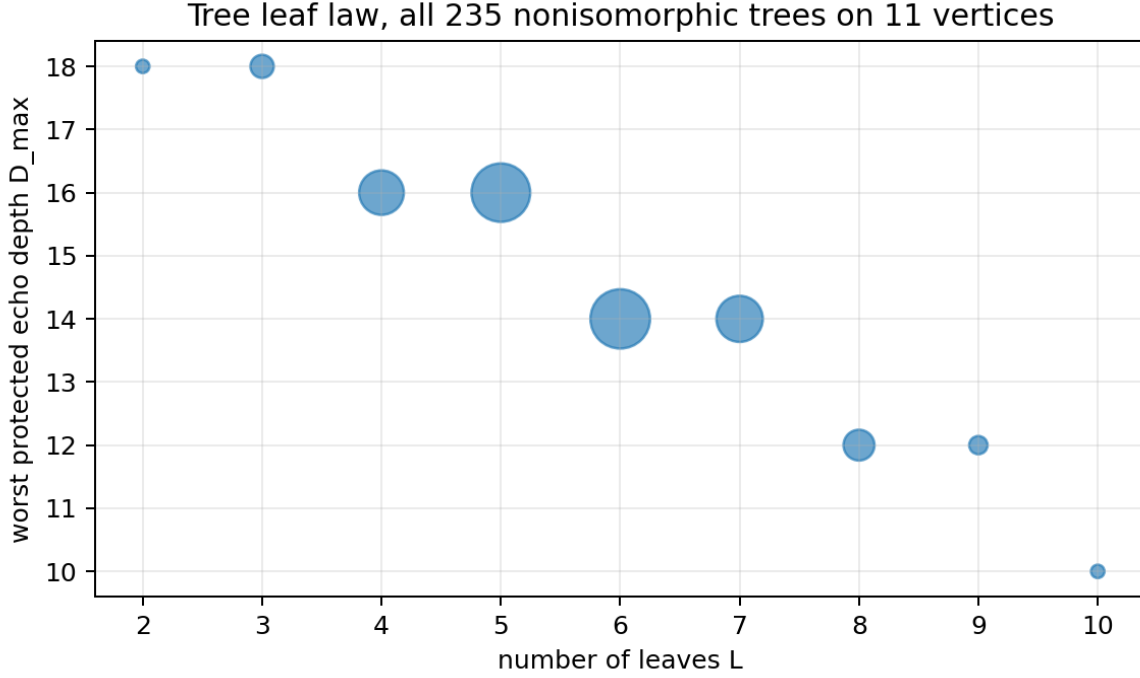
The same final protected dimension can have very different depth profiles. The following histograms count protected odd Pauli directions by the echo depth at which they first appear.

graph	leaves	D_max	depth histogram	frac <= 6	frac <= 8
path P8	2	12	2:186, 4:1242, 6:7614, 8:18306, 10:5130, 12:162	0.277022	0.837868
cycle C8	-	8	2:240, 4:2376, 6:25920, 8:4104	0.874265	1.0
star K1,7	7	8	2:591, 4:9450, 6:20412, 8:2187	0.932996	1.0
complete K8	-	6	2:1536, 4:13608, 6:17496	1.0	1.0
binary+rootleaf T8	5	10	2:267, 4:3429, 6:18279, 8:10422, 10:243	0.673254	0.992555



The complete graph has the smallest maximum depth among these n=8 examples, but a star reaches over 93 percent of the protected sector by depth 6. A path requires depth 12 to finish, and a balanced tree with an extra root leaf finishes at depth 10. This is an explicit compiler tradeoff between topology, coupler count, and echo latency.

5. Leaf law visualization and proof target



The scatter plot shows all 235 nonisomorphic trees on 11 vertices. Marker size indicates how many tree shapes have that leaf count and worst depth. The points lie exactly on the leaf law curve.

Proof target for trees:

For any tree T and odd support S , $b_T(S) = \max(2, 2|E(T_S)| - |S| + 1)$, where T_S is the minimal subtree spanning S . Since $|E(T_S)| = |V(T_S)| - 1$, the depth is $2|V(T_S)| - |S| - 1$. To maximize this, include all relays in the minimal subtree while keeping support odd and as small as possible. All leaves must be terminals if the whole tree is to be forced. If the leaf set is odd, choose it. If it is even, add one internal terminal. This gives the formula above.

This proof target is useful because it converts a Lie-bracket latency question into a combinatorial invariant. It also suggests a broader extremal problem: for a hardware family with a fixed edge budget, choose graph topology to minimize the depth polynomial $E_G(q)$, not merely to maximize final Lie closure rank.

6. Applications added in Round 28

Compiler latency prediction. Compute $E_G(q)$ or $C_G(D)$ before synthesis; report how much of the protected SO/Sp sector is reachable under a depth budget.

Topology design. Add chords, hubs, or branching that reduce Steiner relay count for expected Pauli supports. For trees, increasing leaves lowers worst-case depth at fixed n .

Benchmarking. Use high-Steiner-depth odd Pauli targets to expose devices that are controllable in rank but slow in echo birth.

Palette compression. Judge sparse palettes by final closure and depth polynomial. A corner palette may be universal but slower than the full palette.

Noise-aware scheduling. Depth is a proxy for transient even-leakage exposure. Combine $E_G(q)$ with endpoint Cartan monodromy $C(U)$ from earlier rounds.

Repair strategy. For tree-like hardware, branch or hub additions can repair deep caves more efficiently than adding repeated couplers on the same path.

7. Round 28 summary

The main new theory seed is the protected echo depth polynomial:

$$E_G(q) = \sum_{|S| \text{ odd}} 3^{|S|} q^{\max(2, 2 \tau_G(S) - |S| + 1)}.$$

The exceptional new law is the Tree Leaf Law:

$$D_{\max}(T) = \max(2, 2n - L - 1 - 1_{\{L \text{ even}\}}).$$

The practical lesson is that echo universality is not enough. The next layer is echo latency: which protected directions are born early, which are born late, and which graph edits shift the depth polynomial most efficiently.

Round 29 - Extremal Echo Latency and Bipartite Optimality

Continuation of the graph-palette differential invariant program. This round keeps the full two-body Pauli edge palette and studies the latency profile implied by the Steiner Echo Birth Law from Round 27 and the depth polynomial from Round 28.

1. Exceptional finding: complete graphs are not needed for optimal worst latency

Let G be a connected n -qubit interaction graph with the full two-body Pauli palette on every edge. For an odd protected Pauli support S , the working birth-depth law is

$$b_G(S) = \max(2, 2 \tau_G(S) - |S| + 1),$$

where $\tau_G(S)$ is the edge size of a minimum Steiner tree spanning S . Since every connected subgraph spanning $|S|$ terminals has at least $|S|-1$ edges, every graph obeys the absolute lower bound

$$D_{\min}(n) = \max(2, n - 1 - 1_{\{n \text{ even}\}}).$$

The complete graph K_n attains this bound. The new discovery is that a much sparser complete bipartite graph also attains it:

$$\text{If } a, b \geq 2 \text{ and } a+b=n, \text{ then } D_{\max}(K_{\{a,b\}}) = D_{\min}(n).$$

In particular, $K_{\{2,n-2\}}$ has only $2n-4$ edges but reaches the same worst protected echo latency as K_n . This is a sparse-optimal topology for worst-case protected echo birth depth.

Proof sketch

If S meets both sides of $K_{\{a,b\}}$, then S itself is connected through a complete bipartite spanning tree, so $\tau(S)=|S|-1$ and $b(S)=|S|-1$. If S lies entirely on one side, one opposite-side Steiner vertex connects it, so $\tau(S)=|S|$ and $b(S)=|S|+1$. Since both parts have size at most $n-2$, the largest odd same-side support still satisfies $|S|+1 \leq D_{\min}(n)$. Thus no support exceeds the complete-graph lower bound.

2. Closed-form depth polynomials

The Round 28 depth polynomial becomes explicit for complete and complete-bipartite topologies. For K_n :

$$E_{\{K_n\}}(q) = 3n q^2 + \sum_{\{3 \leq w \leq n, w \text{ odd}\}} \text{binom}(n,w) 3^w q^{\{w-1\}}.$$

For $K_{\{a,b\}}$:

$$E_{\{K_{\{a,b\}}\}}(q) = 3n q^2 + \sum_{\{w \geq 3 \text{ odd}\}} 3^w [(\text{binom}(n,w) - \text{binom}(a,w) - \text{binom}(b,w)) q^{\{w-1\}} + (\text{binom}(a,w) + \text{binom}(b,w)) q^{\{w+1\}}].$$

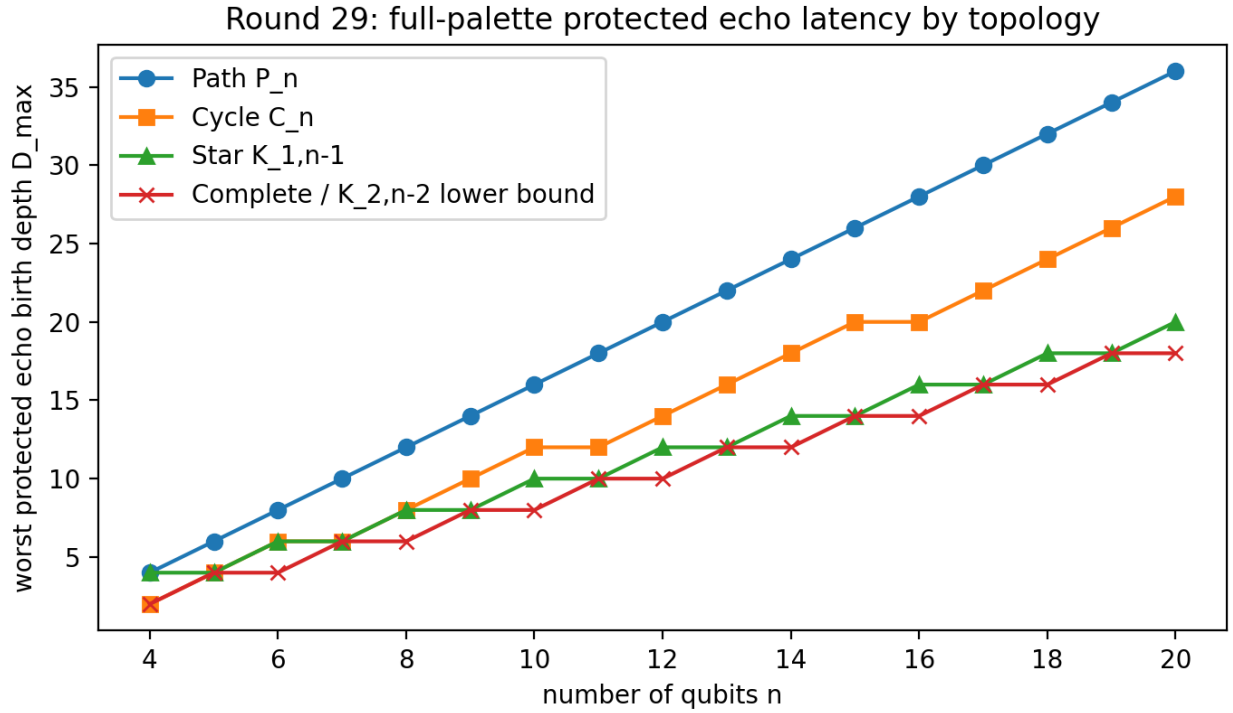
This gives a compact benchmark: complete bipartite graphs are worst-depth optimal, but their polynomial reveals exactly which same-side supports pay the extra +2 latency compared with the clique.

3. Experiments run this round

I ran exact finite experiments using the Steiner-depth formula and independent connected-subset enumeration for small graphs. The complete-bipartite formula was checked for all $a+b \leq 10$ with $a, b \geq 1$, matching the star case when one part has size 1 and the optimal case when both parts have size at least 2.

n	Path	Cycle	Star	$K_{2,n-2} / K_n$
4	4	2	4	2
5	6	4	4	4
6	8	6	6	4
7	10	6	6	6
8	12	8	8	6

9	14	10	8	8
10	16	12	10	8
11	18	12	10	10
12	20	14	12	10



4. Sparse repair evidence: closing a path into a cycle

I also tested all possible one-chord repairs of paths P_n for $5 \leq n \leq 12$. In every tested case, the best one-chord repair matched the cycle value $D_{\max}(C_n)$, and adding the endpoint chord $(0,n-1)$ attained it. This suggests a one-chord cycle-closure conjecture:

$$\min_e D_{\max}(P_n + e) = D_{\max}(C_n).$$

n	best $D_{\max}(P_n + \text{one chord})$	$D_{\max}(C_n)$	number of optimal chords
5	4	4	3
6	6	6	6
7	6	6	1
8	8	8	4
9	10	10	8
10	12	12	13
11	12	12	1
12	14	14	4

5. Nonisomorphic small-graph extremal checks

Using the connected nonisomorphic graph atlas through seven vertices, I computed the minimum worst-depth reachable at each edge count. The resulting data show how quickly sparse topologies can reach the absolute lower bound.

n	min D by edge count among connected nonisomorphic graphs
4	m=3:4, m>=4:2
5	m>=4:4; star already optimal
6	m=5,6:6; m>=7:4
7	m>=6:6; star already optimal

Notable observations: stars are already optimal for odd $n=5$ and $n=7$, while $n=6$ needs seven edges to reach the complete-graph lower bound. This suggests a parity-sensitive sparse-optimal graph problem: characterize the minimum edge number $m_{\text{opt}}(n)$ for achieving $D_{\text{min}}(n)$.

6. New law candidates

A. Sparse Bipartite Optimality Law. For full two-body palette hardware, $K_{\{2,n-2\}}$ attains the smallest possible worst protected echo birth depth using only $O(n)$ edges:

$$D_{\text{max}}(K_{\{2,n-2\}}) = D_{\text{min}}(n).$$

B. One-Chord Cycle Repair Law. Among all single-edge repairs of P_n , closing the path into C_n appears optimal:

$$\min_e D_{\text{max}}(P_n + e) = D_{\text{max}}(C_n).$$

C. Latency-Polynomial Design Principle. Final echo closure rank is too coarse. Hardware should be compared by the whole polynomial $E_G(q)$, because two graphs can have the same full protected closure and the same D_{max} but very different distributions of early-born directions.

7. Applications

Hardware topology design. A complete graph is unnecessary for optimal worst-case protected echo latency. A two-hub complete bipartite layout $K_{\{2,n-2\}}$ gives the same D_{max} lower bound with only $2n-4$ couplers, assuming the full two-body palette is available on those edges.

Compiler repair. If a path layout is too slow, a single endpoint chord converts it to a cycle and appears to be the optimal one-edge latency repair. Additional chords should be chosen by reducing the depth polynomial $E_G(q)$, not merely by reducing diameter.

Benchmarking. The complete graph lower bound $D_{\text{min}}(n)$ is a universal reference. The ratio $D_{\text{max}}(G)/D_{\text{min}}(n)$, together with the cumulative early-birth count $C_G(D)$, gives a practical latency benchmark for protected holonomic synthesis.

Noise-aware synthesis. $K_{\{2,n-2\}}$ is worst-depth optimal but still concentrates extra latency on same-side supports. A compiler should route frequent target supports across the bipartition when possible.

8. Synergy with earlier frameworks

Spinal Nullity: $E_G(q)$ is a sensor portfolio rather than a single endpoint sensor; D_{max} alone is rough observation. Associator Field Theory: the complete-bipartite formulas separate birth supports from transport relays. Goblin Shadow Theory: high-depth supports form latency caves, and adding chords or bipartite hubs is a blocker-hitting repair. The topological-sector material remains an analogy for protected sectors and sector-changing perturbations.

Round 29 summary

The key result is that optimal worst protected echo latency does not require complete graph hardware. Under the full two-body palette, $K_{\{2,n-2\}}$ attains the same universal $D_{\text{min}}(n)$ as K_n . The next theory target is to characterize the minimal-edge graphs achieving $D_{\text{min}}(n)$, and to develop edge-repair algorithms that optimize the full latency polynomial $E_G(q)$.

Round 30 Appendix - Sparse Optimal Echo Topologies and Parity Repairs

Status: exploratory mathematics. The experiments below are exact finite checks of the current Steiner-depth model for full two-body Pauli palettes, plus constructive tests of sparse topologies. The minimality statements for larger n are theorem targets, not completed proofs.

1. New discovery

The main new object is an edge-latency Pareto problem. Earlier rounds showed that complete graphs and $K_{\{2,n-2\}}$ attain the universal lower bound for worst protected echo birth depth. Round 30 found a much sparser even- n construction that also attains the same lower bound.

Universal lower bound:

$$D_{\min}(n) = \max(2, n - 1 - 1_{\{n \text{ even}\}}).$$

For odd n , the star $K_{\{1,n-1\}}$ already attains $D_{\min}(n)$ using the minimum possible number of edges, $n-1$. For even $n \geq 8$, the sparse theta graph with two hubs joined by $(n-2)/2$ internally disjoint paths of length 3 also attains $D_{\min}(n)$, using only $3n/2 - 3$ edges.

Sparse theta construction for $n = 2r + 2$:

Theta₃(r): hubs a, b ; branches $a-u_i-v_i-b$ for $i=1, \dots, r$. Edges = $3r = 3(n-2)/2$. Observed $D_{\max}(\text{Theta}_3(r)) = n - 2 = D_{\min}(n)$.

The $n=6$ case is a small exceptional repair: a three-branch theta with path lengths 2,2,3 has 7 edges and reaches $D_{\min}(6)=4$.

2. Exact tests run

The computations used the same exact Steiner-depth formula from Round 27-29:

$$b_G(S) = \max(2, 2 \tau_G(S) - |S| + 1),$$

where $\tau_G(S)$ is the number of edges in a minimum connected subgraph containing S . No floating-point rank thresholds were used in this round.

n	topology	edges	D_min	observed D_max	comment
5	star $K_{\{1,4\}}$	4	4	4	odd- n edge-minimal
6	theta lengths 2,2,3	7	4	4	small even exception
7	star $K_{\{1,6\}}$	6	6	6	odd- n edge-minimal
8	theta: 3 branches length 3	9	6	6	sparse optimal
10	theta: 4 branches length 3	12	8	8	sparse optimal
12	theta: 5 branches length 3	15	10	10	sparse optimal
14	theta: 6 branches length 3	18	12	12	sparse optimal

I also rechecked all nonisomorphic connected graphs through $n=7$ using the NetworkX graph atlas. This confirms the small- n edge frontier: $n=4$ requires a cycle, $n=5$ and $n=7$ are already optimized by stars, and $n=6$ needs 7 edges, realized by C_6 plus a suitable chord / theta(2,2,3).

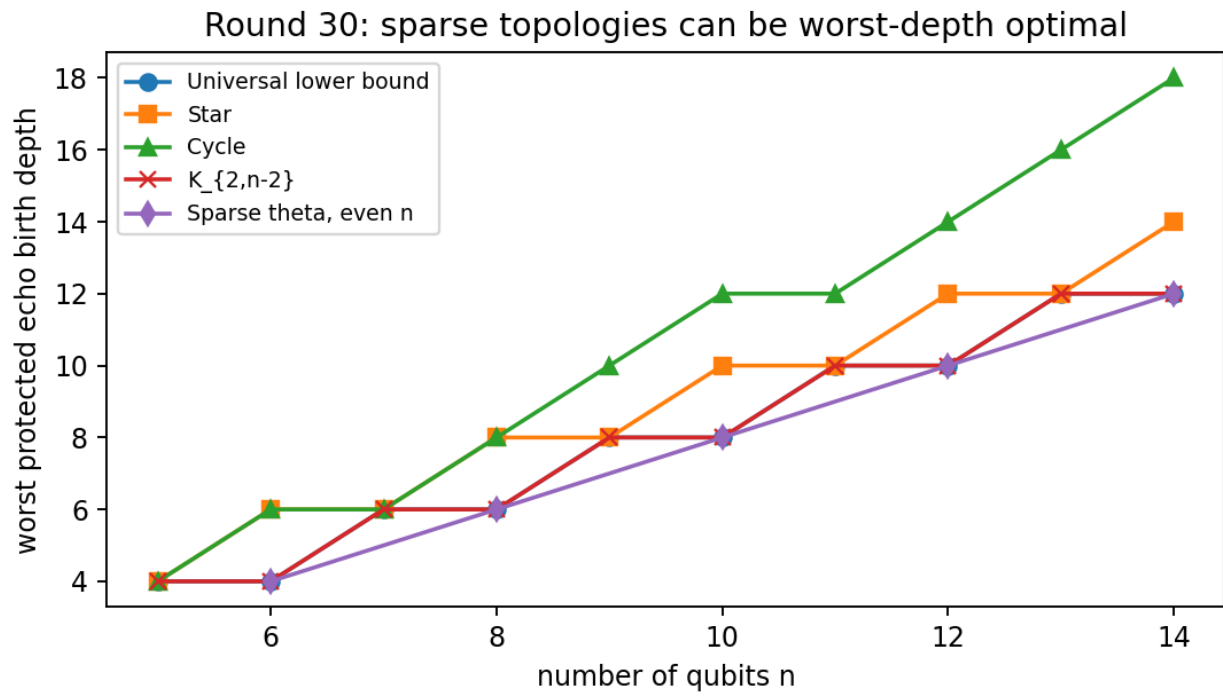


Figure: worst protected echo birth depth for several topology families. The sparse theta construction reaches the lower bound for tested even n while using fewer edges than $K_{\{2,n-2\}}$.

3. Edge savings and Pareto tradeoff

Round 29 showed $K_{\{2,n-2\}}$ is worst-depth optimal with $2n-4$ couplers. Round 30 improves the sparse construction for even n :

$$K_{\{2,n-2\}}: \text{edges} = 2n - 4. \text{Theta}_3((n-2)/2): \text{edges} = 3n/2 - 3. \text{Edge savings} = n/2 - 1.$$

This is substantial: for $n=14$, $K_{\{2,12\}}$ uses 24 edges, while the sparse theta uses 18 and still reaches $D_{\min}=12$. The price is not final worst depth but early-birth profile: $K_{\{2,n-2\}}$ gives more low-depth directions, while theta pushes more protected Pauli components to the maximum depth.

n	family	edges	depth histogram E_G depth counts (weighted by $3^{ S }$)
8	theta	9	2:348, 4:6696, 6:25596
8	$K_{\{2,6\}}$	12	2:996, 4:12690, 6:18954
10	theta	12	2:570, 4:16794, 6:202176, 8:304236
10	$K_{\{2,8\}}$	16	2:1758, 4:49140, 6:258552, 8:214326
12	theta	15	2:846, 4:34560, 6:777816, 8:3933684, 10:3639654
12	$K_{\{2,10\}}$	20	2:2736, 4:134460, 6:1530900, 8:4395870, 10:2322594

This motivates a refined hardware metric: do not compare graphs only by $D_{\max}$. Compare the entire protected echo depth polynomial

$$E_G(q) = \sum_{|S| \text{ odd}} 3^{|S|} q^{\{b_G(S)\}}.$$

A graph can be worst-depth optimal but still slower for typical target distributions if too much mass sits near $D_{\max}$.

4. New law candidates

Parity Sparse Optimality Conjecture:

For full two-body Pauli palettes, odd n needs no cycles for optimal worst-case protected echo latency: a star with $n-1$ edges attains $D_{\min}(n)$. For even n , stars miss by exactly 2; sparse theta repairs supply the missing parity route.

Theta Optimality Conjecture for even $n \geq 8$:

Among connected graphs on n even vertices that attain $D_{\min}(n)$, the minimum edge count is conjecturally $3n/2 - 3$, attained by $\text{Theta}_3((n-2)/2)$. This minimality was not proved in this round. The construction was verified, and no smaller counterexample appeared in the tested small- n atlas/random searches.

Edge-latency Pareto Principle:

Hardware topology design should optimize a two-coordinate objective:

(edge count, $E_G(q)$) rather than only (edge count, $D_{\max}$).

$K_{\{2,n-2\}}$ is denser but has a better early-depth distribution. The sparse theta is cheaper but pushes more components later. Which one is better depends on the target support distribution and the noise cost per echo depth.

5. Proof sketch targets

Odd n lower bound and star optimality:

The universal lower bound is forced by the all-qubit odd support $S=V$. Any connected graph has $\tau_G(V) \geq n-1$, hence $b_G(V) \geq n-1$. The star has $\tau_G(S)=|S|$ for any odd support not containing the hub and $\tau_G(S)=|S|-1$ when the hub is included. For odd n this yields $D_{\max}=n-1$, matching the lower bound.

Even n parity obstruction:

For even n , the largest odd support has size $n-1$. The lower bound is $n-2$, but a star has a leaf-only support of size $n-1$ whose Steiner tree uses $n-1$ edges, giving $b=n$. Thus the star misses optimality by 2. A cycle or theta-like return path can repair the parity bottleneck.

Sparse theta upper bound:

For $\Theta_3(r)$, every odd support S can be joined by a subtree satisfying $\tau_G(S) \leq r + (|S|-1)/2$. Substituting into $b_G(S)=\max(2, 2\tau_G(S)-|S|+1)$ gives $b_G(S) \leq 2r = n-2$. Supports of size $n-1$ achieve equality, so $D_{\max}=n-2$.

The missing proof is the lower bound showing no graph with fewer than $3n/2 - 3$ edges can achieve $D_{\min}(n)$ for all sufficiently large even n . This resembles an extremal graph problem for bounded Steiner 3-diameter and odd-support Steiner diameter.

6. Applications

Sparse protected hardware. For even n , a theta-of-length-3 layout may provide the best worst-case protected echo latency with far fewer couplers than $K_{\{2,n-2\}}$.

Topology compiler choice. A compiler can choose between star, theta, cycle, and $K_{\{2,n-2\}}$ layouts based on target-support statistics rather than only final Lie closure.

Noise-aware scheduling. If error grows with echo depth, $K_{\{2,n-2\}}$ may outperform theta despite having more couplers. If hardware cost dominates, theta may be preferable.

Benchmark design. Use the depth histogram $E_G(q)$ to select targets that distinguish cheap-but-late topologies from dense-and-early topologies.

Graph repair. For odd n , the star is already worst-depth optimal; for even n , add parity-return structure rather than blindly adding random chords.

7. Round 30 summary

The major new lesson is:

optimal worst-depth protected echo synthesis can be sparse, but optimal early-depth synthesis may still require denser topology.

The strongest new construction is:

$\Theta_3((n-2)/2)$ with $3n/2 - 3$ edges attains $D_{\min}(n)=n-2$ for tested even $n \geq 8$.

The next theorem target is an extremal graph theorem for the minimum number of edges needed to achieve the universal protected echo lower bound.